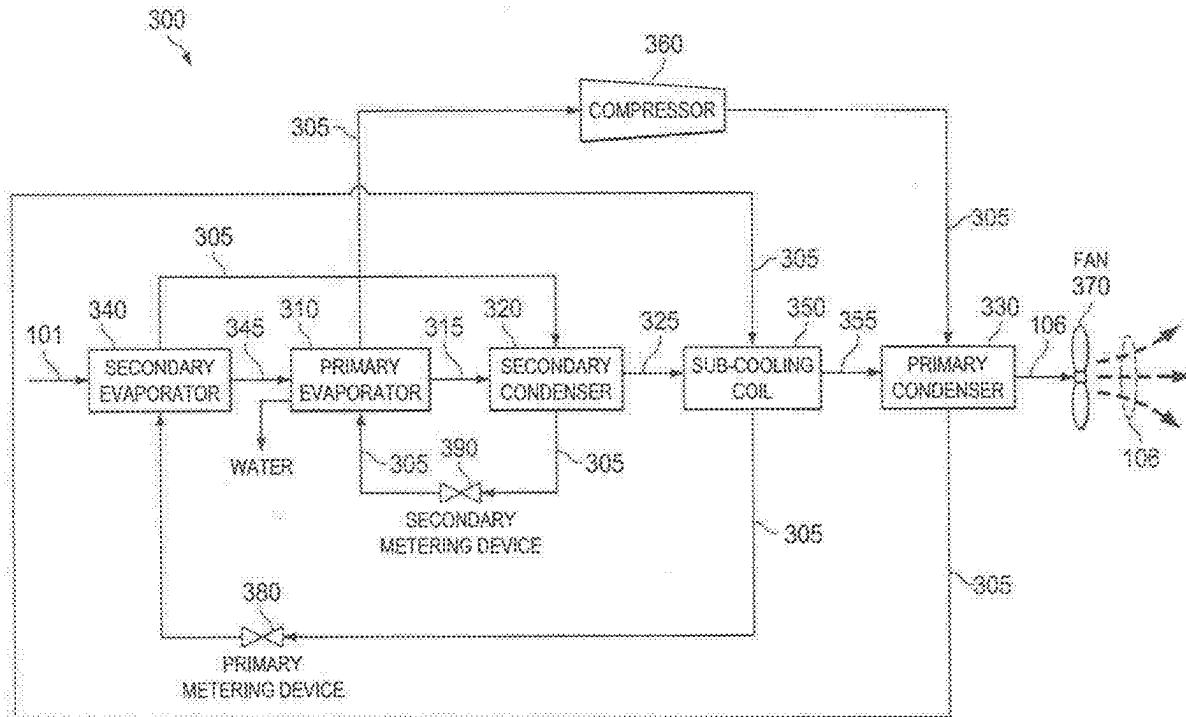
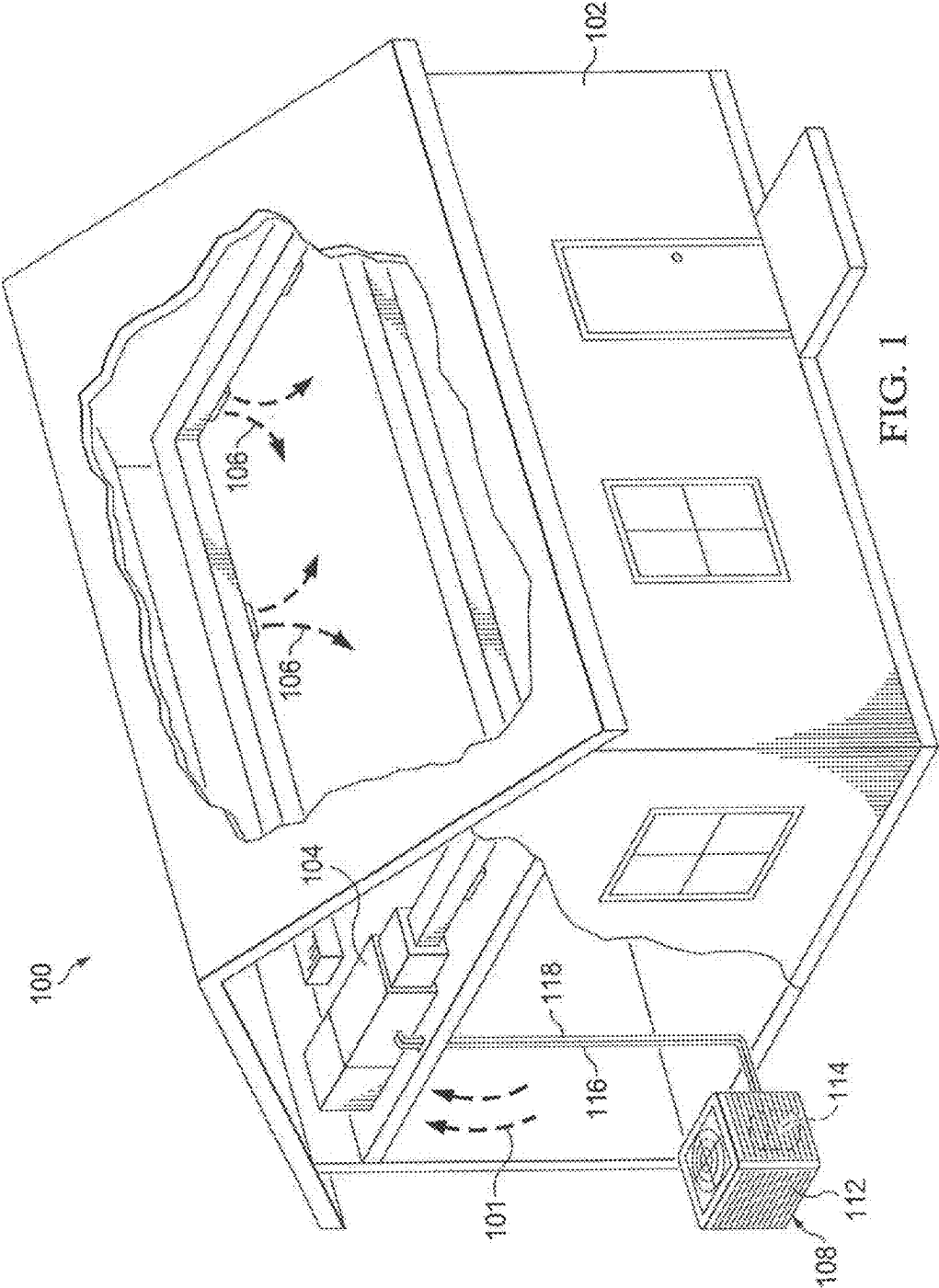




US 20250283617A1

(19) **United States**(12) **Patent Application Publication**
Sloan et al.(10) **Pub. No.: US 2025/0283617 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **PARALLEL EVAPORATOR COILS FOR
SUPERHEAT CONTROL FOR A
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Grove, WI (US); **Timothy S. O'Brien**,
DeForest, WI (US)(21) Appl. No.: **19/217,990**(22) Filed: **May 23, 2025****Related U.S. Application Data**(60) Division of application No. 18/305,662, filed on Apr.
24, 2023, now Pat. No. 12,339,034, which is a con-
tinuation-in-part of application No. 17/197,781, filed
on Mar. 10, 2021, now Pat. No. 11,668,476.**Publication Classification**(51) **Int. Cl.**
F24F 3/14 (2006.01)
F24F 3/153 (2006.01)
(52) **U.S. Cl.**
CPC **F24F 3/1405** (2013.01); **F24F 3/153**
(2013.01); **F24F 2003/144** (2013.01)(57) **ABSTRACT**A dehumidification system includes a primary metering
device, a secondary metering device, an intermixed coil unit,
a primary evaporator, a secondary condenser, a first modu-
lating valve, and a compressor.



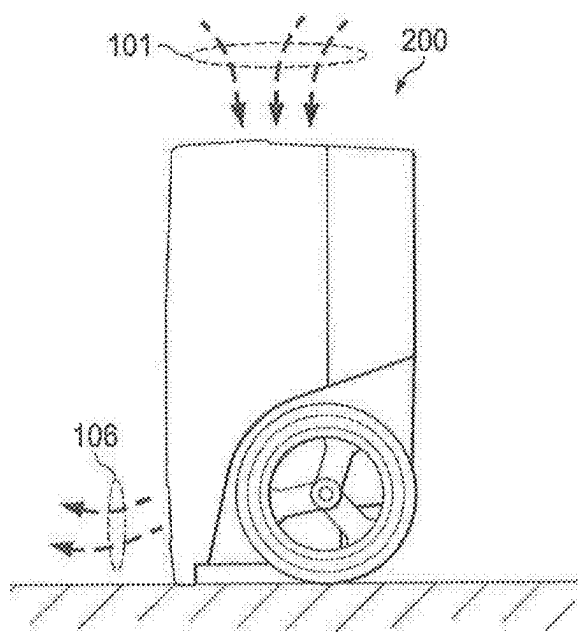


FIG. 2

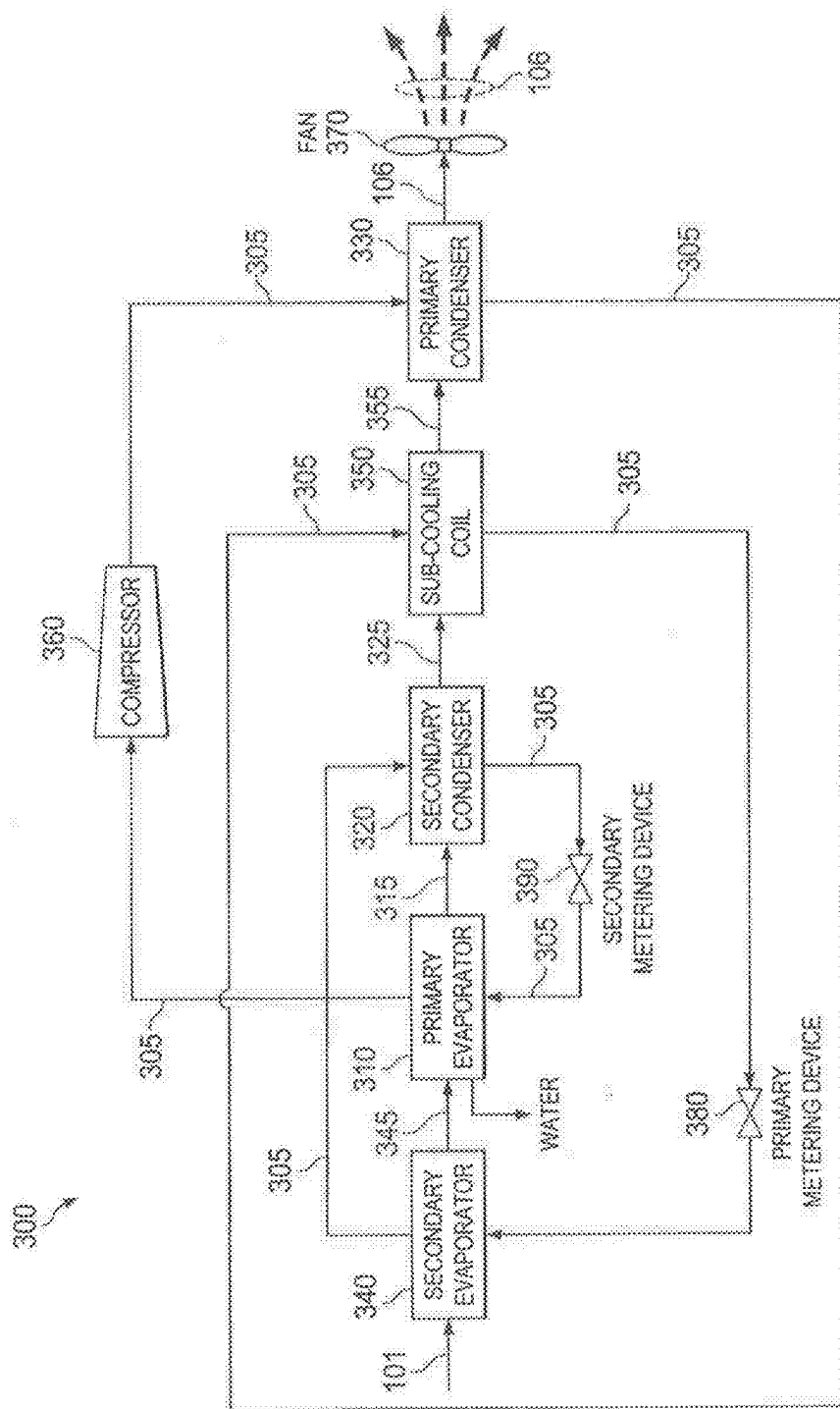
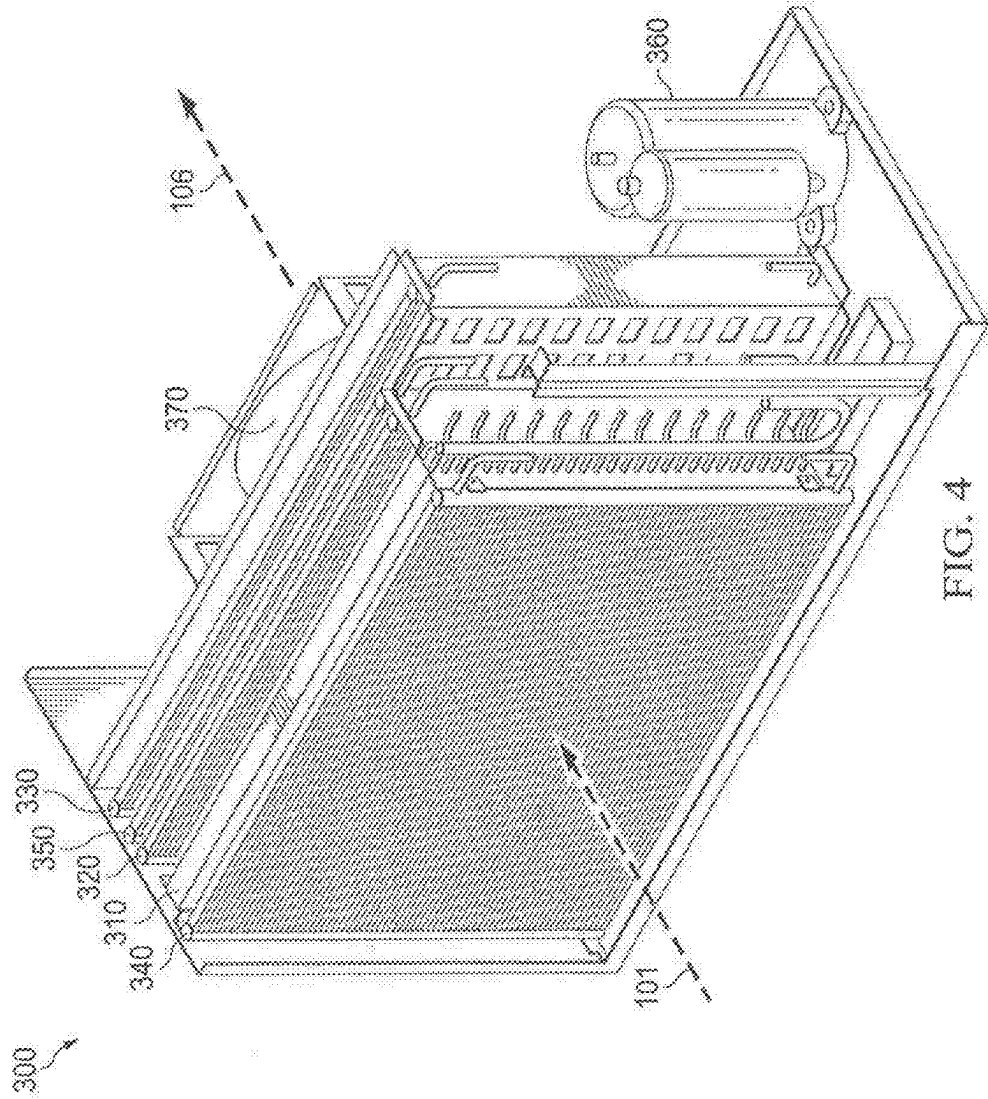


FIG. 3



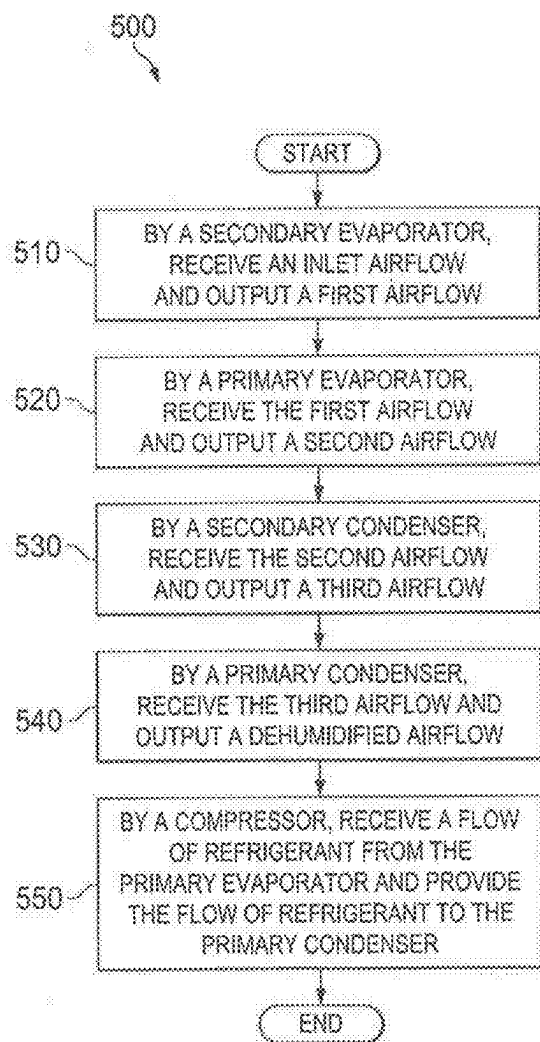


FIG. 5

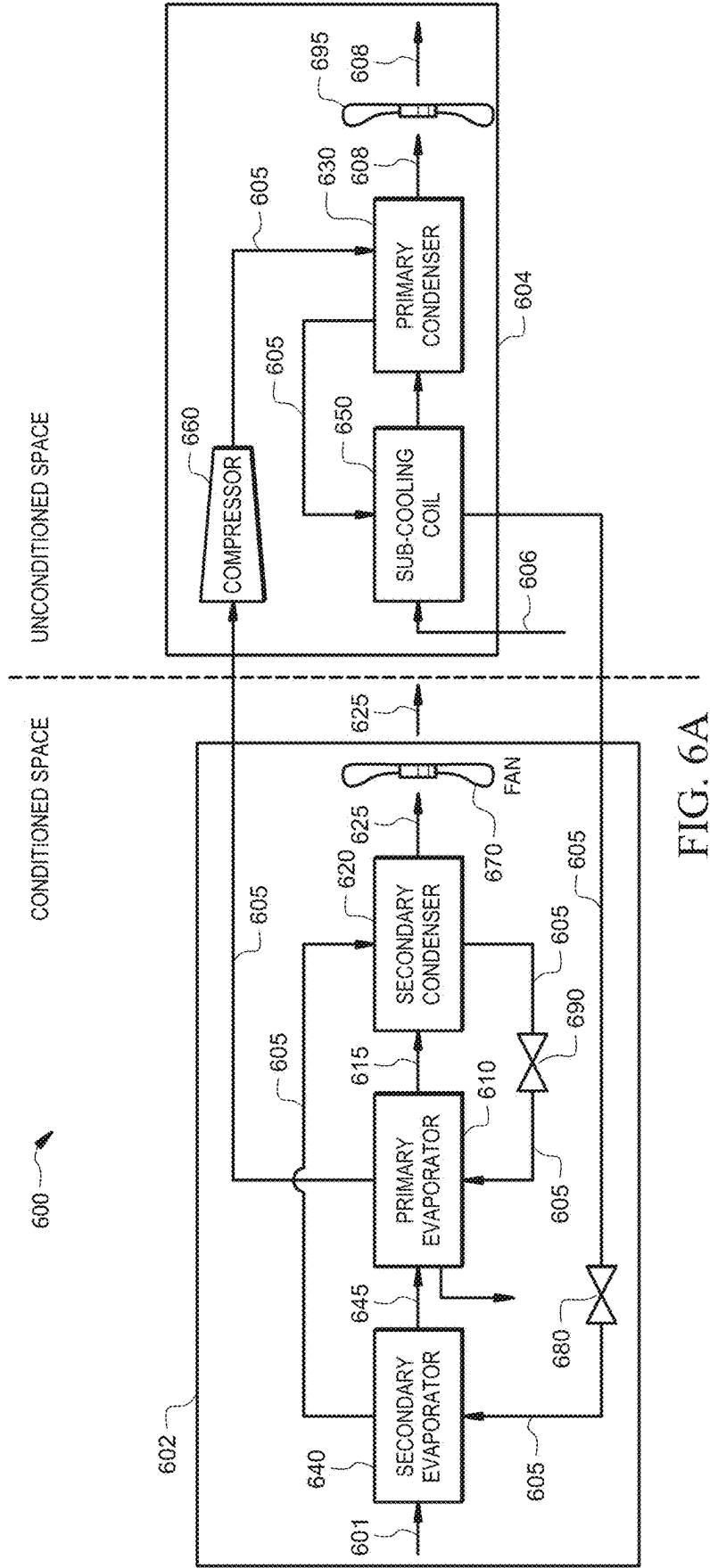


FIG. 6A

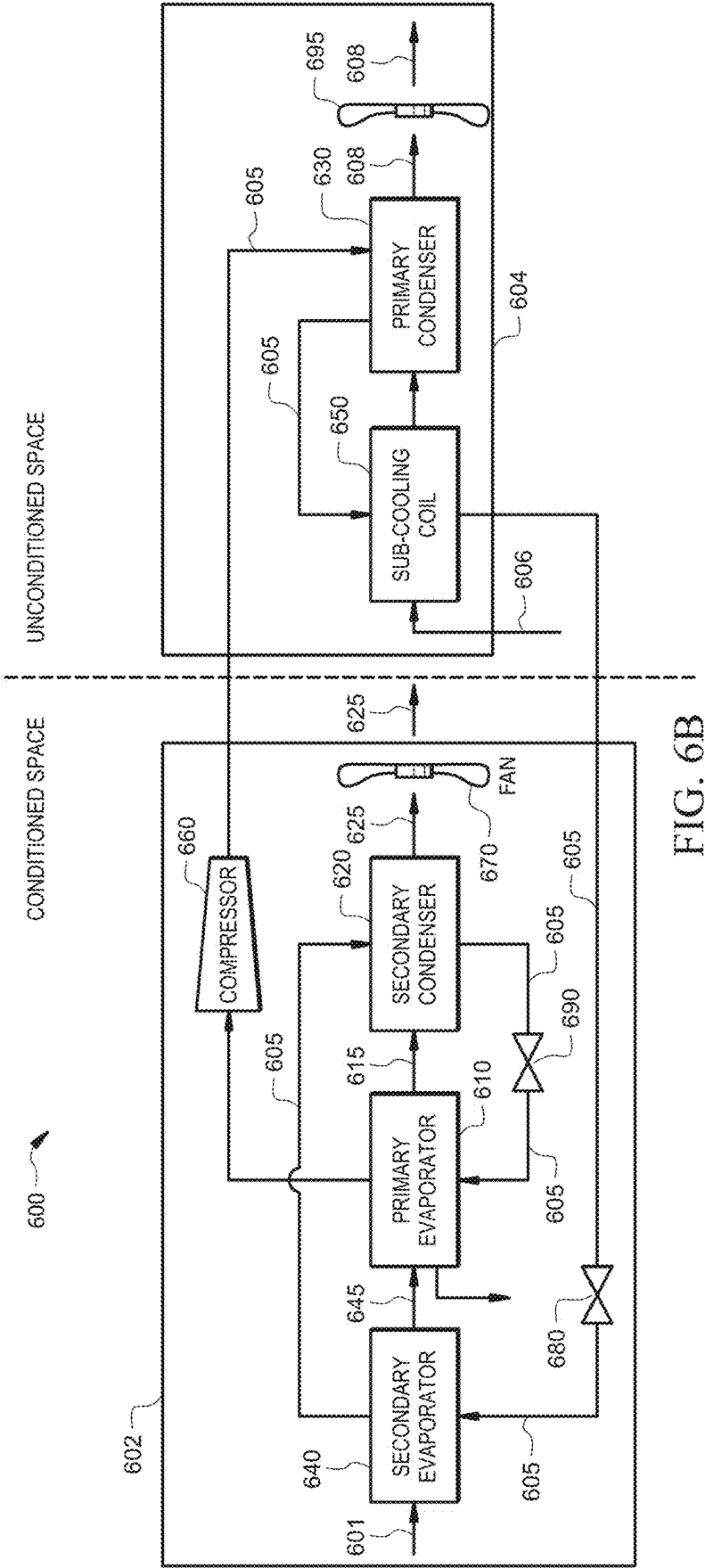


FIG. 6B

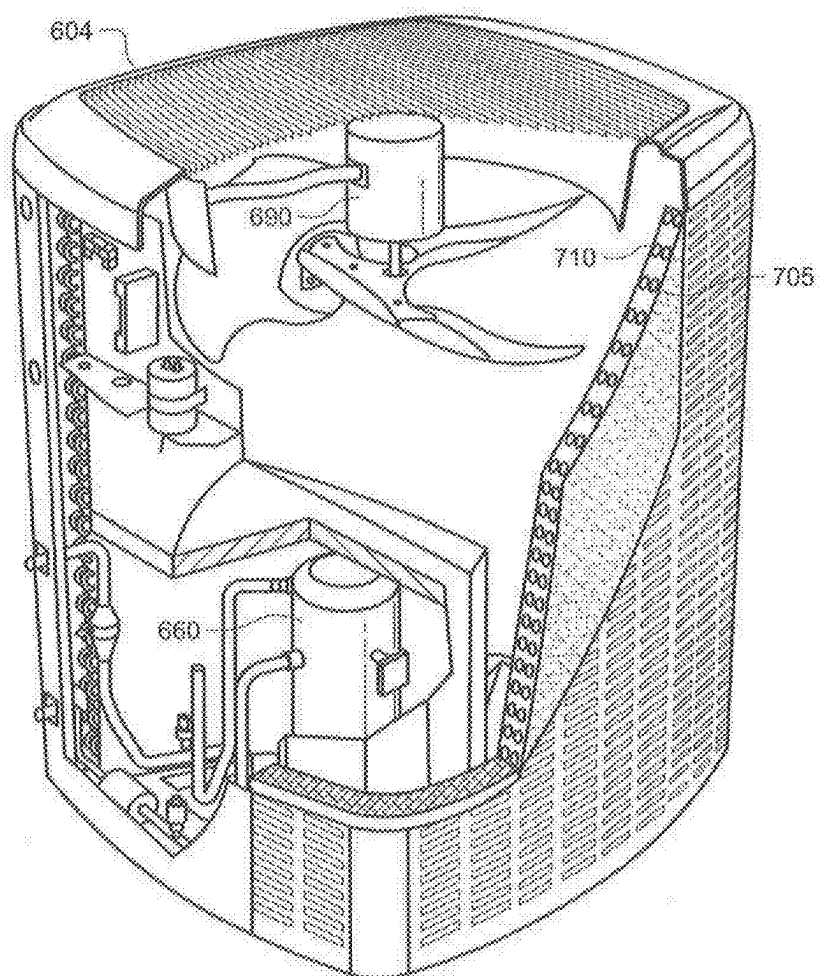
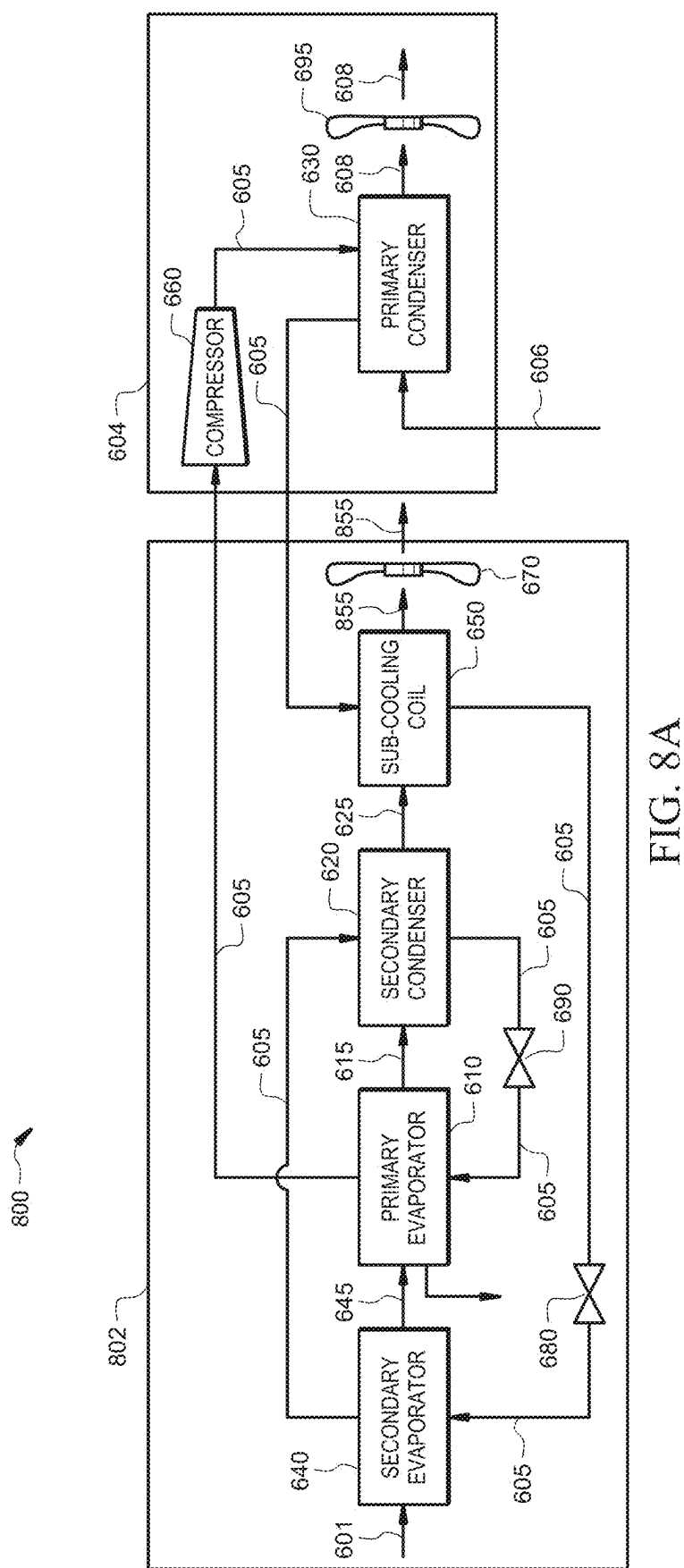
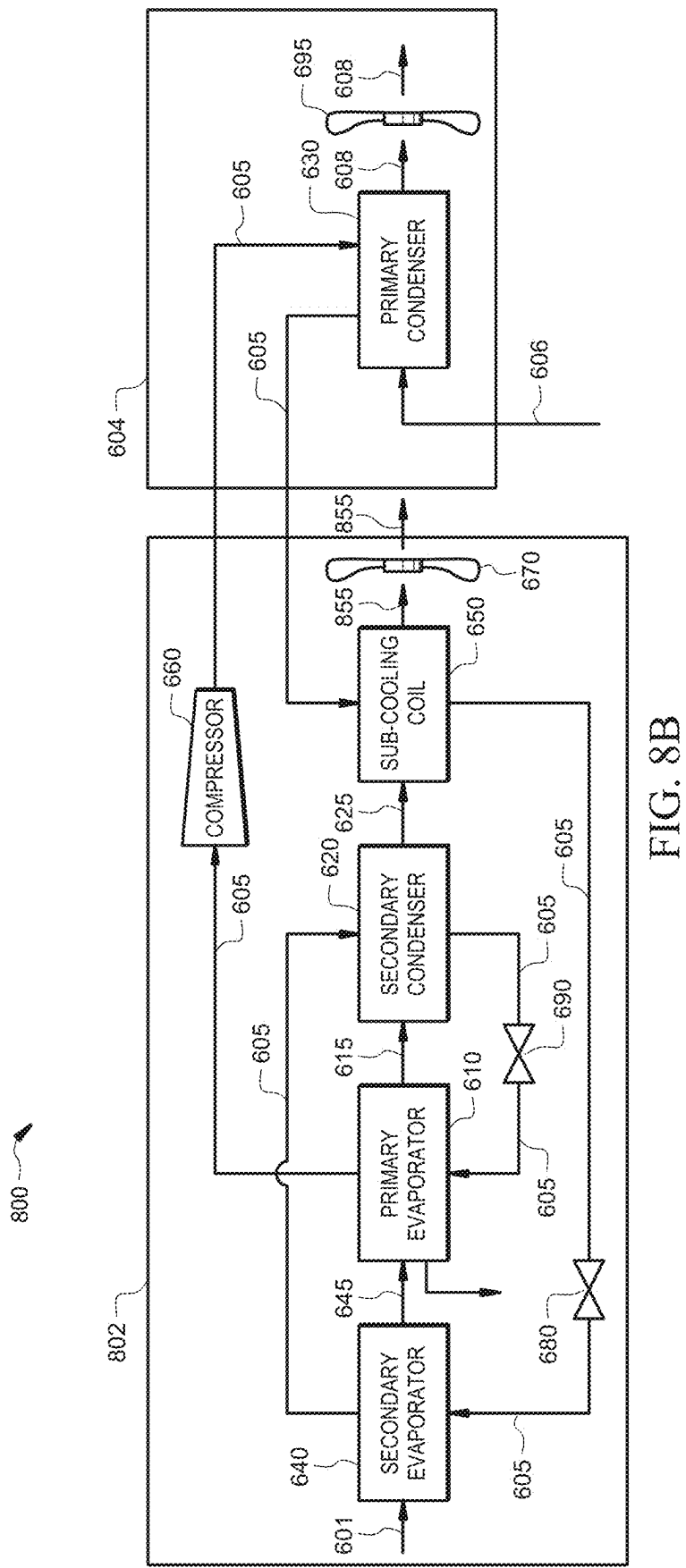
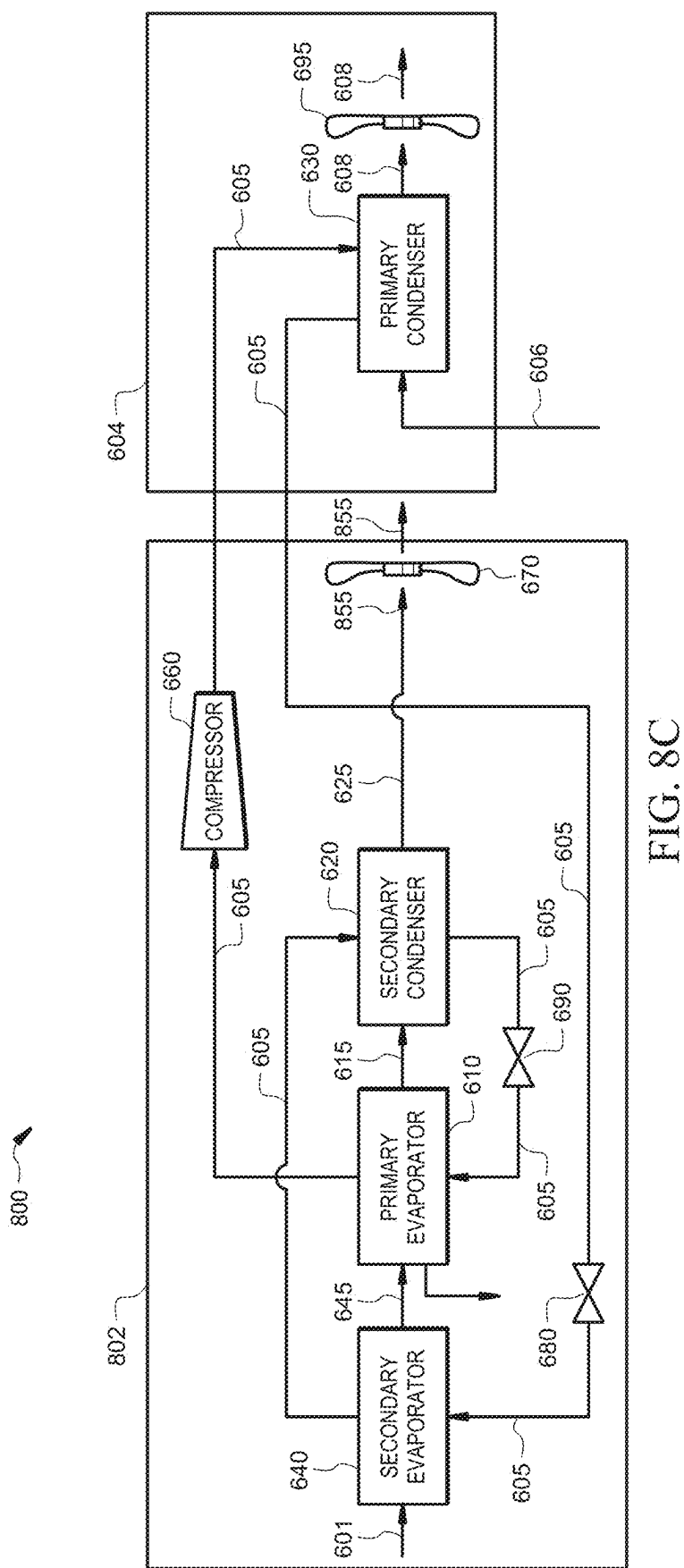
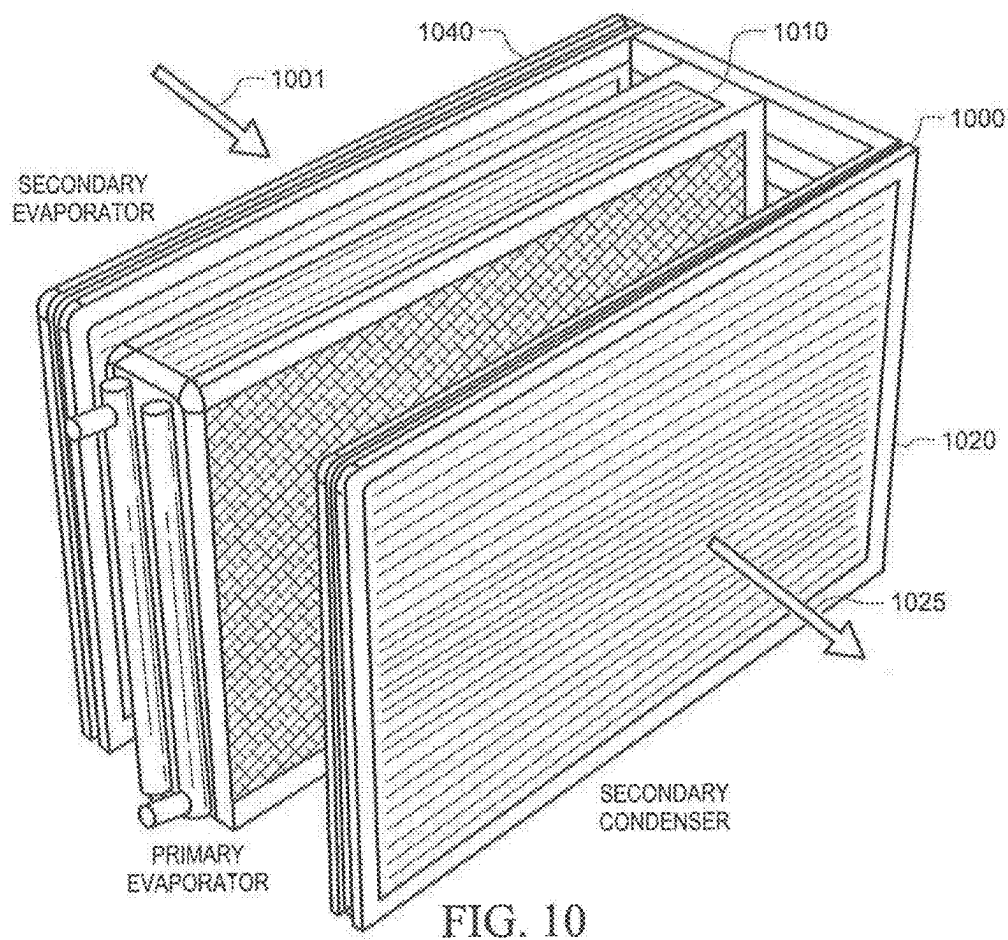


FIG. 7









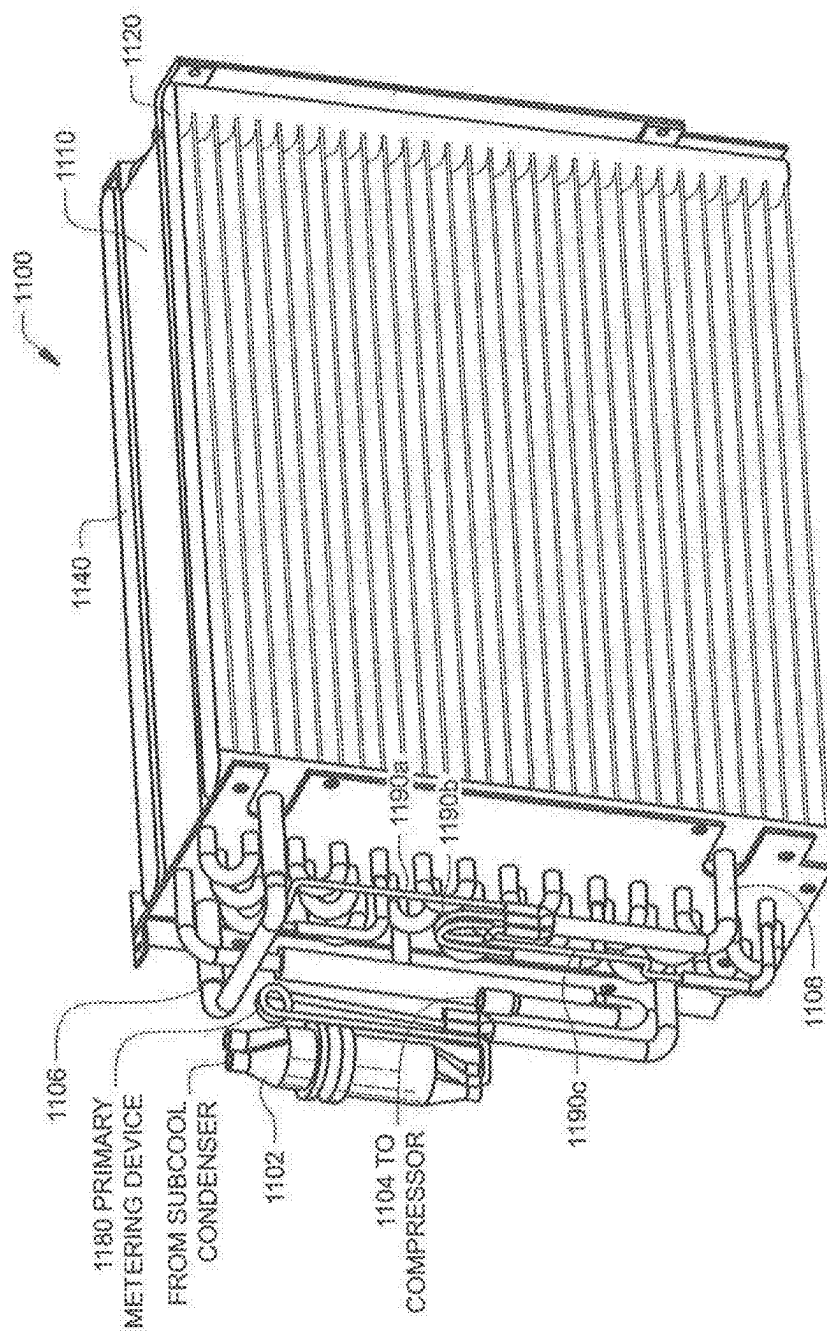


FIG. 11

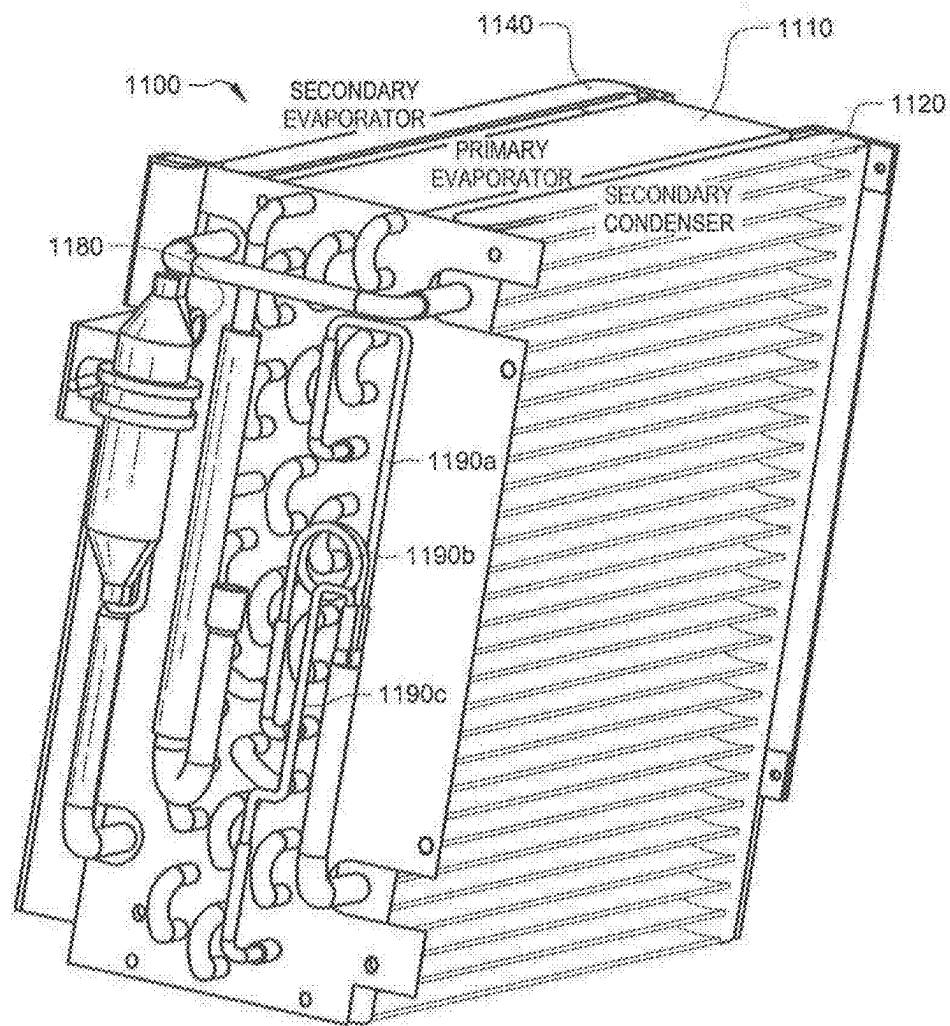
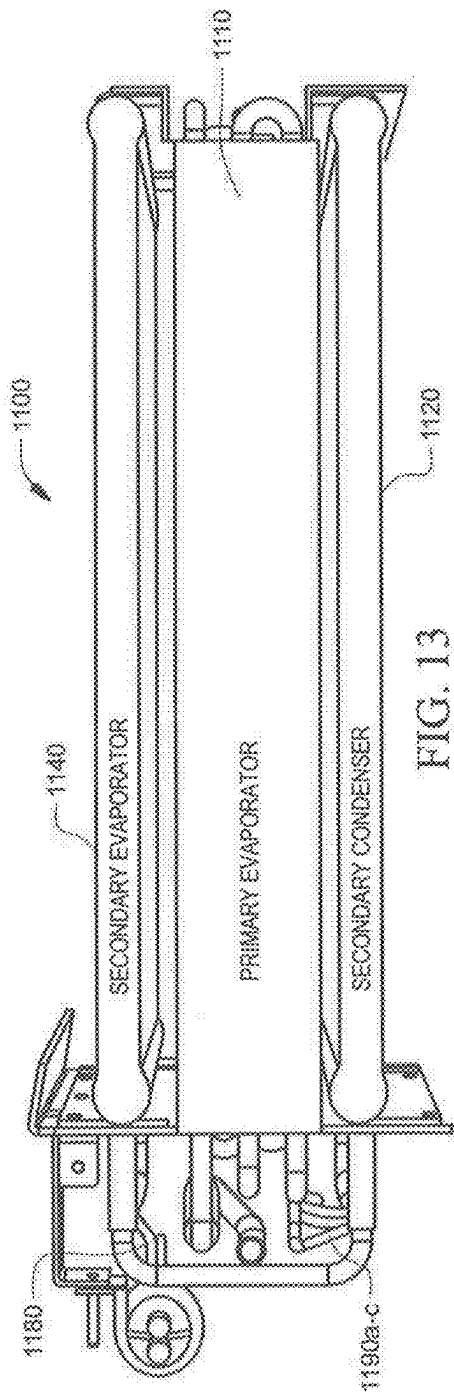


FIG. 12



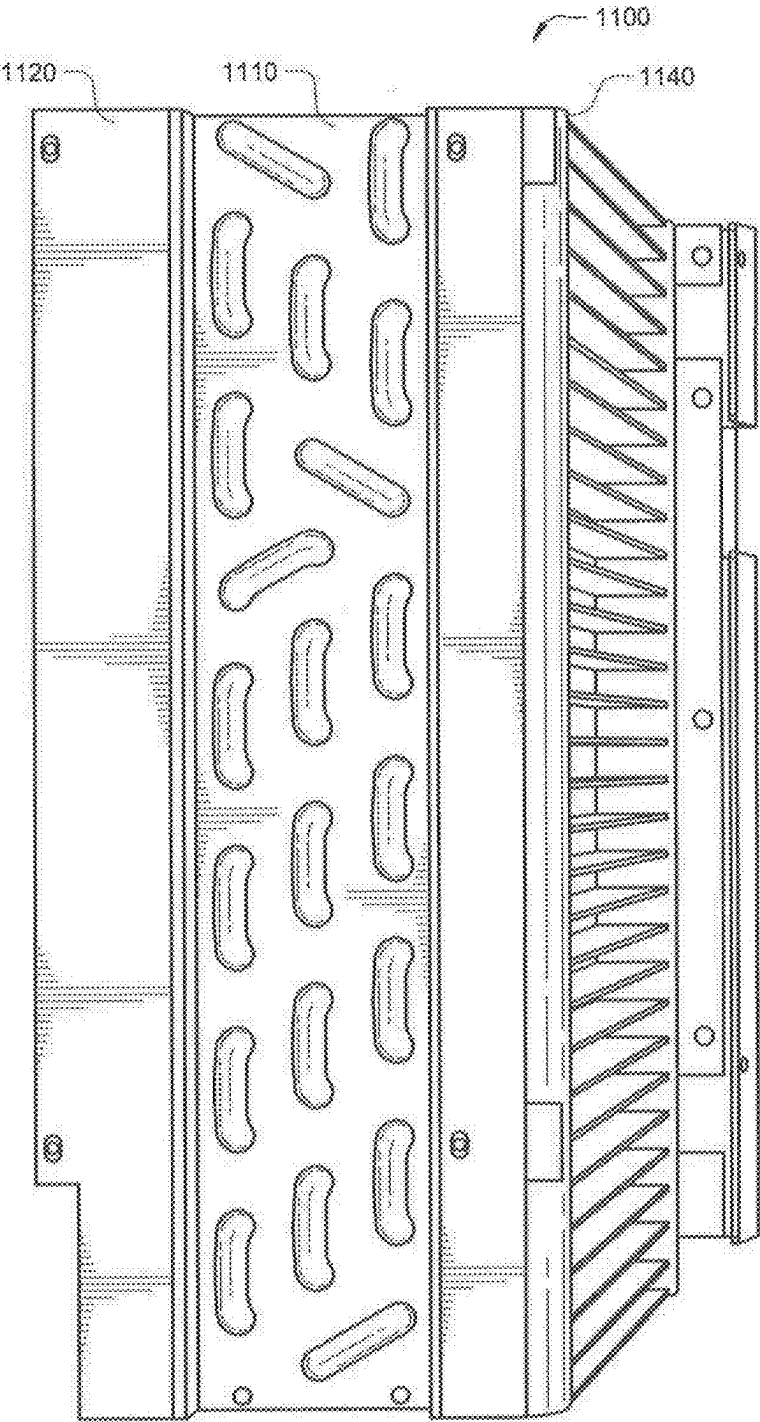


FIG. 14

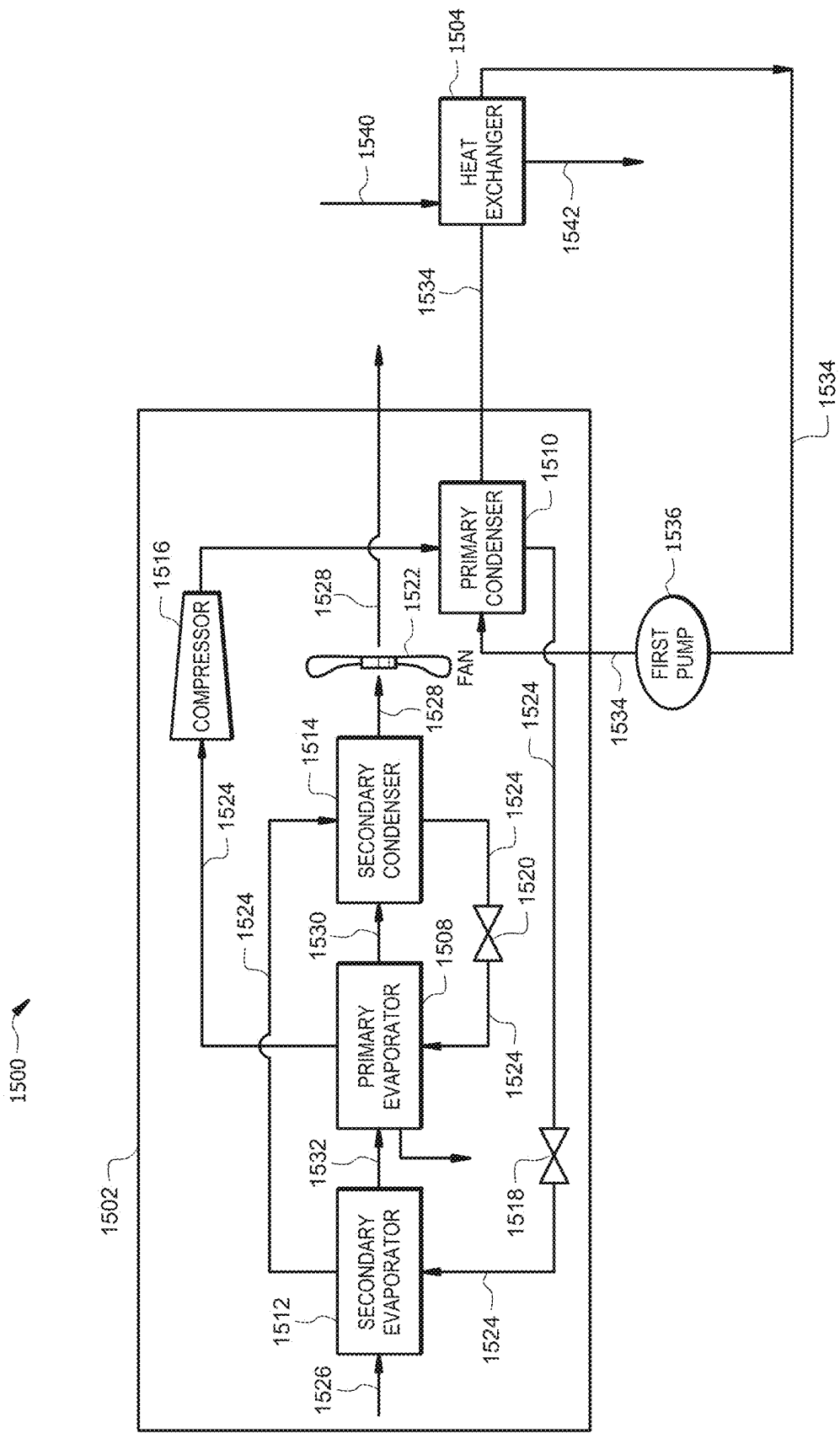


FIG. 15A

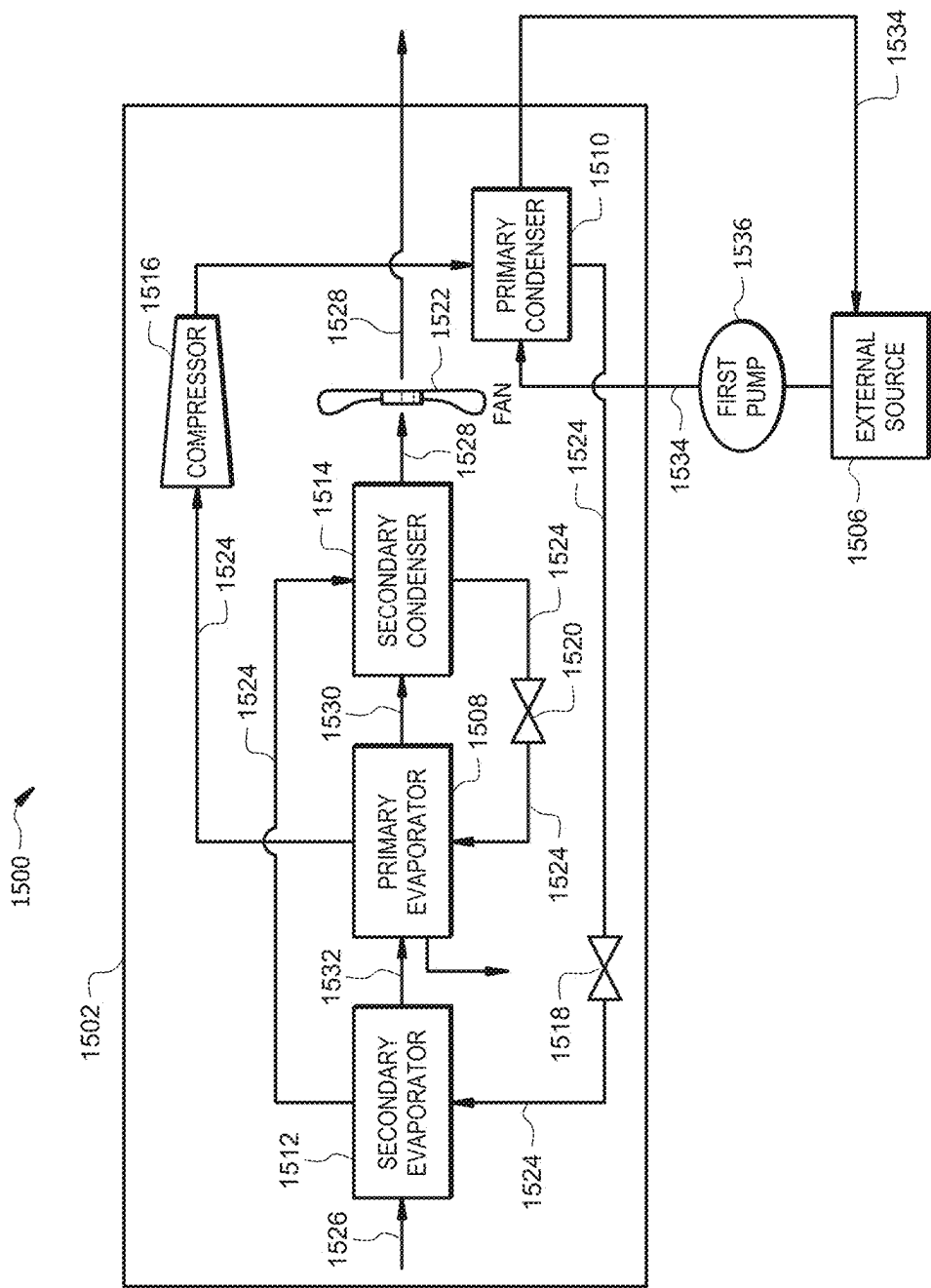


FIG. 15B

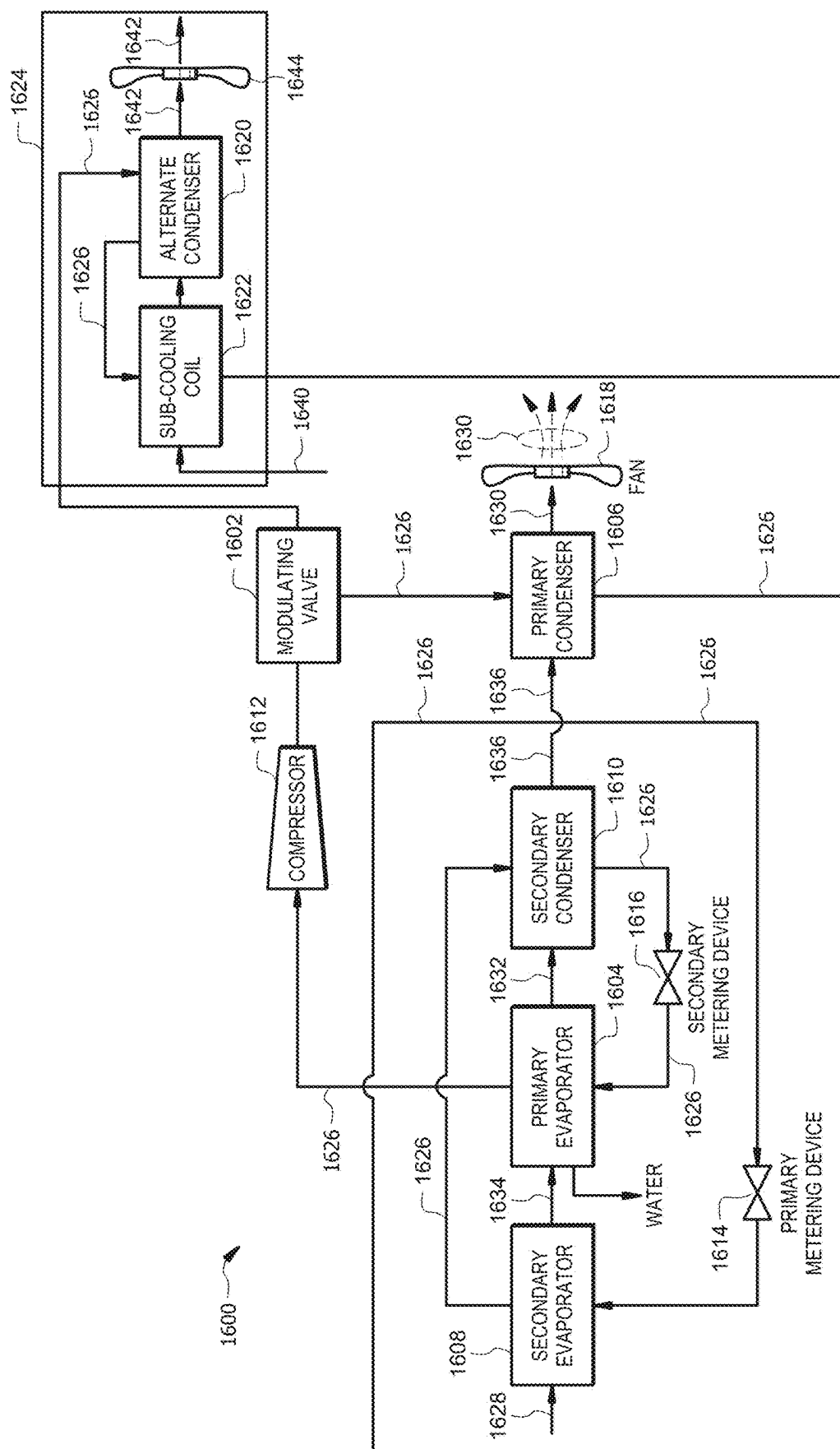


FIG. 16A

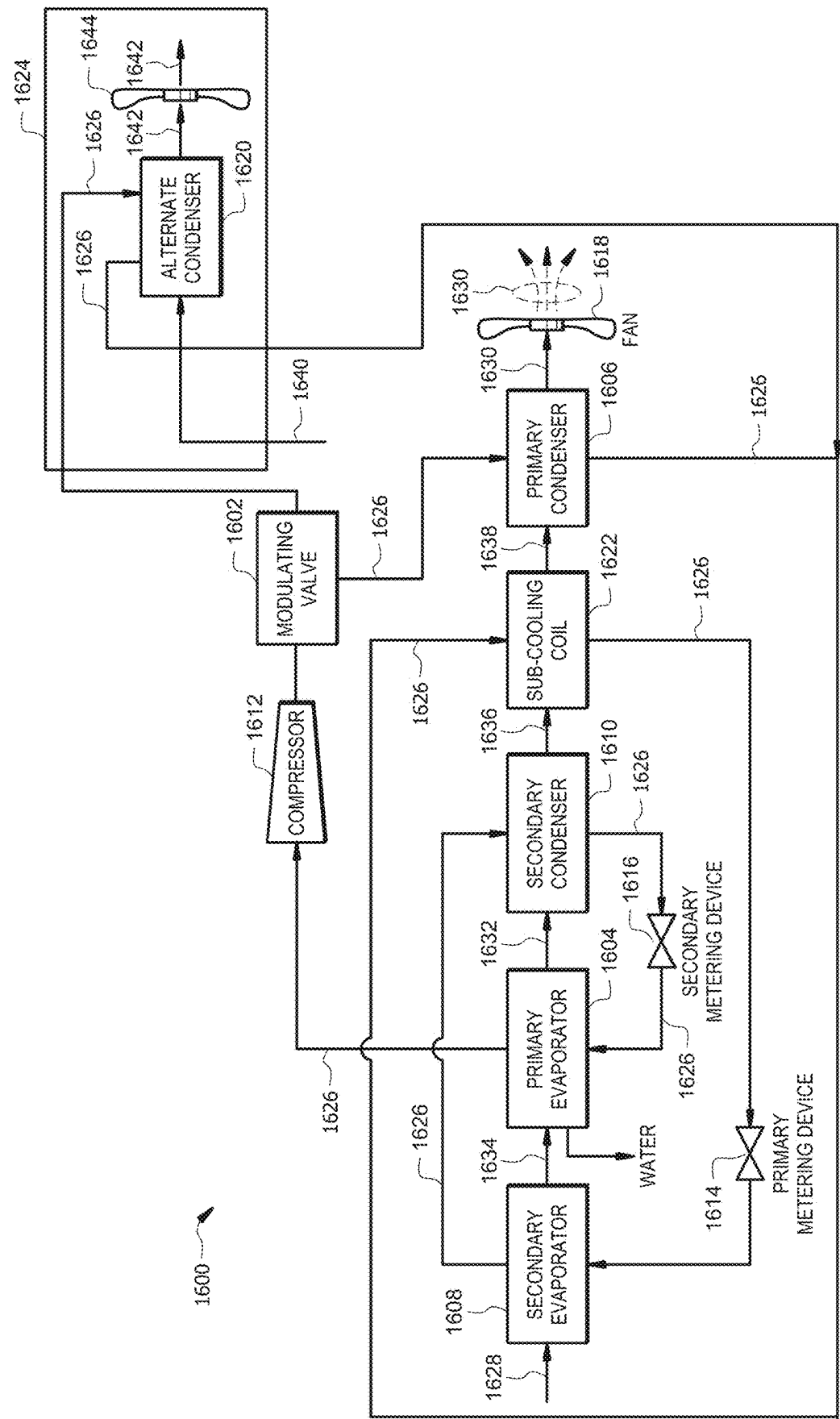


FIG. 16B

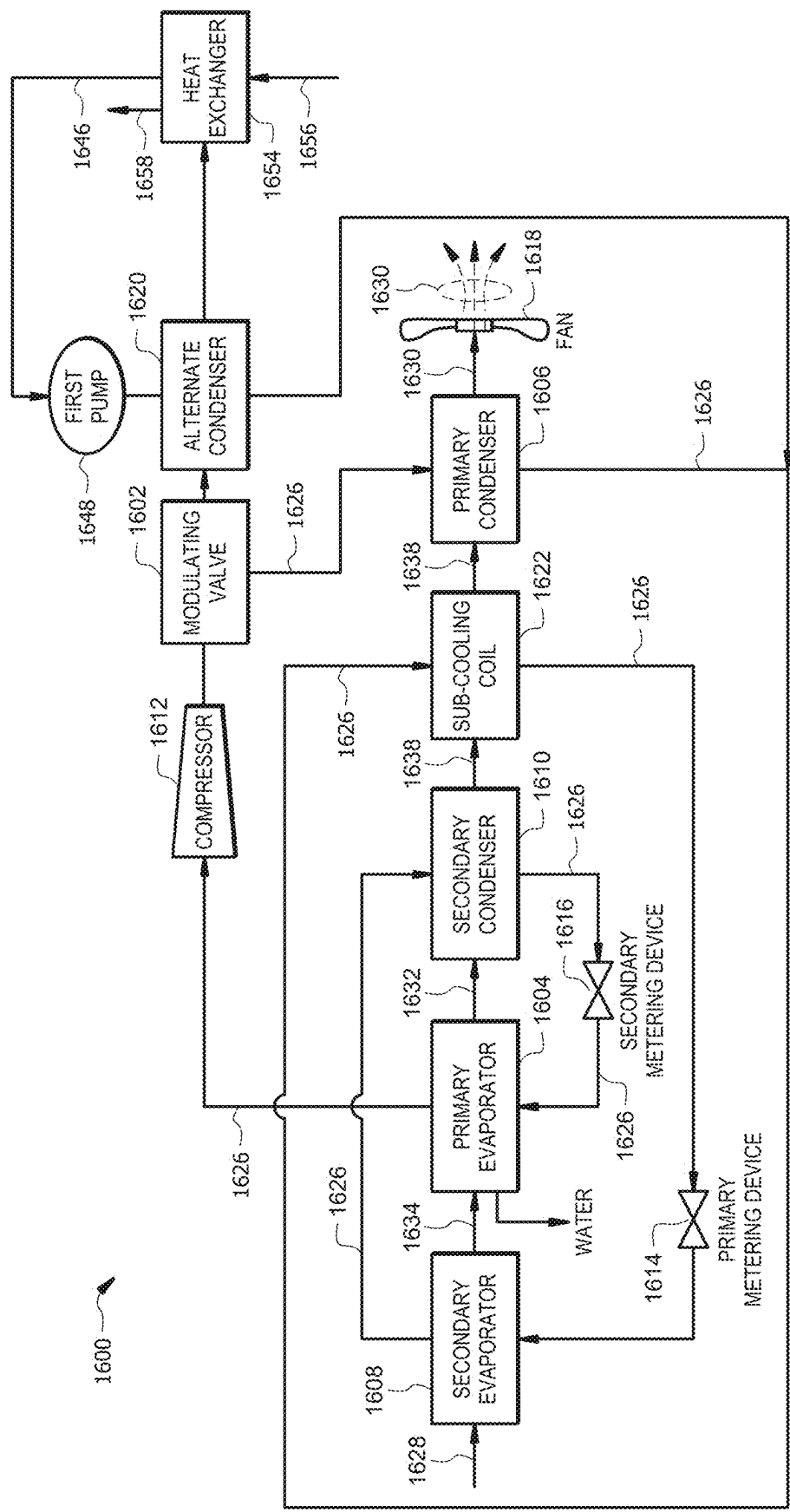
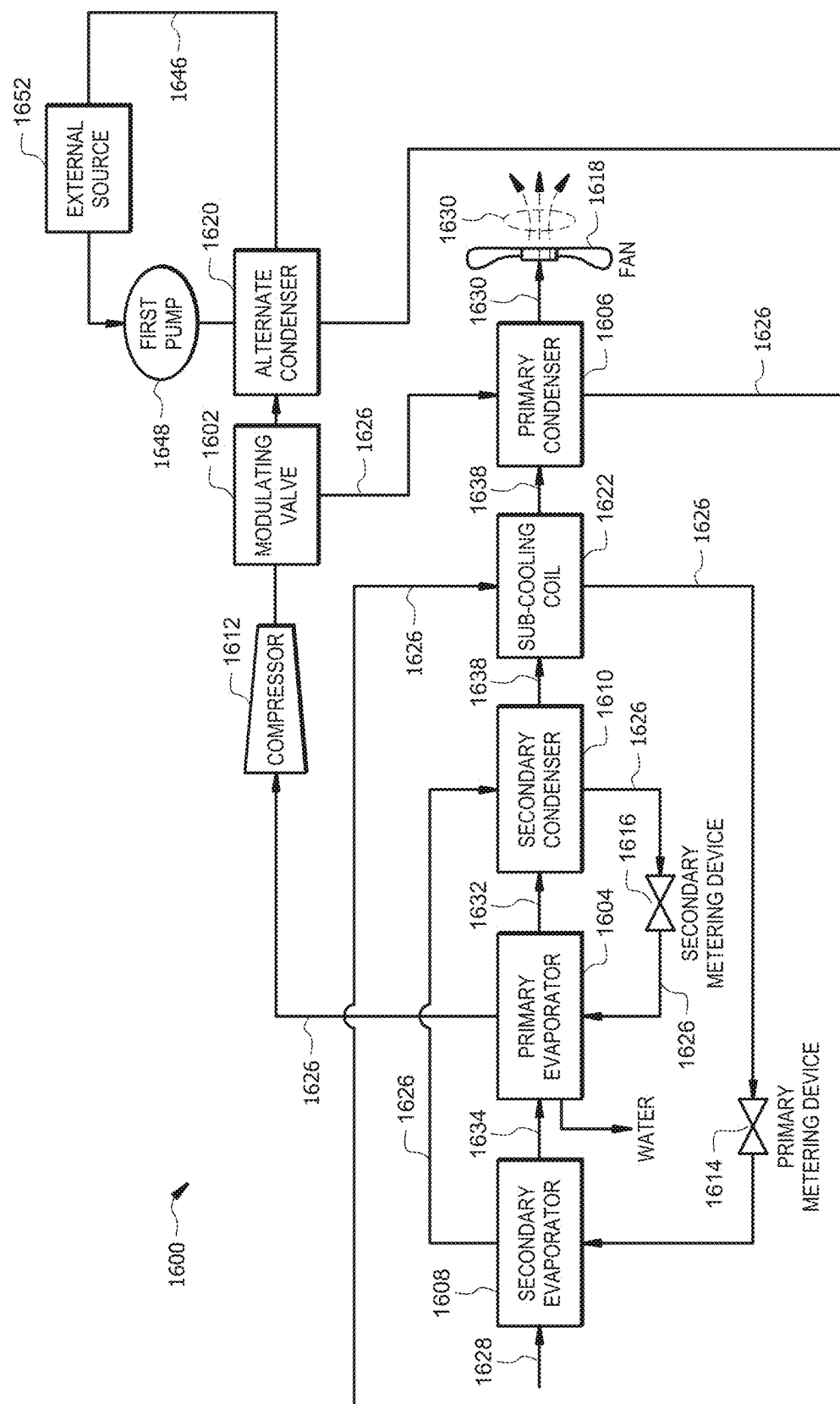


FIG. 16C



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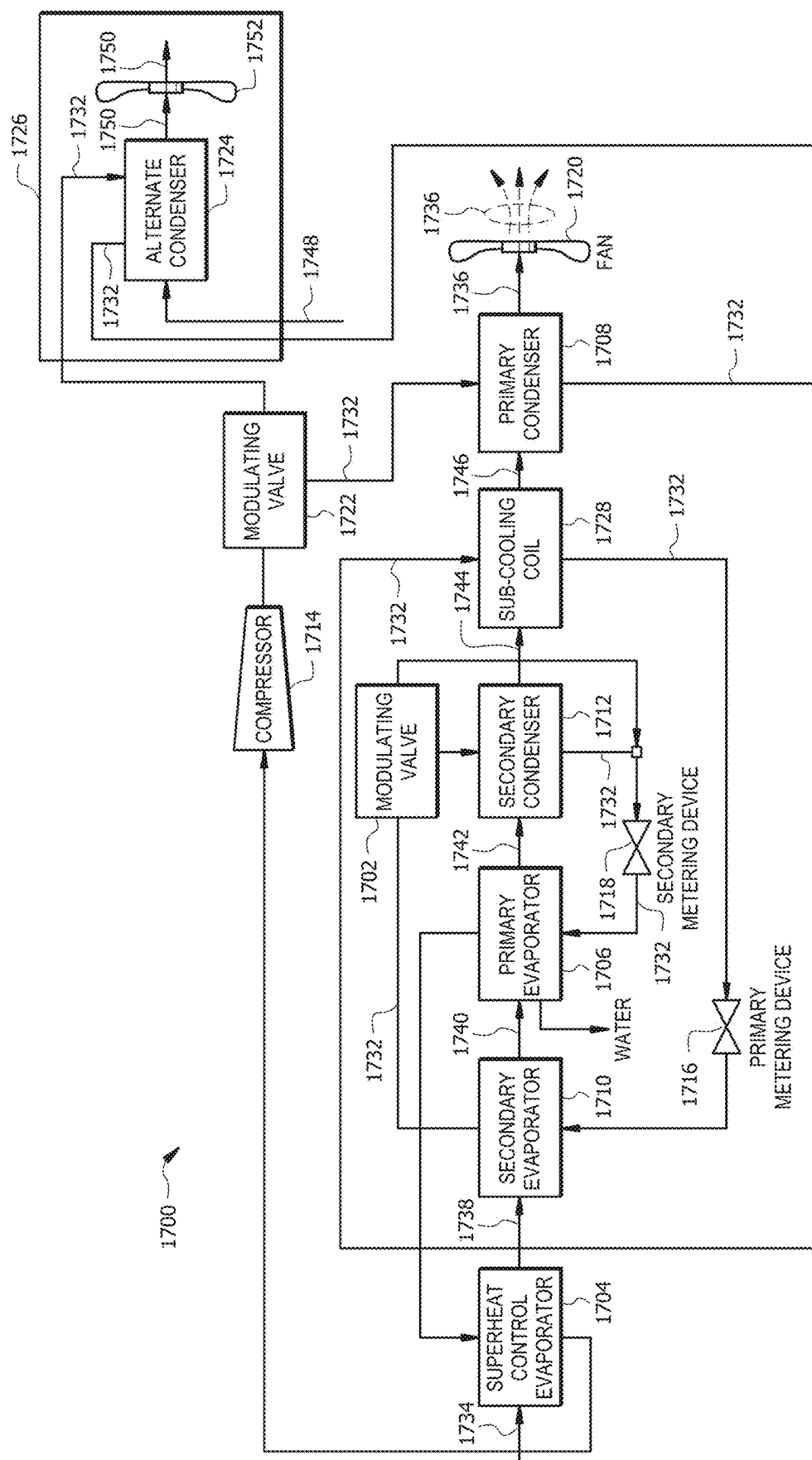


FIG. 17A

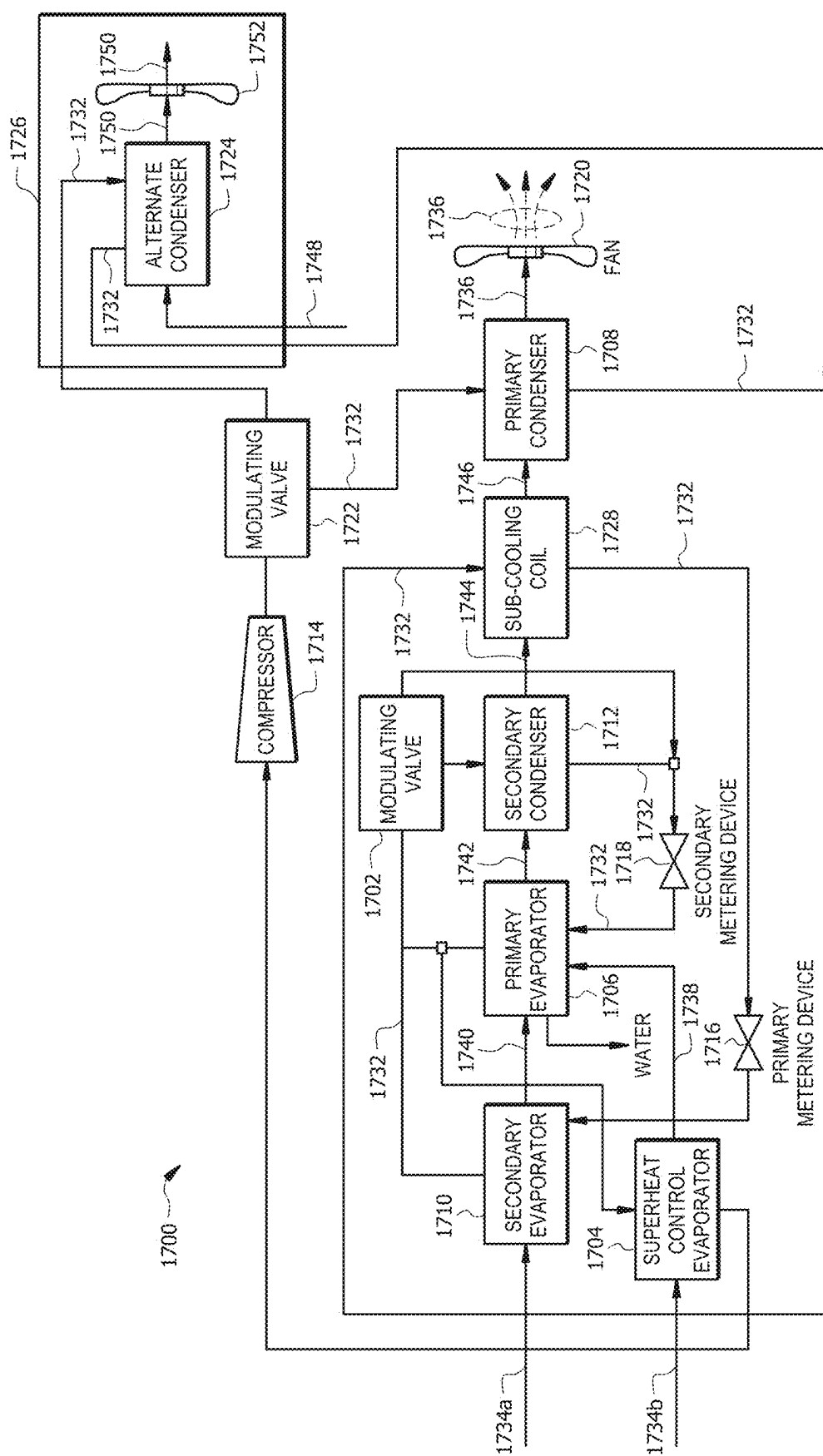


FIG. 17B

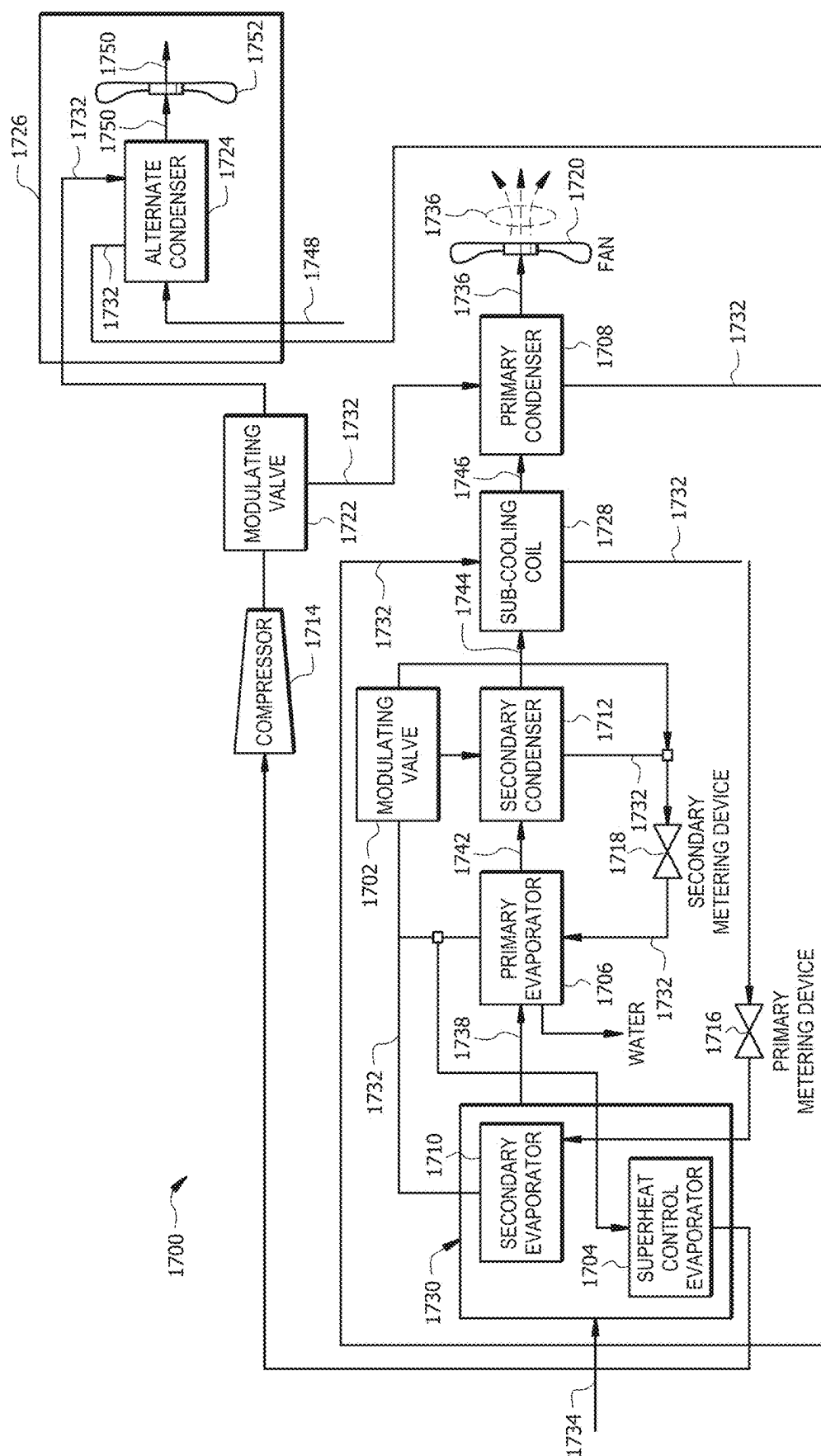


FIG. 17C

PARALLEL EVAPORATOR COILS FOR SUPERHEAT CONTROL FOR A DEHUMIDIFICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of U.S. application Ser. No. 18/305,662 filed Apr. 24, 2023 by Scott E. Sloan et al. and entitled “PARALLEL EVAPORATOR COILS FOR SUPERHEAT CONTROL FOR A DEHUMIDIFICATION SYSTEM”, which is a continuation-in-part and claims priority to U.S. Non-provisional application Ser. No. 17/197,781 filed Mar. 10, 2021 by Weizhong Yu et al. and entitled “HEAT MODULATION DEHUMIDIFICATION SYSTEM”, now U.S. Pat. No. 11,668,476 issued Jun. 6, 2023, which claims priority to U.S. Non-provisional application Ser. No. 16/234,052 filed Dec. 27, 2018 by Steven S. Dingle et al. and entitled “SPLIT DEHUMIDIFICATION SYSTEM WITH SECONDARY EVAPORATOR AND CONDENSER COILS”, now U.S. Pat. No. 10,955,148 issued Mar. 23, 2021, which claims priority to U.S. Non-provisional application Ser. No. 15/460,772 filed Mar. 16, 2017 by Dwaine Walter Tucker et al. and entitled “DEHUMIDIFIER WITH SECONDARY EVAPORATOR AND CONDENSER COILS,” now U.S. Pat. No. 10,168,058 issued Jan. 1, 2019, which are all hereby incorporated by reference as if reproduced in their entirety.

TECHNICAL FIELD

[0002] This invention relates generally to dehumidification and more particularly to a dehumidifier with secondary evaporator and condenser coils.

BACKGROUND OF THE INVENTION

[0003] In certain situations, it is desirable to reduce the humidity of air within a structure. For example, in fire and flood restoration applications, it may be desirable to quickly remove water from areas of a damaged structure. To accomplish this, one or more portable dehumidifiers may be placed within the structure to direct dry air toward water-damaged areas. Current dehumidifiers, however, have proven inefficient in various respects.

SUMMARY OF THE INVENTION

[0004] According to embodiments of the present disclosure, disadvantages and problems associated with previous systems may be reduced or eliminated.

[0005] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and a superheat control evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator and to receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the inlet airflow, the first airflow generated by transferring heat from the inlet airflow to the flow of refrigerant as the inlet airflow passes through the superheat control evaporator. The dehumidification system further comprises a secondary evaporator disposed downstream of and in series with the superheat control evaporator with respect to the airflow. The secondary evaporator is operable to receive the flow of refrigerant from the primary metering device and to receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow,

the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow passes through the secondary evaporator. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive the second airflow and output a third airflow, the third airflow comprising cooler air than the second airflow, the third airflow generated by transferring heat from the second airflow to the flow of refrigerant as the second airflow passes through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the third airflow and output a fourth airflow.

[0006] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0007] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and a superheat control evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator and to receive a first inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the first inlet airflow to the flow of refrigerant as the first inlet airflow passes through the superheat control evaporator. The dehumidification system further comprises a secondary evaporator disposed in parallel to the superheat control evaporator. The secondary evaporator is operable to receive the flow of refrigerant from the primary metering device and to receive a second inlet airflow and output a second airflow, the second airflow comprising cooler air than the second inlet airflow, the second airflow generated by transferring heat from the second inlet airflow to the flow of refrigerant as the second inlet airflow passes through the secondary evaporator. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive both the first airflow and the second airflow and output a third airflow, the third airflow comprising cooler air than both the first airflow and the second airflow, the third airflow generated by transferring heat from both the first airflow and the second airflow to the flow of refrigerant as both the first airflow and the second

airflow pass through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the third airflow and output a fourth airflow.

[0008] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0009] In certain embodiments, a dehumidification system comprises a primary metering device, a secondary metering device, and an intermixed coil unit. The intermixed coil unit is operable to receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the inlet airflow to a flow of refrigerant within a superheat control evaporator and a secondary evaporator as the inlet airflow passes through both the superheat control evaporator and the secondary evaporator. The intermixed coil unit comprises the superheat control evaporator and the secondary evaporator. The superheat control evaporator is operable to receive a flow of refrigerant from a primary evaporator, and the secondary evaporator is operable to receive the flow of refrigerant from the primary metering device. The dehumidification system further comprises a primary evaporator operable to receive the flow of refrigerant from a first modulating valve and to receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow, the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow pass through the primary evaporator. The dehumidification system further comprises a secondary condenser operable to receive the flow of refrigerant from the first modulating valve and to receive the second airflow and output a third airflow.

[0010] The dehumidification system further comprises the first modulating valve operable to receive the flow of refrigerant from the secondary evaporator. During a first mode of operation, the first modulating valve is operable to direct the flow of refrigerant to the secondary condenser. During a second mode of operation, the first modulating valve is operable direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser. The dehumidification system is further configured to operate in

a third mode of operation wherein the first modulating valve is operable to direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator. The dehumidification system further comprises a compressor operable to receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor. The dehumidification system further comprises a primary condenser operable to receive the flow of refrigerant discharged from the compressor. In response to receiving the flow of refrigerant from the compressor, the primary condenser is further operable to output a dischargeable airflow.

[0011] Certain embodiments of the present disclosure may provide one or more technical advantages. The advantages with these embodiments include modulating the amount of sensible heat to latent heat. For example, adjusting the ratio of sensible to latent heat can further decrease the temperature of the surrounding airflow or increase the amount of water removed from the airflow. The secondary condenser may be isolated, wherein energy is not recovered by the airflow via refrigerant flowing through the secondary condenser, thereby providing an air cooling operation rather than dehumidification. Further, modulation is based off controlling the superheat from one or more evaporator coils of the dehumidification system. The addition of another evaporator coil upstream of the primary evaporator provides faster and more efficient control of operations of the dehumidification system. For example, providing increased superheat control can prevent the cooling of an airflow already determined to be at a control or set temperature.

[0012] In further examples, certain embodiments include two evaporators, two condensers, and two metering devices that utilize a closed refrigeration loop. This configuration causes part of the refrigerant within the system to evaporate and condense twice in one refrigeration cycle, thereby increasing the compressor capacity over typical systems without adding any additional power to the compressor. This, in turn, increases the overall efficiency of the system by providing more dehumidification per kilowatt of power used. The lower humidity of the output airflow may allow for increased drying potential, which may be beneficial in certain applications (e.g., fire and flood restoration).

[0013] Certain embodiments of the present disclosure may include some, all, or none of the above advantages. One or more other technical advantages may be readily apparent to those skilled in the art from the figures, descriptions, and claims included herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] To provide a more complete understanding of the present invention and the features and advantages thereof, reference is made to the following description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 illustrates an example split system for reducing the humidity of air within a structure, according to certain embodiments;

[0016] FIG. 2 illustrates an example portable system for reducing the humidity of air within a structure, according to certain embodiments;

[0017] FIGS. 3 and 4 illustrate an example dehumidification system that may be used by the systems of FIGS. 1 and 2 to reduce the humidity of air within a structure, according to certain embodiments;

[0018] FIG. 5 illustrates an example dehumidification method that may be used by the systems of FIGS. 1 and 2 to reduce the humidity of air within a structure, according to certain embodiments;

[0019] FIGS. 6A and 6B illustrate an example air conditioning and dehumidification system, according to certain embodiments;

[0020] FIG. 7 illustrates an example condenser system for use in the system described herein, according to certain embodiments;

[0021] FIGS. 8A, 8B, and 8C illustrate an example air conditioning and dehumidification system, according to certain embodiments;

[0022] FIGS. 9 and 10 illustrate examples of single coil packs for use in the system described herein, according to certain embodiments;

[0023] FIGS. 11, 12, 13, and 14 illustrate an example of a primary evaporator comprising three circuits for use in the system described herein, according to certain embodiments;

[0024] FIGS. 15A and 15B illustrate an example dehumidification system with a liquid cooled condenser, according to certain embodiments; and

[0025] FIGS. 16A, 16B, 16C, and 16D illustrate an example dehumidification system with a modulating valve, according to certain embodiments.

[0026] FIGS. 17A, 17B, and 17C illustrate an example dehumidification system with a superheat control evaporator, according to certain embodiments.

DETAILED DESCRIPTION OF THE DRAWINGS

[0027] In certain situations, it is desirable to reduce the humidity of air within a structure. For example, in fire and flood restoration applications, it may be desirable to remove water from a damaged structure by placing one or more portable dehumidifiers unit within the structure. As another example, in areas that experience weather with high humidity levels, or in buildings where low humidity levels are required (e.g., libraries), it may be desirable to install a dehumidification unit within a central air conditioning system. Furthermore, it may be necessary to hold a desired humidity level in some commercial applications. Current dehumidifiers, however, have proven inadequate or inefficient in various respects.

[0028] To address the inefficiencies and other issues with current dehumidification systems, the disclosed embodiments provide a dehumidification system that includes a secondary evaporator and a secondary condenser, which causes part of the refrigerant within the multi-stage system to evaporate and condense twice in one refrigeration cycle. This increases the compressor capacity over typical systems without adding any additional power to the compressor. This, in turn, increases the overall efficiency of the system by providing more dehumidification per kilowatt of power used.

[0029] FIG. 1 illustrates an example dehumidification system 100 for supplying dehumidified air 106 to a structure 102, according to certain embodiments. Dehumidification system 100 includes an evaporator system 104 located within structure 102. Structure 102 may include all or a portion of a building or other suitable enclosed space, such as an apartment building, a hotel, an office space, a commercial building, or a private dwelling (e.g., a house). Evaporator system 104 receives inlet air 101 from within structure 102, reduces the moisture in received inlet air 101,

and supplies dehumidified air 106 back to structure 102. Evaporator system 104 may distribute dehumidified air 106 throughout structure 102 via air ducts, as illustrated.

[0030] In general, dehumidification system 100 is a split system wherein evaporator system 104 is coupled to a remote condenser system 108 that is located external to structure 102. Remote condenser system 108 may include a condenser unit 112 and a compressor unit 114 that facilitate the functions of evaporator system 104 by processing a flow of refrigerant as part of a refrigeration cycle. The flow of refrigerant may include any suitable cooling material, such as R410a refrigerant. In certain embodiments, compressor unit 114 may receive the flow of refrigerant vapor from evaporator system 104 via a refrigerant line 116. Compressor unit 114 may pressurize the flow of refrigerant, thereby increasing the temperature of the refrigerant. The speed of the compressor may be modulated to effectuate desired operating characteristics. Condenser unit 112 may receive the pressurized flow of refrigerant vapor from compressor unit 114 and cool the pressurized refrigerant by facilitating heat transfer from the flow of refrigerant to the ambient air exterior to structure 102. In certain embodiments, remote condenser system 108 may utilize a heat exchanger, such as a microchannel heat exchanger to remove heat from the flow of refrigerant. Remote condenser system 108 may include a fan that draws ambient air from outside structure 102 for use in cooling the flow of refrigerant. In certain embodiments, the speed of this fan is modulated to effectuate desired operating characteristics. An illustrative embodiment of an example condenser system is shown, for example, in FIG. 7 (described in further detail below).

[0031] After being cooled and condensed to liquid by condenser unit 112, the flow of refrigerant may travel by a refrigerant line 118 to evaporator system 104. In certain embodiments, the flow of refrigerant may be received by an expansion device (described in further detail below) that reduces the pressure of the flow of refrigerant, thereby reducing the temperature of the flow of refrigerant. An evaporator unit (described in further detail below) of evaporator system 104 may receive the flow of refrigerant from the expansion device and use the flow of refrigerant to dehumidify and cool an incoming airflow. The flow of refrigerant may then flow back to remote condenser system 108 and repeat this cycle.

[0032] In certain embodiments, evaporator system 104 may be installed in series with an air mover. An air mover may include a fan that blows air from one location to another. An air mover may facilitate distribution of outgoing air from evaporator system 104 to various parts of structure 102. An air mover and evaporator system 104 may have separate return inlets from which air is drawn. In certain embodiments, outgoing air from evaporator system 104 may be mixed with air produced by another component (e.g., an air conditioner) and blown through air ducts by the air mover. In other embodiments, evaporator system 104 may perform both cooling and dehumidifying and thus may be used without a conventional air conditioner.

[0033] Although a particular implementation of dehumidification system 100 is illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system 100, according to particular needs. Moreover, although various components of dehumidification system 100 have been depicted as being located at particular positions, the present disclosure contemplates

those components being positioned at any suitable location, according to particular needs.

[0034] FIG. 2 illustrates an example portable dehumidification system 200 for reducing the humidity of air within structure 102, according to certain embodiments of the present disclosure. Dehumidification system 200 may be positioned anywhere within structure 102 in order to direct dehumidified air 106 towards areas that require dehumidification (e.g., water-damaged areas). In general, dehumidification system 200 receives inlet airflow 101, removes water from the inlet airflow 101, and discharges dehumidified air 106 air back into structure 102. In certain embodiments, structure 102 includes a space that has suffered water damage (e.g., as a result of a flood or fire). In order to restore the water-damaged structure 102, one or more dehumidification systems 200 may be strategically positioned within structure 102 in order to quickly reduce the humidity of the air within the structure 102 and thereby dry the portions of structure 102 that suffered water damage.

[0035] Although a particular implementation of portable dehumidification system 200 is illustrated and primarily described, the present disclosure contemplates any suitable implementation of portable dehumidification system 200, according to particular needs. Moreover, although various components of portable dehumidification system 200 have been depicted as being located at particular positions within structure 102, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0036] FIGS. 3 and 4 illustrate an example dehumidification system 300 that may be used by dehumidification system 100 and portable dehumidification system 200 of FIGS. 1 and 2 to reduce the humidity of air within structure 102. Dehumidification system 300 includes a primary evaporator 310, a primary condenser 330, a secondary evaporator 340, a secondary condenser 320, a compressor 360, a primary metering device 380, a secondary metering device 390, and a fan 370. In some embodiments, dehumidification system 300 may additionally include a sub-cooling coil 350. In certain embodiments, sub-cooling coil 350 and primary condenser 330 are combined into a single coil. A flow of refrigerant 305 is circulated through dehumidification system 300 as illustrated. In general, dehumidification system 300 receives inlet airflow 101, removes water from inlet airflow 101, and discharges dehumidified air 106. Water is removed from inlet air 101 using a refrigeration cycle of flow of refrigerant 305. By including secondary evaporator 340 and secondary condenser 320, however, dehumidification system 300 causes at least part of the flow of refrigerant 305 to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0037] In general, dehumidification system 300 attempts to match the saturating temperature of secondary evaporator 340 to the saturating temperature of secondary condenser 320. The saturating temperature of secondary evaporator 340 and secondary condenser 320 generally is controlled according to the equation: (temperature of inlet air 101 + temperature of second airflow 315)/2. As the saturating temperature of secondary evaporator 340 is lower than inlet air 101, evaporation happens in secondary evaporator 340. As the saturating temperature of secondary condenser 320 is

higher than second airflow 315, condensation happens in the secondary condenser 320. The amount of refrigerant 305 evaporating in secondary evaporator 340 is substantially equal to that condensing in secondary condenser 320.

[0038] Primary evaporator 310 receives flow of refrigerant 305 from secondary metering device 390 and outputs flow of refrigerant 305 to compressor 360. Primary evaporator 310 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator 310 receives first airflow 345 from secondary evaporator 340 and outputs second airflow 315 to secondary condenser 320. Second airflow 315, in general, is at a cooler temperature than first airflow 345. To cool incoming first airflow 345, primary evaporator 310 transfers heat from first airflow 345 to flow of refrigerant 305, thereby causing flow of refrigerant 305 to evaporate at least partially from liquid to gas. This transfer of heat from first airflow 345 to flow of refrigerant 305 also removes water from first airflow 345.

[0039] Secondary condenser 320 receives flow of refrigerant 305 from secondary evaporator 340 and outputs flow of refrigerant 305 to secondary metering device 390. Secondary condenser 320 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser 320 receives second airflow 315 from primary evaporator 310 and outputs third airflow 325. Third airflow 325 is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow 315. Secondary condenser 320 generates third airflow 325 by transferring heat from flow of refrigerant 305 to second airflow 315, thereby causing flow of refrigerant 305 to condense at least partially from gas to liquid.

[0040] Primary condenser 330 receives flow of refrigerant 305 from compressor 360 and outputs flow of refrigerant 305 to either primary metering device 380 or sub-cooling coil 350. Primary condenser 330 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser 330 receives either third airflow 325 or fourth airflow 355 and outputs dehumidified air 106. Dehumidified air 106 is, in general, warmer and drier (i.e., have a lower relative humidity) than third airflow 325 and fourth airflow 355. Primary condenser 330 generates dehumidified air 106 by transferring heat from flow of refrigerant 305, thereby causing flow of refrigerant 305 to condense at least partially from gas to liquid. In some embodiments, primary condenser 330 completely condenses flow of refrigerant 305 to a liquid (i.e., 100% liquid). In other embodiments, primary condenser 330 partially condenses flow of refrigerant 305 to a liquid (i.e., less than 100% liquid). In certain embodiments, as shown in FIG. 4, a portion of primary condenser 330 receives a separate airflow in addition to airflow 101. For example, the right-most edge of primary condenser 330 of FIG. 4 extends beyond, or overhangs, the right-most edges of secondary evaporator 340, primary evaporator 310, secondary condenser 320, and sub-cooling coil 350. This overhanging portion of primary condenser 330 may receive an additional separate airflow.

[0041] Secondary evaporator 340 receives flow of refrigerant 305 from primary metering device 380 and outputs flow of refrigerant 305 to secondary condenser 320. Secondary evaporator 340 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator 340 receives inlet air 101 and outputs first airflow 345 to primary evaporator 310. First airflow 345, in general, is at a cooler temperature than inlet air 101. To cool incoming inlet air

101, secondary evaporator **340** transfers heat from inlet air **101** to flow of refrigerant **305**, thereby causing flow of refrigerant **305** to evaporate at least partially from liquid to gas.

[0042] Sub-cooling coil **350**, which is an optional component of dehumidification system **300**, sub-cools the liquid refrigerant **305** as it leaves primary condenser **330**. This, in turn, supplies primary metering device **380** with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil **350**. For example, if flow of refrigerant **305** entering sub-cooling coil **350** is 340 psig/105° F./60% vapor, flow of refrigerant **305** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **350**. The sub-cooled refrigerant **305** has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant **305**. This results in greater efficiency and less energy use of dehumidification system **300**. Embodiments of dehumidification system **300** may or may not include a sub-cooling coil **350**. For example, embodiments of dehumidification system **300** utilized within portable dehumidification system **200** that have a micro-channel condenser **330** or **320** may include a sub-cooling coil **350**, while embodiments of dehumidification system **300** that utilize another type of condenser **330** or **320** may not include a sub-cooling coil **350**. As another example, dehumidification system **300** utilized within a split system such as dehumidification system **100** may not include a sub-cooling coil **350**.

[0043] Compressor **360** pressurizes flow of refrigerant **305**, thereby increasing the temperature of refrigerant **305**. For example, if flow of refrigerant **305** entering compressor **360** is 128 psig/52° F./100% vapor, flow of refrigerant **305** may be 340 psig/150° F./100% vapor as it leaves compressor **360**. Compressor **360** receives flow of refrigerant **305** from primary evaporator **310** and supplies the pressurized flow of refrigerant **305** to primary condenser **330**.

[0044] Fan **370** may include any suitable components operable to draw inlet air **101** into dehumidification system **300** and through secondary evaporator **340**, primary evaporator **310**, secondary condenser **320**, sub-cooling coil **350**, and primary condenser **330**. Fan **370** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **370** may be a backward inclined impeller positioned adjacent to primary condenser **330** as illustrated in FIG. 3. While fan **370** is depicted in FIG. 3 as being located adjacent to primary condenser **330**, it should be understood that fan **370** may be located anywhere along the airflow path of dehumidification system **300**. For example, fan **370** may be positioned in the airflow path of any one of airflows **101**, **345**, **315**, **325**, **355**, or **106**. Moreover, dehumidification system **300** may include one or more additional fans positioned within any one or more of these airflow paths.

[0045] Primary metering device **380** and secondary metering device **390** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **380** is a thermostatic expansion valve (TXV) and secondary metering device **390** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **380** and **390** remove pressure from flow of refrigerant **305** to allow expansion or change of state from a liquid to a vapor in evaporators **310** and **340**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **380** and **390** is at a higher temperature than the liquid refrigerant **305**

leaving metering devices **380** and **390**. For example, if flow of refrigerant **305** entering primary metering device **380** is 340 psig/80° F./0% vapor, flow of refrigerant **305** may be 196 psig/68° F./5% vapor as it leaves primary metering device **380**. As another example, if flow of refrigerant **305** entering secondary metering device **390** is 196 psig/68° F./4% vapor, flow of refrigerant **305** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **390**.

[0046] Refrigerant **305** may be any suitable refrigerant such as R410a. In general, dehumidification system **300** utilizes a closed refrigeration loop of refrigerant **305** that passes from compressor **360** through primary condenser **330**, (optionally) sub-cooling coil **350**, primary metering device **380**, secondary evaporator **340**, secondary condenser **320**, secondary metering device **390**, and primary evaporator **310**. Compressor **360** pressurizes flow of refrigerant **305**, thereby increasing the temperature of refrigerant **305**. Primary and secondary condensers **330** and **320**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **305** by facilitating heat transfer from the flow of refrigerant **305** to the respective airflows passing through them (i.e., fourth airflow **355** and second airflow **315**). The cooled flow of refrigerant **305** leaving primary and secondary condensers **330** and **320** may enter a respective expansion device (i.e., primary metering device **380** and secondary metering device **390**) that is operable to reduce the pressure of flow of refrigerant **305**, thereby reducing the temperature of flow of refrigerant **305**. Primary and secondary evaporators **310** and **340**, which may include any suitable heat exchanger, receive flow of refrigerant **305** from secondary metering device **390** and primary metering device **380**, respectively. Primary and secondary evaporators **310** and **340** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **101** and first airflow **345**) to flow of refrigerant **305**. Flow of refrigerant **305**, after leaving primary evaporator **310**, passes back to compressor **360**, and the cycle is repeated.

[0047] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **310** and **340** operate in a flooded state. In other words, flow of refrigerant **305** may enter evaporators **310** and **340** in a liquid state, and a portion of flow of refrigerant **305** may still be in a liquid state as it exits evaporators **310** and **340**. Accordingly, the phase change of flow of refrigerant **305** (liquid to vapor as heat is transferred to flow of refrigerant **305**) occurs across evaporators **310** and **340**, resulting in nearly constant pressure and temperature across the entire evaporators **310** and **340** (and, as a result, increased cooling capacity).

[0048] In operation of example embodiments of dehumidification system **300**, inlet air **101** may be drawn into dehumidification system **300** by fan **370**. Inlet air **101** passes through secondary evaporator **340** in which heat is transferred from inlet air **101** to the cool flow of refrigerant **305** passing through secondary evaporator **340**. As a result, inlet air **101** may be cooled. As an example, if inlet air **101** is 80° F./60% humidity, secondary evaporator **340** may output first airflow **345** at 70° F./84% humidity. This may cause flow of refrigerant **305** to partially vaporize within secondary evaporator **340**. For example, if flow of refrigerant **305** entering secondary evaporator **340** is 196 psig/68° F./5% vapor, flow of refrigerant **305** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **340**.

[0049] The cooled inlet air 101 leaves secondary evaporator 340 as first airflow 345 and enters primary evaporator 310. Like secondary evaporator 340, primary evaporator 310 transfers heat from first airflow 345 to the cool flow of refrigerant 305 passing through primary evaporator 310. As a result, first airflow 345 may be cooled to or below its dew point temperature, causing moisture in first airflow 345 to condense (thereby reducing the absolute humidity of first airflow 345). As an example, if first airflow 345 is 70° F./84% humidity, primary evaporator 310 may output second airflow 315 at 54° F./98% humidity. This may cause flow of refrigerant 305 to partially or completely vaporize within primary evaporator 310. For example, if flow of refrigerant 305 entering primary evaporator 310 is 128 psig/44° F./14% vapor, flow of refrigerant 305 may be 128 psig/52° F./100% vapor as it leaves primary evaporator 310. In certain embodiments, the liquid condensate from first airflow 345 may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system 300 (e.g., via a drain hose) to a suitable drainage or storage location.

[0050] The cooled first airflow 345 leaves primary evaporator 310 as second airflow 315 and enters secondary condenser 320. Secondary condenser 320 facilitates heat transfer from the hot flow of refrigerant 305 passing through the secondary condenser 320 to second airflow 315. This reheats second airflow 315, thereby decreasing the relative humidity of second airflow 315. As an example, if second airflow 315 is 54° F./98% humidity, secondary condenser 320 may output third airflow 325 at 65° F./68% humidity. This may cause flow of refrigerant 305 to partially or completely condense within secondary condenser 320. For example, if flow of refrigerant 305 entering secondary condenser 320 is 196 psig/68° F./38% vapor, flow of refrigerant 305 may be 196 psig/68° F./4% vapor as it leaves secondary condenser 320.

[0051] In some embodiments, the dehumidified second airflow 315 leaves secondary condenser 320 as third airflow 325 and enters primary condenser 330. Primary condenser 330 facilitates heat transfer from the hot flow of refrigerant 305 passing through the primary condenser 330 to third airflow 325. This further heats third airflow 325, thereby further decreasing the relative humidity of third airflow 325. As an example, if third airflow 325 is 65° F./68% humidity, secondary condenser 320 may output dehumidified air 106 at 102° F./19% humidity. This may cause flow of refrigerant 305 to partially or completely condense within primary condenser 330. For example, if flow of refrigerant 305 entering primary condenser 330 is 340 psig/150° F./100% vapor, flow of refrigerant 305 may be 340 psig/105° F./60% vapor as it leaves primary condenser 330.

[0052] As described above, some embodiments of dehumidification system 300 may include a sub-cooling coil 350 in the airflow between secondary condenser 320 and primary condenser 330. Sub-cooling coil 350 facilitates heat transfer from the hot flow of refrigerant 305 passing through sub-cooling coil 350 to third airflow 325. This further heats third airflow 325, thereby further decreasing the relative humidity of third airflow 325. As an example, if third airflow 325 is 65° F./68% humidity, sub-cooling coil 350 may output fourth airflow 355 at 81° F./37% humidity. This may cause flow of refrigerant 305 to partially or completely condense

within sub-cooling coil 350. For example, if flow of refrigerant 305 entering sub-cooling coil 350 is 340 psig/150° F./60% vapor, flow of refrigerant 305 may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil 350.

[0053] Some embodiments of dehumidification system 300 may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0054] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system 300, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0055] Although particular implementations of dehumidification system 300 are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system 300, according to particular needs. Moreover, although various components of dehumidification system 300 have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0056] FIG. 5 illustrates an example dehumidification method 500 that may be used by dehumidification system 100 and portable dehumidification system 200 of FIGS. 1 and 2 to reduce the humidity of air within structure 102. Method 500 may begin in step 510 where a secondary evaporator receives an inlet airflow and outputs a first airflow. In some embodiments, the secondary evaporator is secondary evaporator 340. In some embodiments, the inlet airflow is inlet air 101 and the first airflow is first airflow 345. In some embodiments, the secondary evaporator of step 510 receives a flow of refrigerant from a primary metering device such as primary metering device 380 and supplies the flow of refrigerant (in a changed state) to a secondary condenser such as secondary condenser 320. In some embodiments, the flow of refrigerant of method 500 is flow of refrigerant 305 described above.

[0057] At step 520, a primary evaporator receives the first airflow of step 510 and outputs a second airflow. In some embodiments, the primary evaporator is primary evaporator

310 and the second airflow is second airflow **315**. In some embodiments, the primary evaporator of step **520** receives the flow of refrigerant from a secondary metering device such as secondary metering device **390** and supplies the flow of refrigerant (in a changed state) to a compressor such as compressor **360**.

[0058] At step **530**, a secondary condenser receives the second airflow of step **520** and outputs a third airflow. In some embodiments, the secondary condenser is secondary condenser **320** and the third airflow is third airflow **325**. In some embodiments, the secondary condenser of step **530** receives a flow of refrigerant from the secondary evaporator of step **510** and supplies the flow of refrigerant (in a changed state) to a secondary metering device such as secondary metering device **390**.

[0059] At step **540**, a primary condenser receives the third airflow of step **530** and outputs a dehumidified airflow. In some embodiments, the primary condenser is primary condenser **330** and the dehumidified airflow is dehumidified air **106**. In some embodiments, the primary condenser of step **540** receives a flow of refrigerant from the compressor of step **520** and supplies the flow of refrigerant (in a changed state) to the primary metering device of step **510**. In alternate embodiments, the primary condenser of step **540** supplies the flow of refrigerant (in a changed state) to a sub-cooling coil such as sub-cooling coil **350** which in turn supplies the flow of refrigerant (in a changed state) to the primary metering device of step **510**.

[0060] At step **550**, a compressor receives the flow of refrigerant from the primary evaporator of step **520** and provides the flow of refrigerant (in a changed state) to the primary condenser of step **540**. After step **550**, method **500** may end.

[0061] Particular embodiments may repeat one or more steps of method **500** of FIG. **5**, where appropriate. Although this disclosure describes and illustrates particular steps of the method of FIG. **5** as occurring in a particular order, this disclosure contemplates any suitable steps of the method of FIG. **5** occurring in any suitable order. Moreover, although this disclosure describes and illustrates an example dehumidification method for reducing the humidity of air within a structure including the particular steps of the method of FIG. **5**, this disclosure contemplates any suitable method for reducing the humidity of air within a structure including any suitable steps, which may include all, some, or none of the steps of the method of FIG. **5**, where appropriate. Furthermore, although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of the method of FIG. **5**, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of the method of FIG. **5**.

[0062] While the example method of FIG. **5** is described at times above with respect to dehumidification system **300** of FIG. **3**, it should be understood that the same or similar methods can be carried out using any of the dehumidification systems described herein, including dehumidification systems **600** and **800** of FIGS. **6A-6B** and **8** (described below). Moreover, it should be understood that, with respect to the example method of FIG. **5**, reference to an evaporator or condenser can refer to an evaporator portion or condenser portion of a single coil pack operable to perform the functions of these components, for example, as described above with respect to examples of FIGS. **9** and **10**.

[0063] FIGS. **6A** and **6B** illustrate an example air conditioning and dehumidification system **600** that may be used in accordance with split dehumidification system **100** of FIG. **1** to reduce the humidity of air within structure **102**. Dehumidification system **600** includes a dehumidification unit **602**, which is generally indoors, and a condenser system **604** (e.g., condenser system **108** of FIG. **1**). As illustrated in FIG. **6A**, dehumidification unit **602** includes a primary evaporator **610**, a secondary evaporator **640**, a secondary condenser **620**, a primary metering device **680**, a secondary metering device **690**, and a first fan **670**, while condenser system **604** includes a primary condenser **630**, a compressor **660**, an optional sub-cooling coil **650** and a second fan **695**. In the embodiment illustrated in FIG. **6B**, the compressor **660** may be disposed within the dehumidification unit **602** rather than disposed within the condenser system **604**.

[0064] With reference to both FIGS. **6A** and **6B**, a flow of refrigerant **605** is circulated through dehumidification system **600** as illustrated. In general, dehumidification unit **602** receives inlet airflow **601**, removes water from inlet airflow **601**, and discharges dehumidified air **625** into a conditioned space. Water is removed from inlet air **601** using a refrigeration cycle of flow of refrigerant **605**. The flow of refrigerant **605** through system **600** of FIGS. **6A** AND **6B** proceeds in a similar manner to that of the flow of refrigerant **305** through dehumidification system **300** of FIG. **3**. However, the path of airflow through system **600** is different than that through system **300**, as described herein. By including secondary evaporator **640** and secondary condenser **620**, however, dehumidification system **600** causes at least part of the flow of refrigerant **605** to evaporate and condense twice in a single refrigeration cycle. This increases refrigerating capacity over typical systems without requiring any additional power to the compressor, thereby increasing the overall efficiency of the system.

[0065] The split configuration of system **600**, which includes dehumidification unit **602** and condenser system **604**, allows heat from the cooling and dehumidification process to be rejected outdoors or to an unconditioned space (e.g., external to a space being dehumidified). This allows dehumidification system **600** to have a similar footprint to that of typical central air conditioning systems or heat pumps. In general, the temperature of third airflow **625** output to the conditioned space from system **600** is significantly decreased compared to that of airflow **106** output from system **300** of FIG. **3**. Thus, the configuration of system **600** allows dehumidified air to be provided to the conditioned space at a decreased temperature. Accordingly, system **600** may perform functions of both a dehumidifier (dehumidifying air) and a central air conditioner (cooling air).

[0066] In general, dehumidification system **600** attempts to match the saturating temperature of secondary evaporator **640** to the saturating temperature of secondary condenser **620**. The saturating temperature of secondary evaporator **640** and secondary condenser **620** generally is controlled according to the equation: (temperature of inlet air **601**+ temperature of second airflow **615**)/2. As the saturating temperature of secondary evaporator **640** is lower than inlet air **601**, evaporation happens in secondary evaporator **640**. As the saturating temperature of secondary condenser **620** is higher than second airflow **615**, condensation happens in secondary condenser **620**. The amount of refrigerant **605**

evaporating in secondary evaporator **640** is substantially equal to that condensing in secondary condenser **620**.

[0067] Primary evaporator **610** receives flow of refrigerant **605** from secondary metering device **690** and outputs flow of refrigerant **605** to compressor **660**. Primary evaporator **610** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator **610** receives first airflow **645** from secondary evaporator **640** and outputs second airflow **615** to secondary condenser **620**. Second airflow **615**, in general, is at a cooler temperature than first airflow **645**. To cool incoming first airflow **645**, primary evaporator **610** transfers heat from first airflow **645** to flow of refrigerant **605**, thereby causing flow of refrigerant **605** to evaporate at least partially from liquid to gas. This transfer of heat from first airflow **645** to flow of refrigerant **605** also removes water from first airflow **645**.

[0068] Secondary condenser **620** receives flow of refrigerant **605** from secondary evaporator **640** and outputs flow of refrigerant **605** to secondary metering device **690**. Secondary condenser **620** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser **620** receives second airflow **615** from primary evaporator **610** and outputs third airflow **625**. Third airflow **625** is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow **615**. Secondary condenser **620** generates third airflow **625** by transferring heat from flow of refrigerant **605** to second airflow **615**, thereby causing flow of refrigerant **605** to condense at least partially from gas to liquid. As described above, third airflow **625** is output into the conditioned space. In other embodiments (e.g., as shown in FIGS. **8A** and **8B**), third airflow **625** may first pass through and/or over sub-cooling coil **650** before being output into the conditioned space at a further decreased relative humidity.

[0069] As shown in FIG. **6A**, refrigerant **605** flows outdoors or to an unconditioned space to compressor **660** of condenser system **604**. Alternatively, the refrigerant **605** may continue to flow to the compressor **660** within the dehumidification unit **602** prior to flowing outdoors or to an unconditioned space, as seen in FIG. **6B**. In both FIGS. **6A** and **6B**, compressor **660** pressurizes flow of refrigerant **605**, thereby increasing the temperature of refrigerant **605**. For example, if flow of refrigerant **605** entering compressor **660** is 128 psig/52° F./100% vapor, flow of refrigerant **605** may be 340 psig/150° F./100% vapor as it leaves compressor **660**. Compressor **660** receives flow of refrigerant **605** from primary evaporator **610** and supplies the pressurized flow of refrigerant **605** to primary condenser **630**.

[0070] Primary condenser **630** receives flow of refrigerant **605** from compressor **660** and outputs flow of refrigerant **605** to sub-cooling coil **650**. Primary condenser **630** may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser **630** and sub-cooling coil **650** receive first outdoor airflow **606** and output second outdoor airflow **608**. Second outdoor airflow **608** is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow **606**. Primary condenser **630** transfers heat from flow of refrigerant **605**, thereby causing flow of refrigerant **605** to condense at least partially from gas to liquid. In some embodiments, primary condenser **630** completely condenses flow of refrigerant **605** to a liquid (i.e., 100% liquid). In other embodiments, primary condenser **630** partially condenses flow of refrigerant **605** to a liquid (i.e., less than 100% liquid).

[0071] Sub-cooling coil **650**, which is an optional component of dehumidification system **600**, sub-cools the liquid refrigerant **605** as it leaves primary condenser **630**. This, in turn, supplies primary metering device **680** with a liquid refrigerant that is 30 degrees (or more) cooler than before it enters sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/105° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**. The sub-cooled refrigerant **605** has a greater heat enthalpy factor as well as a greater density, which improves energy transfer between airflow and evaporator resulting in the removal of further latent heat from refrigerant **605**. This further results in greater efficiency and less energy use of dehumidification system **600**. Embodiments of dehumidification system **600** may or may not include a sub-cooling coil **650**.

[0072] In certain embodiments, sub-cooling coil **650** and primary condenser **630** are combined into a single coil. Such a single coil includes appropriate circuiting for flow of airflows **606** and **608** and refrigerant **605**. An illustrative example of a condenser system **604** comprising a single coil condenser and sub-cooling coil is shown in FIG. **7**. The single unit coil comprises interior tubes **710** corresponding to the condenser and exterior tubes **705** corresponding to the sub-cooling coil. Refrigerant may be directed through the interior tubes **710** before flowing through exterior tubes **705**. In the illustrative example shown in FIG. **7**, airflow is drawn through the single unit coil by fan **695** and expelled upwards. It should be understood, however, that condenser systems of other embodiments can include a condenser, compressor, optional sub-cooling coil, and fan with other configurations known in the art.

[0073] Secondary evaporator **640** receives flow of refrigerant **605** from primary metering device **680** and outputs flow of refrigerant **605** to secondary condenser **620**. Secondary evaporator **640** may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator **640** receives inlet air **601** and outputs first airflow **645** to primary evaporator **610**. First airflow **645**, in general, is at a cooler temperature than inlet air **601**. To cool incoming inlet air **601**, secondary evaporator **640** transfers heat from inlet air **601** to flow of refrigerant **605**, thereby causing flow of refrigerant **605** to evaporate at least partially from liquid to gas.

[0074] Fan **670** may include any suitable components operable to draw inlet air **601** into dehumidification unit **602** and through secondary evaporator **640**, primary evaporator **610**, and secondary condenser **620**. Fan **670** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **670** may be a backward inclined impeller positioned adjacent to secondary condenser **620**.

[0075] While fan **670** is depicted in FIGS. **6A** and **6B** as being located adjacent to condenser **620**, it should be understood that fan **670** may be located anywhere along the airflow path of dehumidification unit **602**. For example, fan **670** may be positioned in the airflow path of any one of airflows **601**, **645**, **615**, or **625**. Moreover, dehumidification unit **602** may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, while fan **695** of condenser system **604** is depicted in FIGS. **6A** and **6B** as being located above primary condenser **630**, it should be understood that fan **695** may be located anywhere (e.g., above, below, beside) with respect to condenser

630 and sub-cooling coil 650, so long fan 695 is appropriately positioned and configured to facilitate flow of airflow 606 towards primary condenser 630 and sub-cooling coil 650.

[0076] The rate of airflow generated by fan 670 may be different than that generated by fan 695. For example, the flow rate of airflow 606 generated by fan 695 may be higher than the flow rate of airflow 601 generated by fan 670. This difference in flow rates may provide several advantages for the dehumidification systems described herein. For example, a large airflow generated by fan 695 may provide for improved heat transfer at the sub-cooling coil 650 and primary condenser 630 of the condenser system 604. In general, the rate of airflow generated by second fan 695 is between about 2-times to 5-times that of the rate of airflow generated by first fan 670. For example, the rate of airflow generated by first fan 670 may be from about 200 to 400 cubic feet per minute (cfm). For example, the rate of airflow generated by second fan 695 may be from about 900 to 1200 cubic feet per minute (cfm).

[0077] Primary metering device 680 and secondary metering device 690 are any appropriate type of metering/expansion device. In some embodiments, primary metering device 680 is a thermostatic expansion valve (TXV) and secondary metering device 690 is a fixed orifice device (or vice versa). In certain embodiments, metering devices 680 and 690 remove pressure from flow of refrigerant 605 to allow expansion or change of state from a liquid to a vapor in evaporators 610 and 640. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices 680 and 690 is at a higher temperature than the liquid refrigerant 605 leaving metering devices 680 and 690. For example, if flow of refrigerant 605 entering primary metering device 680 is 340 psig/80° F./0% vapor, flow of refrigerant 605 may be 196 psig/68° F./5% vapor as it leaves primary metering device 680. As another example, if flow of refrigerant 605 entering secondary metering device 690 is 196 psig/68° F./4% vapor, flow of refrigerant 605 may be 128 psig/44° F./14% vapor as it leaves secondary metering device 690.

[0078] In certain embodiments, secondary metering device 690 is operated in a substantially open state (referred to herein as a “fully open” state) such that the pressure of refrigerant 605 entering metering device 690 is substantially the same as the pressure of refrigerant 605 exiting metering device 605. For example, the pressure of refrigerant 605 may be 80%, 90%, 95%, 99%, or up to 100% of the pressure of refrigerant 605 entering metering device 690. With the secondary metering device 690 operated in a “fully open” state, primary metering device 680 is the primary source of pressure drop in dehumidification system 600. In this configuration, airflow 615 is not substantially heated when it passes through secondary condenser 620, and the secondary evaporator 640, primary evaporator 610, and secondary condenser 620 effectively act as a single evaporator. Although, less water may be removed from airflow 601 when the secondary metering device 690 is operated in a “fully open” state, airflow 606 will be output to the conditioned space at a lower temperature than when secondary metering device 690 is not in a “fully open” state. This configuration corresponds to a relatively high sensible heat ratio (SHR) operating mode such that dehumidification system 600 may produce a cool airflow 625 with properties similar to those of an airflow produced by a central air conditioner. If the rate of airflow 601 is increased to a

threshold value (e.g., by increasing the speed of fan 670 or one or more other fans of dehumidification system 600), dehumidification system 600 may perform sensible cooling without removing water from airflow 601.

[0079] Refrigerant 605 may be any suitable refrigerant such as R410a. In general, dehumidification system 600 utilizes a closed refrigeration loop of refrigerant 605 that passes from compressor 660 through primary condenser 630, (optionally) sub-cooling coil 650, primary metering device 680, secondary evaporator 640, secondary condenser 620, secondary metering device 690, and primary evaporator 610. Compressor 660 pressurizes flow of refrigerant 605, thereby increasing the temperature of refrigerant 605. Primary and secondary condensers 630 and 620, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant 605 by facilitating heat transfer from the flow of refrigerant 605 to the respective airflows passing through them (i.e., first outdoor airflow 606 and second airflow 615). The cooled flow of refrigerant 605 leaving primary and secondary condensers 630 and 620 may enter a respective expansion device (i.e., primary metering device 680 and secondary metering device 690) that is operable to reduce the pressure of flow of refrigerant 605, thereby reducing the temperature of flow of refrigerant 605. Primary and secondary evaporators 610 and 640, which may include any suitable heat exchanger, receive flow of refrigerant 605 from secondary metering device 690 and primary metering device 680, respectively. Primary and secondary evaporators 610 and 640 facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air 601 and first airflow 645) to flow of refrigerant 605. Flow of refrigerant 605, after leaving primary evaporator 610, passes back to compressor 660, and the cycle is repeated.

[0080] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators 610 and 640 operate in a flooded state. In other words, flow of refrigerant 605 may enter evaporators 610 and 640 in a liquid state, and a portion of flow of refrigerant 605 may still be in a liquid state as it exits evaporators 610 and 640. Accordingly, the phase change of flow of refrigerant 605 (liquid to vapor as heat is transferred to flow of refrigerant 605) occurs across evaporators 610 and 640, resulting in nearly constant pressure and temperature across the entire evaporators 610 and 640 (and, as a result, increased cooling capacity).

[0081] In operation of example embodiments of dehumidification system 600, inlet air 601 may be drawn into dehumidification system 600 by fan 670. Inlet air 601 passes through secondary evaporator 640 in which heat is transferred from inlet air 601 to the cool flow of refrigerant 605 passing through secondary evaporator 640. As a result, inlet air 601 may be cooled. As an example, if inlet air 601 is 80° F./60% humidity, secondary evaporator 640 may output first airflow 645 at 70° F./84% humidity. This may cause flow of refrigerant 605 to partially vaporize within secondary evaporator 640. For example, if flow of refrigerant 605 entering secondary evaporator 640 is 196 psig/68° F./5% vapor, flow of refrigerant 605 may be 196 psig/68° F./38% vapor as it leaves secondary evaporator 640.

[0082] The cooled inlet air 601 leaves secondary evaporator 640 as first airflow 645 and enters primary evaporator 610. Like secondary evaporator 640, primary evaporator 610 transfers heat from first airflow 645 to the cool flow of refrigerant 605 passing through primary evaporator 610. As

a result, first airflow **645** may be cooled to or below its dew point temperature, causing moisture in first airflow **645** to condense (thereby reducing the absolute humidity of first airflow **645**). As an example, if first airflow **645** is 70° F./84% humidity, primary evaporator **610** may output second airflow **615** at 54° F./98% humidity. This may cause flow of refrigerant **605** to partially or completely vaporize within primary evaporator **610**. For example, if flow of refrigerant **605** entering primary evaporator **610** is 128 psig/44° F./14% vapor, flow of refrigerant **605** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **610**. In certain embodiments, the liquid condensate from first airflow **645** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system **600** (e.g., via a drain hose) to a suitable drainage or storage location.

[0083] The cooled first airflow **645** leaves primary evaporator **610** as second airflow **615** and enters secondary condenser **620**. Secondary condenser **620** facilitates heat transfer from the hot flow of refrigerant **605** passing through the secondary condenser **620** to second airflow **615**. This reheats second airflow **615**, thereby decreasing the relative humidity of second airflow **615**. As an example, if second airflow **615** is 54° F./98% humidity, secondary condenser **620** may output dehumidified airflow **625** at 65° F./68% humidity. This may cause flow of refrigerant **605** to partially or completely condense within secondary condenser **620**. For example, if flow of refrigerant **605** entering secondary condenser **620** is 196 psig/68° F./38% vapor, flow of refrigerant **605** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **620**. In some embodiments, second airflow **615** leaves secondary condenser **620** as dehumidified airflow **625** and is output to a conditioned space.

[0084] Primary condenser **630** facilitates heat transfer from the hot flow of refrigerant **605** passing through the primary condenser **630** to a first outdoor airflow **606**. This heats outdoor airflow **606**, which is output to the unconditioned space (e.g., outdoors) as second outdoor airflow **608**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, primary condenser **630** may output second outdoor airflow **608** at 102° F./19% humidity. This may cause flow of refrigerant **605** to partially or completely condense within primary condenser **630**. For example, if flow of refrigerant **605** entering primary condenser **630** is 340 psig/150° F./100% vapor, flow of refrigerant **605** may be 340 psig/105° F./60% vapor as it leaves primary condenser **630**.

[0085] As described above, some embodiments of dehumidification system **600** may include a sub-cooling coil **650** in the airflow between an inlet of the condenser system **604** and primary condenser **630**. Sub-cooling coil **650** facilitates heat transfer from the hot flow of refrigerant **605** passing through sub-cooling coil **650** to first outdoor airflow **606**. This heats first outdoor airflow **606**, thereby increasing the temperature of first outdoor airflow **606**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, sub-cooling coil **650** may output an airflow at 81° F./37% humidity. This may cause flow of refrigerant **605** to partially or completely condense within sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/150° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**.

[0086] In the embodiment depicted in FIGS. 6A and 6B, sub-cooling coil **650** is within condenser system **604**. This configuration minimizes the temperature of third airflow **625**, which is output into the conditioned space. An alternative embodiment is shown as dehumidification system **800** of FIGS. 8A and 8B in which dehumidification unit **802** includes sub-cooling coil **650**. In these embodiments, airflow **625** first passes through sub-cooling coil **650** before being output to the conditioned space as airflow **855** via fan **670**. As described herein, fan **670** can alternatively be located anywhere along the path of airflow in dehumidification unit **802**, and one or more additional fans can be included in dehumidification unit **802**.

[0087] Without wishing to be bound to any particular theory, the configuration of dehumidification system **800** is believed to be more energy efficient under common operating conditions than that of dehumidification system **600** of FIGS. 6A-6B. For example, if the temperature of third airflow **625** is less than the outdoor temperature (i.e., the temperature of airflow **606**), then refrigerant **605** will be more effectively cooled, or sub-cooled, with sub-cooling coil **650** placed in the dehumidification unit **802**. Such operating conditions may be common, for example, in locations with warm climates and/or during summer months. As illustrated in FIG. 8B, indoor dehumidification unit **802** also includes compressor **660**, which may, for example, be located near secondary evaporator **640**, primary evaporator **610**, and/or secondary condenser **620**. In certain embodiments, the dehumidification unit **802** may comprise the compressor **660**, but the dehumidification system **800** may lack the optional sub-cooling coil **650**, as illustrated in FIG. 8C. The dehumidification system **800** of FIG. 8C may not require the sub-cooling coil **650** if, for example, the primary condenser **630** is operable to facilitate heat transfer from the flow of refrigerant **605** to a first outdoor airflow **606** in order to effectively condense the refrigerant prior to the flow of refrigerant entering a primary metering device **680**.

[0088] In operation of example embodiments of dehumidification system **800**, as illustrated in each of FIGS. 8A-8C, inlet air **601** may be drawn into dehumidification system **800** by fan **670**. Inlet air **601** passes through secondary evaporator **640** in which heat is transferred from inlet air **601** to the cool flow of refrigerant **605** passing through secondary evaporator **640**. As a result, inlet air **601** may be cooled. As an example, if inlet air **601** is 80° F./60% humidity, secondary evaporator **640** may output first airflow **645** at 70° F./84% humidity. This may cause flow of refrigerant **605** to partially vaporize within secondary evaporator **640**. For example, if flow of refrigerant **605** entering secondary evaporator **640** is 196 psig/68° F./5% vapor, flow of refrigerant **605** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **640**.

[0089] The cooled inlet air **601** leaves secondary evaporator **640** as first airflow **645** and enters primary evaporator **610**. Like secondary evaporator **640**, primary evaporator **610** transfers heat from first airflow **645** to the cool flow of refrigerant **605** passing through primary evaporator **610**. As a result, first airflow **645** may be cooled to or below its dew point temperature, causing moisture in first airflow **645** to condense (thereby reducing the absolute humidity of first airflow **645**). As an example, if first airflow **645** is 70° F./84% humidity, primary evaporator **610** may output second airflow **615** at 54° F./98% humidity. This may cause flow of refrigerant **605** to partially or completely vaporize

within primary evaporator **610**. For example, if flow of refrigerant **605** entering primary evaporator **610** is 128 psig/44° F./14% vapor, flow of refrigerant **605** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **610**. In certain embodiments, the liquid condensate from first airflow **645** may be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of dehumidification system **800** (e.g., via a drain hose) to a suitable drainage or storage location.

[0090] The cooled first airflow **645** leaves primary evaporator **610** as second airflow **615** and enters secondary condenser **620**. Secondary condenser **620** facilitates heat transfer from the hot flow of refrigerant **605** passing through the secondary condenser **620** to second airflow **615**. This reheats second airflow **615**, thereby decreasing the relative humidity of second airflow **615**. As an example, if second airflow **615** is 54° F./98% humidity, secondary condenser **620** may output dehumidified airflow **625** at 65° F./68% humidity. This may cause flow of refrigerant **605** to partially or completely condense within secondary condenser **620**. For example, if flow of refrigerant **605** entering secondary condenser **620** is 196 psig/68° F./38% vapor, flow of refrigerant **605** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **620**. In some embodiments, second airflow **615** leaves secondary condenser **620** as dehumidified airflow **625** and is output to a conditioned space.

[0091] In both FIGS. 8A and 8B, dehumidified airflow **625** enters sub-cooling coil **650**, which facilitates heat transfer from the hot flow of refrigerant **605** passing through sub-cooling coil **650** to dehumidified airflow **625**. This heats dehumidified airflow **625**, thereby further decreasing the humidity of dehumidified airflow **625**. As an example, if dehumidified airflow **625** is 65° F./68% humidity, sub-cooling coil **650** may output an airflow **855** at 81° F./37% humidity. This may cause flow of refrigerant **605** to partially or completely condense within sub-cooling coil **650**. For example, if flow of refrigerant **605** entering sub-cooling coil **650** is 340 psig/150° F./60% vapor, flow of refrigerant **605** may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil **650**.

[0092] With reference back to each of FIGS. 8A-8C, primary condenser **630** facilitates heat transfer from the hot flow of refrigerant **605** passing through the primary condenser **630** to a first outdoor airflow **606**. This heats outdoor airflow **606**, which is output to the unconditioned space as second outdoor airflow **608**. As an example, if first outdoor airflow **606** is 65° F./68% humidity, primary condenser **630** may output second outdoor airflow **608** at 102° F./19% humidity. This may cause flow of refrigerant **605** to partially or completely condense within primary condenser **630**. For example, if flow of refrigerant **605** entering primary condenser **630** is 340 psig/150° F./100% vapor, flow of refrigerant **605** may be 340 psig/105° F./60% vapor as it leaves primary condenser **630**.

[0093] Some embodiments of dehumidification systems **600** and **800** of FIGS. 6A-6B and 8A-8C may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicat-

ing data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0094] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification systems **600** and **800**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0095] Although particular implementations of dehumidification systems **600** and **800** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification systems **600** and **800**, according to particular needs. Moreover, although various components of dehumidification systems **600** and **800** have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0096] In certain embodiments, the secondary evaporator (**340**, **640**), primary evaporator (**310**, **610**), and secondary condenser (**320**, **620**) of FIG. 3, 6A-6B, or 8A-8C are combined in a single coil pack. The single coil pack may include portions (e.g., separate refrigerant circuits) to accommodate the respective functions of secondary evaporator, primary evaporator, and secondary condenser, described above. An illustrative example of such a single coil pack is shown in FIG. 9. FIG. 9 shows a single coil pack **900** which includes a plurality of coils (represented by circles in FIG. 9). Coil pack **900** includes a secondary evaporator portion **940**, primary evaporator portion **910**, and secondary condenser portion **920**. The coil pack may include and/or be fluidly connectable to metering devices **980** and **990** as shown in the exemplary case of FIG. 9. In certain embodiments, metering devices **980** and **990** correspond to primary metering device **380** and secondary metering device **390** of FIG. 3.

[0097] In general, metering devices **980** and **990** may be any appropriate type of metering/expansion device. In some embodiments, metering device **980** is a thermostatic expansion valve (TXV) and secondary metering device **990** is a fixed orifice device (or vice versa). In general, metering devices **980** and **990** remove pressure from flow of refrigerant **905** to allow expansion or change of state from a liquid to a vapor in evaporator portions **910** and **940**. The high-pressure liquid (or mostly liquid) refrigerant **905** entering metering devices **980** and **990** is at a higher temperature than the liquid refrigerant **905** leaving metering devices **980** and

990. For example, if flow of refrigerant **905** entering metering device **980** is 340 psig/80° F./0% vapor, flow of refrigerant **905** may be 196 psig/68° F./5% vapor as it leaves primary metering device **980**. As another example, if flow of refrigerant **905** entering secondary metering device **990** is 196 psig/68° F./4% vapor, flow of refrigerant **905** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **990**. Refrigerant **905** may be any suitable refrigerant, as described above with respect to refrigerant **305** of FIG. 3.

[0098] In operation of example embodiments of the single coil pack **900**, inlet airflow **901** passes through secondary evaporator portion **940** in which heat is transferred from inlet air **901** to the cool flow of refrigerant **905** passing through secondary evaporator portion **940**. As a result, inlet air **901** may be cooled. As an example, if inlet air **901** is 80° F./60% humidity, secondary evaporator portion **940** may output first airflow at 70° F./84% humidity. This may cause flow of refrigerant **905** to partially vaporize within secondary evaporator portion **940**. For example, if flow of refrigerant **905** entering secondary evaporator portion **940** is 196 psig/68° F./5% vapor, flow of refrigerant **905** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator portion **940**.

[0099] The cooled inlet air **901** proceeds through coil pack **900**, reaching primary evaporator portion **910**. Like secondary evaporator portion **940**, primary evaporator portion **910** transfers heat from airflow **901** to the cool flow of refrigerant **905** passing through primary evaporator portion **910**. As a result, airflow **901** may be cooled to or below its dew point temperature, causing moisture in airflow **901** to condense (thereby reducing the absolute humidity of airflow **901**). As an example, if airflow **901** is 70° F./84% humidity, primary evaporator portion **910** may cool airflow **901** to 54° F./98% humidity. This may cause flow of refrigerant **905** to partially or completely vaporize within primary evaporator portion **910**. For example, if flow of refrigerant **905** entering primary evaporator portion **910** is 128 psig/44° F./14% vapor, flow of refrigerant **905** may be 128 psig/52° F./100% vapor as it leaves primary evaporator portion **910**. In certain embodiments, the liquid condensate from airflow through primary evaporator portion **910** may be collected in a drain pan connected to a condensate reservoir (e.g., as illustrated in FIG. 4 and described herein). Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of coil pack **900** (e.g., via a drain hose) to a suitable drainage or storage location.

[0100] The cooled airflow **901** leaving primary evaporator portion **910** enters secondary condenser portion **920**. Secondary condenser portion **920** facilitates heat transfer from the hot flow of refrigerant **905** passing through the secondary condenser portion **920** to airflow **901**. This reheats airflow **901**, thereby decreasing its relative humidity. As an example, if airflow **901** is 54° F./98% humidity, secondary condenser portion **920** may output an outlet airflow **925** at 65° F./68% humidity. This may cause flow of refrigerant **905** to partially or completely condense within secondary condenser portion **920**. For example, if flow of refrigerant **905** entering secondary condenser portion **920** is 196 psig/68° F./38% vapor, flow of refrigerant **905** may be 196 psig/68° F./4% vapor as it leaves secondary condenser portion **920**. Outlet airflow **925** may, for example, enter primary condenser portion **330** or sub-cooling coil **350** of FIG. 3.

[0101] Although a particular implementation of coil pack **900** is illustrated and primarily described, the present disclosure contemplates any suitable implementation of coil pack **900**, according to particular needs. Moreover, although various components of coil pack **900** have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0102] In certain embodiments, secondary evaporator (**340**, **640**) and secondary condenser (**320**, **620**) of FIG. 3, **6A-6B**, or **8A-8C** are combined in a single coil pack such that the single coil pack includes portions (e.g., separate refrigerant circuits) to accommodate the respective functions of the secondary evaporator and secondary condenser. An illustrative example of such an embodiment is shown in FIG. 10. FIG. 10 shows a single coil pack **1000** which includes a secondary evaporator portion **1040** and secondary condenser portion **1020**. As shown in the illustrative example of FIG. 10, a primary evaporator **1010** is located between the secondary evaporator portion **1040** and secondary condenser portion **1020** of the single coil pack **1000**. In this exemplary embodiment, the single coil pack **1000** is shown as a “U”-shaped coil. However, alternate embodiments may be used as long as flow airflow **1001** passes sequentially through secondary evaporator portion **1040**, primary evaporator **1010**, and secondary condenser portion **1020**. In general, single coil pack **1000** can include the same or a different coil type compared to that of primary evaporator **1010**. For example, single coil pack **1000** may include a microchannel coil type, while primary evaporator **1010** may include a fin tube coil type. This may provide further flexibility for optimizing a dehumidification system in which single coil pack **1000** and primary evaporator **1010** are used.

[0103] In operation of example embodiments of the single coil pack **1000**, inlet air **1001** passes through secondary evaporator portion **1040** in which heat is transferred from inlet air **1001** to the cool flow of refrigerant passing through secondary evaporator portion **1040**. As a result, inlet air **1001** may be cooled. As an example, if inlet air **1001** is 80° F./60% humidity, secondary evaporator portion **1040** may output airflow at 70° F./84% humidity. This may cause flow of refrigerant to partially vaporize within secondary evaporator portion **1040**. For example, if flow of refrigerant entering secondary evaporator **1040** is 196 psig/68° F./5% vapor, flow of refrigerant **1005** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator portion **1040**.

[0104] The cooled inlet air **1001** leaves secondary evaporator portion **1040** and enters primary evaporator **1010**. Like secondary evaporator portion **1040**, primary evaporator **1010** transfers heat from airflow **1001** to the cool flow of refrigerant passing through primary evaporator **1010**. As a result, airflow **1001** may be cooled to or below its dew point temperature, causing moisture in airflow **1001** to condense (thereby reducing the absolute humidity of airflow **1001**). As an example, if airflow **1001** entering primary evaporator **1010** is 70° F./84% humidity, primary evaporator **1010** may output airflow at 54° F./98% humidity. This may cause flow of refrigerant to partially or completely vaporize within primary evaporator **1010**. For example, if flow of refrigerant entering primary evaporator **1010** is 128 psig/44° F./14% vapor, flow of refrigerant may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1010**. In certain embodiments, the liquid condensate from airflow **1010** may

be collected in a drain pan connected to a condensate reservoir, as illustrated in FIG. 4. Additionally, the condensate reservoir may include a condensate pump that moves collected condensate, either continually or at periodic intervals, out of primary evaporator 1010, and the associated dehumidification system (e.g., via a drain hose) to a suitable drainage or storage location.

[0105] The cooled airflow 1001 leaves primary evaporator 1010 and enters secondary condenser portion 1020. Secondary condenser portion 1020 facilitates heat transfer from the hot flow of refrigerant passing through the secondary condenser 1020 to airflow 1001. This reheats airflow 1001, thereby decreasing its relative humidity. As an example, if airflow 1001 entering secondary condenser portion 1020 is 54° F./98% humidity, secondary condenser 1020 may output airflow 1025 at 65° F./68% humidity. This may cause flow of refrigerant to partially or completely condense within secondary condenser 1020. For example, if flow of refrigerant entering secondary condenser portion 1020 is 196 psig/68° F./38% vapor, flow of refrigerant may be 196 psig/68° F./4% vapor as it leaves secondary condenser 1020. Outlet airflow 925 may, for example, enter primary condenser 330 or sub-cooling cooling 350 of FIG. 3.

[0106] Although a particular implementation of coil pack 1000 is illustrated and primarily described, the present disclosure contemplates any suitable implementation of coil pack 1000, according to particular needs. Moreover, although various components of coil pack 1000 have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0107] In certain embodiments, one or both of the secondary evaporator (340, 640) and primary evaporator (310, 610) of FIG. 3, 6A-6B, or 8A-8C are subdivided into two or more circuits. In such embodiments, each circuit of the subdivided evaporator(s) is fed refrigerant by a corresponding metering device. The metering devices may include passive metering devices, active metering devices, or combinations thereof. For example, metering device 380 (or 690) may be an active thermostatic expansion valve (TXV) and secondary metering device 390 (or 690) may be a passive fixed orifice device (or vice versa). The metering devices may be configured to feed refrigerant to each circuit within the evaporators at a desired mass flow rate. Metering devices for feeding refrigerant to each circuit of the subdivided evaporator(s) may be used in combination with metering devices 380 and 390 or may replace one or both of metering devices 380 and 390.

[0108] FIGS. 11, 12, 13, and 14 show an illustrative example of a portion 1100 of a dehumidification system in which the primary evaporator 1110 comprises three circuits for flow of refrigerant, according to certain embodiments. Portion 1100 includes a primary metering device 1180, secondary metering devices 1190a-c, a secondary evaporator 1140, a primary evaporator 1110, and a secondary condenser 1120. Primary evaporator 1110 includes three circuits for receiving flow of refrigerant from secondary metering devices 1190a-c. In the example of FIGS. 11, 12, 13, and 14, each of secondary metering devices 1190a-c is a passive metering device (i.e., with an orifice of a fixed inner diameter and length). It should, however be understood that one or more (up to all) of the secondary metering devices 1190a-c may be active metering devices (e.g., thermostatic expansion valves).

[0109] In operation of example embodiments of portion 1100 of a dehumidification system, flow of cooled (or sub-cooled) refrigerant is received at inlet 1102, for example, from sub-cooling coil 350 or primary condenser 330 of dehumidification system 300 of FIG. 3. Primary metering device 1180 determines the flow rate of refrigerant into secondary evaporator 1140. While FIGS. 11, 12, 13, and 14 are shown to have a single primary metering device 1180, other embodiments can include multiple primary metering devices in parallel (e.g., if the secondary evaporator 1140 comprises two or more circuits for flow of refrigerant).

[0110] As the cooled refrigerant passes through secondary evaporator 1140, heat is exchanged between the refrigerant and airflow passing through secondary evaporator 1140, cooling the inlet air. As an example, if inlet air is 80° F./60% humidity, secondary evaporator 1140 may output airflow at 70° F./84% humidity. This may cause flow of refrigerant to partially vaporize within secondary evaporator 1140. For example, if flow of refrigerant entering secondary evaporator 1140 is 196 psig/68° F./5% vapor, flow of refrigerant may be 196 psig/68° F./38% vapor as it leaves secondary evaporator 1140.

[0111] Secondary condenser 1120 receives warmed refrigerant from secondary evaporator 1140 via tube 1106. Secondary condenser 1120 facilitates heat transfer from the hot flow of refrigerant passing through the secondary condenser 1120 to the airflow. This reheats the airflow, thereby decreasing its relative humidity. As an example, if the airflow is 54° F./98% humidity, secondary condenser 1120 may output an airflow at 65° F./68% humidity. This may cause flow of refrigerant to partially or completely condense within secondary condenser 1120. For example, if flow of refrigerant entering secondary condenser 1120 is 196 psig/68° F./38% vapor, flow of refrigerant may be 196 psig/68° F./4% vapor as it leaves secondary condenser 1120.

[0112] The cooled refrigerant exits the secondary condenser at 1108 and is received by metering devices 1190a-c, which distributes the flow of refrigerant into the three circuits of primary evaporator 1110. FIG. 14 shows a view which includes the circuiting of primary evaporator 1110. Airflow passing through primary evaporator 1110 may be cooled to or below its dew point temperature, causing moisture in the airflow to condense (thereby reducing the absolute humidity of the air). As an example, if the airflow is 70° F./84% humidity, primary evaporator 1110 may output airflow at 54° F./98% humidity. This may cause flow of refrigerant to partially or completely vaporize within primary evaporator 1110.

[0113] Each of secondary metering devices 1190a, 1190b, and 1190c is configured to provide flow of refrigerant to each circuit of primary evaporator 1110 at a desired flow rate. For example, the flow rate provided to each circuit may be optimized to improve performance of the primary evaporator 1110. For example, under certain operating conditions, it may be beneficial to prevent the entire flow of refrigerant from passing through the entire evaporator, as occurs in a traditional evaporator coil. Refrigerant flowing through such an evaporator might undergo a change from liquid to gas phase before exiting the coil, resulting in poor performance in the portion of the evaporator that only contacts gaseous refrigerant. To significantly reduce or eliminate this problem, the present disclosure provides for refrigerant flow at a desired flow rate through each circuit. The desired flow rate may be predetermined (e.g., based on known design criteria

and/or operating conditions) and/or variable (e.g., manually and/or automatically adjustable in real time) during operation. The flow rate may be configured such that the flow of refrigerant exits its respective circuit just after transitioning to a gas. For example, the rate of airflow near the edges of an evaporator may be less than near the center of the evaporator. Therefore, a lower rate of refrigerant flow may be supplied by secondary metering devices 1190a-c to the circuits corresponding to the edge of primary evaporator 1110.

[0114] While the example of FIGS. 11, 12, 13, and 14 include a primary evaporator that is subdivided into two or more circuits. In other embodiments, secondary evaporator 1110 may also, or alternatively, be subdivided into two or more circuits. It should also be appreciated that the circuiting exemplified by FIGS. 11, 12, 13, and 14 can also be achieved in single coil packs such as those shown in FIGS. 9 and 10.

[0115] Although a particular implementation of portion 1100 of a dehumidification system is illustrated and primarily described, the present disclosure contemplates any suitable implementation of portion 1100 of a dehumidification system, according to particular needs. Moreover, although various components of portion 1100 of a dehumidification system have been depicted as being located at particular positions, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0116] FIGS. 15A-15B illustrate an example dehumidification system 1500 that may be used in accordance with dehumidification system 300 of FIG. 3 to reduce the humidity of air within a structure. Dehumidification system 1500 includes a dehumidification unit 1502, which is generally indoors, and a heat exchanger 1504 or an external source 1506 configured to contain a volume of a fluid operable to be used by the dehumidification system 1500 to cool a separate fluid flow within the dehumidification unit 1502. FIG. 15A illustrates the dehumidification system 1500 comprising the heat exchanger 1504, and FIG. 15B illustrates the dehumidification system comprising the external source 1506. With reference to both FIGS. 15A-15B, dehumidification unit 1502 includes a primary evaporator 1508, a primary condenser 1510, a secondary evaporator 1512, a secondary condenser 1514, a compressor 1516, a primary metering device 1518, a secondary metering device 1520, and a fan 1522.

[0117] With continued reference to both FIGS. 15A-15B, a flow of refrigerant 1524 is circulated through dehumidification unit 1502 as illustrated. In general, dehumidification unit 1502 receives an inlet airflow 1526, removes water from inlet airflow 1526, and discharges dehumidified air 1528. Water is removed from inlet air 1526 using a refrigeration cycle of flow of refrigerant 1524. By including secondary evaporator 1512 and secondary condenser 1514, however, dehumidification system 1500 causes at least part of the flow of refrigerant 1524 to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0118] In general, dehumidification system 1500 attempts to match the saturating temperature of secondary evaporator 1512 to the saturating temperature of secondary condenser 1514. The saturating temperature of secondary evaporator

1512 and secondary condenser 1514 generally is controlled according to the equation: $(\text{temperature of inlet air } 1526 + \text{temperature of a second airflow } 1530)/2$. As the saturating temperature of secondary evaporator 1512 is lower than inlet air 1526, evaporation happens in secondary evaporator 1512. As the saturating temperature of secondary condenser 1514 is higher than second airflow 1530, condensation happens in the secondary condenser 1514. The amount of refrigerant 1524 evaporating in secondary evaporator 1512 is substantially equal to that condensing in secondary condenser 1514.

[0119] Primary evaporator 1508 receives flow of refrigerant 1524 from secondary metering device 1520 and outputs flow of refrigerant 1524 to compressor 1516. Primary evaporator 1508 may be any suitable type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator 1508 receives a first airflow 1532 from secondary evaporator 1512 and outputs second airflow 1530 to secondary condenser 1514. Second airflow 1530, in general, is at a cooler temperature than first airflow 1532. To cool incoming first airflow 1532, primary evaporator 1508 transfers heat from first airflow 1532 to flow of refrigerant 1524, thereby causing flow of refrigerant 1524 to evaporate at least partially from liquid to gas. This transfer of heat from first airflow 1532 to flow of refrigerant 1524 also removes water from first airflow 1532.

[0120] Secondary condenser 1514 receives flow of refrigerant 1524 from secondary evaporator 1512 and outputs flow of refrigerant 1524 to secondary metering device 1520. Secondary condenser 1514 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser 1514 receives second airflow 1530 from primary evaporator 1508 and outputs dehumidified airflow 1528. Dehumidified airflow 1528 is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow 1530. Secondary condenser 1514 generates dehumidified airflow 1528 by transferring heat from flow of refrigerant 1524 to second airflow 1530, thereby causing flow of refrigerant 1524 to condense at least partially from gas to liquid.

[0121] Primary condenser 1510 receives flow of refrigerant 1524 from compressor 1516 and outputs flow of refrigerant 1524 to primary metering device 1518. Primary condenser 1510 may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant 1524 to the flow of a fluid 1534. In embodiments, the fluid 1534 may be any suitable fluid, such as water or a mixture of water and glycol. Primary condenser 1510 receives both the flow of fluid 1534 and the flow of refrigerant 1524 during operation of dehumidification system 1500, wherein the primary condenser 1510 is operable to transfer heat from the flow of refrigerant 1524, thereby causing flow of refrigerant 1524 to condense at least partially from gas to liquid. In some embodiments, primary condenser 1510 completely condenses flow of refrigerant 1524 to a liquid (i.e., 100% liquid). In other embodiments, primary condenser 1510 partially condenses flow of refrigerant 1524 to a liquid (i.e., less than 100% liquid).

[0122] As illustrated, the dehumidification system 1500 may further comprise a first water pump 1536. The first water pump 1536 may be disposed internal or external to the dehumidification unit 1502. The first water pump 1536 may be any suitable device operable to provide for the flow of fluid 1534. As depicted in FIG. 15A, the first water pump 1536 may be disposed at any suitable position in relation to

the primary condenser 1510 and the heat exchanger 1504 operable to cycle the flow of fluid 1534 between the heat exchanger 1504 and the primary condenser 1510. As depicted in FIG. 15B, the first water pump 1536 may be disposed at any suitable position in relation to the primary condenser 1510 and the external source 1506 operable to cycle the flow of fluid 1534 between the external source 1506 and the primary condenser 1510.

[0123] With reference to FIG. 15A, heat exchanger 1504 may receive the flow of fluid 1534 from primary condenser 1510 at a first temperature and output flow of fluid 1534 to primary condenser 1510 at a second temperature after transferring heat away from the flow of fluid 1534, wherein the second temperature is lower than the first temperature. Heat exchanger 1504 may be any suitable type of heat exchanger, such as, for example, a cooling tower or a dry cooler. Heat exchanger 1504 receives the flow of fluid 1534 and a first outdoor airflow 1540, wherein heat is transferred between the flow of fluid 1534 and the first outdoor airflow 1540. Heat exchanger 1504 may further output the flow of fluid 1534 and a second outdoor airflow 1542, wherein the flow of fluid 1534 leaving the heat exchanger 1504 is at a lower temperature than the flow of fluid 1534 received by the heat exchanger 1504, and the second outdoor airflow 1542 is at a greater temperature than the first outdoor airflow 1540.

[0124] In embodiments wherein the heat exchanger 1504 is a cooling tower, the heat exchanger 1504 may be operable to dispense the flow of fluid 1534 within its internal structure, wherein the fluid 1534 directly contacts the first outdoor airflow 1540 as the fluid 1534 flows through the heat exchanger 1504 and transfers heat to the first outdoor airflow 1540. At least a portion of the fluid 1534 may evaporate and exit to the atmosphere as the heat transfers from the fluid 1534 to the first outdoor airflow 1540, and the heat exchanger 1504 may collect a remaining portion of the fluid 1534 after transferring heat to the first outdoor airflow 1540, wherein the remaining portion of the fluid 1534 is at a lower temperature. In embodiments wherein the heat exchanger 1504 is a dry cooler, the heat exchanger 1504 may be operable to induce the first outdoor airflow 1540 to flow through the heat exchanger 1504 where heat transfers indirectly between the first outdoor airflow 1540 and the flow of fluid 1534. In these embodiments, heat transfer would not result in loss of a portion of the fluid 1534 through evaporation to the atmosphere.

[0125] With reference now to FIG. 15B, external source 1506 may receive the flow of fluid 1534 from the primary condenser 1510 and output flow of fluid 1534 to the primary condenser 1510 via first water pump 1536. External source 1506 may be configured to contain and/or store a volume of fluid 1534 to be used by primary condenser 1510 to lower the temperature of the flow of refrigerant 1524 in the dehumidification unit 1502. The external source 1506 may be configured to receive the flow of fluid 1534 from primary condenser 1510 at a first temperature and output flow of fluid 1534 to primary condenser 1510 at a second temperature after transferring heat away from the flow of fluid 1534, wherein the second temperature is lower than the first temperature. Without limitations, the external source 1506 may be any suitable number and combination of a ground reservoir, a natatorium, and an outdoor body of water, among others. In embodiments wherein the external source 1506 is a ground reservoir, the external source 1506 may implement an open or closed ground water system, wherein

the conduit providing for the flow of fluid 1534 within the ground reservoir may be disposed substantially parallel to a horizontal plane of the ground surface, substantially perpendicular to the horizontal plane of the ground surface, or combinations thereof.

[0126] With reference to both FIGS. 15A-15B, secondary evaporator 1512 receives flow of refrigerant 1524 from primary metering device 1518 and outputs flow of refrigerant 1524 to secondary condenser 1514. Secondary evaporator 1512 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator 1512 receives inlet air 1526 and outputs first airflow 1532 to primary evaporator 1508. First airflow 1532, in general, is at a cooler temperature than inlet air 1526. To cool incoming inlet air 1526, secondary evaporator 1512 transfers heat from inlet air 1526 to flow of refrigerant 1524, thereby causing flow of refrigerant 1524 to evaporate at least partially from liquid to gas.

[0127] Compressor 1516 pressurizes flow of refrigerant 1524, thereby increasing the temperature of refrigerant 1524. For example, if flow of refrigerant 1524 entering compressor 1516 is 128 psig/52° F./100% vapor, flow of refrigerant 1524 may be 340 psig/150° F./100% vapor as it leaves compressor 1516. Compressor 1516 receives flow of refrigerant 1524 from primary evaporator 1508 and supplies the pressurized flow of refrigerant 1524 to primary condenser 1510.

[0128] Fan 1522 may include any suitable components operable to draw inlet air 1526 into dehumidification unit 1502 and through secondary evaporator 1512, primary evaporator 1508, and secondary condenser 1514. Fan 1522 may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan 1522 may be a backward inclined impeller positioned adjacent to secondary condenser 1514. While fan 1522 is depicted as being located adjacent to secondary condenser 1514, it should be understood that fan 1522 may be located anywhere along the airflow path of dehumidification unit 1502. For example, fan 1522 may be positioned in the airflow path of any one of airflows 1526, 1532, 1530, or 1528. Moreover, dehumidification unit 1502 may include one or more additional fans positioned within any one or more of these airflow paths.

[0129] Primary metering device 1518 and secondary metering device 1520 are any appropriate type of metering/expansion device. In some embodiments, primary metering device 1518 is a thermostatic expansion valve (TXV) and secondary metering device 1520 is a fixed orifice device (or vice versa). In certain embodiments, metering devices 1518 and 1520 remove pressure from flow of refrigerant 1524 to allow expansion or change of state from a liquid to a vapor in evaporators 1512 and 1508. The high-pressure liquid (or mostly liquid) refrigerant 1524 entering metering devices 1518 and 1520 is at a higher temperature than the liquid refrigerant 1524 leaving metering devices 1518 and 1520. For example, if flow of refrigerant 1524 entering primary metering device 1518 is 340 psig/80° F./0% vapor, flow of refrigerant 1524 may be 196 psig/68° F./5% vapor as it leaves primary metering device 1518. As another example, if flow of refrigerant 1524 entering secondary metering device 1520 is 196 psig/68° F./4% vapor, flow of refrigerant 1524 may be 128 psig/44° F./14% vapor as it leaves secondary metering device 1520.

[0130] Refrigerant 1524 may be any suitable refrigerant such as R410a. In general, dehumidification system 1500

utilizes a closed refrigeration loop of refrigerant **1524** that passes from compressor **1516** through primary condenser **1510**, primary metering device **1518**, secondary evaporator **1512**, secondary condenser **1514**, secondary metering device **1520**, and primary evaporator **1508**. Compressor **1516** pressurizes flow of refrigerant **1524**, thereby increasing the temperature of refrigerant **1524**. Primary condenser **1510**, which may include any suitable water-cooled heat exchanger, cools the pressurized flow of refrigerant **1524** by facilitating heat transfer from the flow of refrigerant **1524** to the flow of fluid provided by the external source **1506** passing through it (i.e., flow of fluid **1534**). Secondary condenser, which may include any suitable air-cooled heat exchanger, cools the pressurized flow of refrigerant **1524** by facilitating heat transfer from the flow of refrigerant **1524** to the respective airflow passing through it (i.e., second airflow **1530**).

[0131] The cooled flow of refrigerant **1524** leaving primary and secondary condensers **1510** and **1514** may enter a respective expansion device (i.e., primary metering device **1518** and secondary metering device **1520**) that is operable to reduce the pressure of flow of refrigerant **1524**, thereby reducing the temperature of flow of refrigerant **1524**. Primary and secondary evaporators **1508** and **1512**, which may include any suitable heat exchanger, receive flow of refrigerant **1524** from secondary metering device **1520** and primary metering device **1518**, respectively. Primary and secondary evaporators **1508** and **1512** facilitate the transfer of heat from the respective airflows passing through them (i.e., inlet air **1526** and first airflow **1532**) to flow of refrigerant **1524**. Flow of refrigerant **1524**, after leaving primary evaporator **1508**, passes back to compressor **1516**, and the cycle is repeated.

[0132] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **1508** and **1512** operate in a flooded state. In other words, flow of refrigerant **1524** may enter evaporators **1508** and **1512** in a liquid state, and a portion of flow of refrigerant **1524** may still be in a liquid state as it exits evaporators **1508** and **1512**. Accordingly, the phase change of flow of refrigerant **1524** (liquid to vapor as heat is transferred to flow of refrigerant **1524**) occurs across evaporators **1508** and **1512**, resulting in nearly constant pressure and temperature across the entire evaporators **1508** and **1512** (and, as a result, increased cooling capacity).

[0133] In operation of example embodiments of dehumidification system **1500**, inlet air **1526** may be drawn into dehumidification unit **1502** by fan **1522**. Inlet air **1526** passes through secondary evaporator **1512** in which heat is transferred from inlet air **1526** the cool flow of refrigerant **1524** passing through secondary evaporator **1512**. As a result, inlet air **1526** may be cooled. As an example, if inlet air **1526** is 80° F./60% humidity, secondary evaporator **1512** may output first airflow **1532** at 70° F./84% humidity. This may cause flow of refrigerant **1524** to partially vaporize within secondary evaporator **1512**. For example, if flow of refrigerant **1524** entering secondary evaporator **1512** is 196 psig/68° F./5% vapor, flow of refrigerant **1524** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **1512**.

[0134] The cooled inlet air **1526** leaves secondary evaporator **1512** as first airflow **1532** and enters primary evaporator **1508**. Like secondary evaporator **1512**, primary evaporator **1508** transfers heat from first airflow **1532** to the cool

flow of refrigerant **1524** passing through primary evaporator **1508**. As a result, first airflow **1532** may be cooled to or below its dew point temperature, causing moisture in first airflow **1532** to condense (thereby reducing the absolute humidity of first airflow **1532**). As an example, if first airflow **1532** is 70° F./84% humidity, primary evaporator **1508** may output second airflow **1530** at 54° F./98% humidity. This may cause flow of refrigerant **1524** to partially or completely vaporize within primary evaporator **1508**. For example, if flow of refrigerant **1524** entering primary evaporator **1508** is 128 psig/44° F./14% vapor, flow of refrigerant **1524** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1508**.

[0135] The cooled first airflow **1532** leaves primary evaporator **1508** as second airflow **1530** and enters secondary condenser **1514**. Secondary condenser **1514** facilitates heat transfer from the hot flow of refrigerant **1524** passing through the secondary condenser **1514** to second airflow **1530**. This reheats second airflow **1530**, thereby decreasing the relative humidity of second airflow **1530**. As an example, if second airflow **1530** is 54° F./98% humidity, secondary condenser **1514** may output dehumidified airflow **1528** at 65° F./68% humidity. This may cause flow of refrigerant **1524** to partially or completely condense within secondary condenser **1514**. For example, if flow of refrigerant **1524** entering secondary condenser **1514** is 196 psig/68° F./38% vapor, flow of refrigerant **1524** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1514**.

[0136] Some embodiments of dehumidification system **1500** may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0137] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system **1500**, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0138] Although particular implementations of dehumidification system **1500** are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system **1500**, according to par-

ticular needs. Moreover, although various components of dehumidification system 1500 have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0139] FIGS. 16A, 16B, 16C, and 16D illustrate an example dehumidification system 1600 with a modulating valve 1602 that may be used in accordance with split dehumidification system 600 of FIGS. 6A-6B to reduce humidity of an airflow. Dehumidification system 1600 includes the modulating valve 1602, a primary evaporator 1604, a primary condenser 1606, a secondary evaporator 1608, a secondary condenser 1610, a compressor 1612, a primary metering device 1614, a secondary metering device 1616, a fan 1618, and an alternate condenser 1620. In some embodiments, dehumidification system 1600 may additionally include an optional sub-cooling coil 1622. As illustrated in FIGS. 16A-16B, the alternate condenser 1620 may be disposed in an external condenser unit 1624. With reference to FIG. 16A, the optional sub-cooling coil 1622 may be disposed in the external condenser unit 1624 with the alternate condenser 1620, wherein the sub-cooling coil 1622 and the alternate condenser 1620 may be combined into a single coil. With reference to FIG. 16B, the optional sub-cooling coil 1622 may be disposed adjacent to the primary condenser 1606, wherein sub-cooling coil 1620 and primary condenser 1606 may be combined into a single coil. FIGS. 16C-16D illustrate an embodiment of dehumidification system 1600 wherein both optional sub-cooling coil 1622 and alternate condenser 1620 are not in the external condenser unit 1624 and where alternate condenser 1620 is liquid-cooled.

[0140] With reference to each of FIGS. 16A-16D, a flow of refrigerant 1626 is circulated through dehumidification system 1600 as illustrated. In general, dehumidification system 1600 receives inlet airflow 1628, removes water from inlet airflow 1628, and discharges dehumidified air 1630. Water is removed from inlet air 1628 using a refrigeration cycle of flow of refrigerant 1626. By including secondary evaporator 1608 and secondary condenser 1610, however, dehumidification system 1600 causes at least part of the flow of refrigerant 1626 to evaporate and condense twice in a single refrigeration cycle. This increases the refrigeration capacity over typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0141] In general, dehumidification system 1600 attempts to match the saturating temperature of secondary evaporator 1608 to the saturating temperature of secondary condenser 1610. The saturating temperature of secondary evaporator 1608 and secondary condenser 1610 generally is controlled according to the equation: $(\text{temperature of inlet air } 1628 + \text{temperature of a second airflow } 1632)/2$. As the saturating temperature of secondary evaporator 1608 is lower than inlet air 1628, evaporation happens in secondary evaporator 1608. As the saturating temperature of secondary condenser 1610 is higher than second airflow 1632, condensation happens in the secondary condenser 1610. The amount of refrigerant 1626 evaporating in secondary evaporator 1608 is substantially equal to that condensing in secondary condenser 1610.

[0142] Primary evaporator 1604 receives flow of refrigerant 1626 from secondary metering device 1616 and outputs

flow of refrigerant 1626 to compressor 1612. Primary evaporator 1604 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator 1604 receives a first airflow 1634 from secondary evaporator 1608 and outputs second airflow 1632 to secondary condenser 1610. Second airflow 1632, in general, is at a cooler temperature than first airflow 1634. To cool incoming first airflow 1634, primary evaporator 1604 transfers heat from first airflow 1634 to flow of refrigerant 1626, thereby causing flow of refrigerant 1626 to evaporate at least partially from liquid to gas. This transfer of heat from first airflow 1634 to flow of refrigerant 1626 also removes water from first airflow 1634.

[0143] Secondary condenser 1610 receives flow of refrigerant 1626 from secondary evaporator 1608 and outputs flow of refrigerant 1626 to secondary metering device 1616. Secondary condenser 1610 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser 1610 receives second airflow 1632 from primary evaporator 1604 and outputs a third airflow 1636. Third airflow 1636 is, in general, warmer and drier (i.e., the dew point will be the same but relative humidity will be lower) than second airflow 1632. Secondary condenser 1610 generates third airflow 1632 by transferring heat from flow of refrigerant 1626 to second airflow 1632, thereby causing flow of refrigerant 1626 to condense at least partially from gas to liquid.

[0144] Primary condenser 1606 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser 1606 is operable to receive flow of refrigerant 1626 from modulating valve 1602 and outputs flow of refrigerant 1626 to either primary metering device 1614 or sub-cooling coil 1622. As shown in FIG. 16A, primary condenser 1606 outputs flow of refrigerant 1626 to primary metering device 1614. In these embodiments, primary condenser 1606 receives third airflow 1636 and outputs dehumidified air 1630. But with reference to FIGS. 16B-16D, primary condenser 1606 outputs flow of refrigerant 1626 to the optional sub-cooling coil 1622 before the flow of refrigerant 1626 flows to primary metering device 1614. In these embodiments, primary condenser 1606 receives a fourth airflow 1638 generated by the sub-cooling coil 1622 and outputs dehumidified air 1630. With reference to each of FIGS. 16A-16D, dehumidified air 1630 is, in general, warmer and drier (i.e., have a lower relative humidity) than either third airflow 1636 or fourth airflow 1638. Primary condenser 1606 generates dehumidified air 1630 by transferring heat away from flow of refrigerant 1626, thereby causing flow of refrigerant 1626 to condense at least partially from gas to liquid. In some embodiments, primary condenser 1606 completely condenses flow of refrigerant 1626 to a liquid (i.e., 100% liquid). In other embodiments, primary condenser 1606 partially condenses flow of refrigerant 1626 to a liquid (i.e., less than 100% liquid).

[0145] Secondary evaporator 1608 receives flow of refrigerant 1626 from primary metering device 1614 and outputs flow of refrigerant 1626 to secondary condenser 1610. Secondary evaporator 1608 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator 1608 receives inlet air 1628 and outputs first airflow 1634 to primary evaporator 1604. First airflow 1634, in general, is at a cooler temperature than inlet air 1628. To cool incoming inlet air 1628, secondary evaporator 1608 transfers heat

from inlet air 1608 to flow of refrigerant 1626, thereby causing flow of refrigerant 1626 to evaporate at least partially from liquid to gas.

[0146] Sub-cooling coil 1622, which is an optional component of dehumidification system 1600, sub-cools the liquid refrigerant 1626 as it leaves the primary condenser 1606, the alternate condenser 1620, or combinations thereof. In embodiments wherein the sub-cooling coil 1622 is disposed within the external condenser unit 1624, the sub-cooling coil 1622 may receive refrigerant 1626 as it leaves the alternate condenser 1620, as seen in FIG. 16A. In embodiments wherein the sub-cooling coil 1622 is disposed adjacent to the primary condenser 1606, the sub-cooling coil 1622 may receive refrigerant 1626 as it leaves the primary condenser 1606 and/or the alternate condenser 1620, as seen in FIGS. 16B-16D. With reference to each of FIGS. 16A-16D, this, in turn, supplies primary metering device 1614 with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil 1622. For example, if flow of refrigerant 1626 entering sub-cooling coil 1622 is 340 psig/105° F./60% vapor, flow of refrigerant 1626 may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil 1622. The sub-cooled refrigerant 1626 has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant 1626. This results in greater efficiency and less energy use of dehumidification system 1600.

[0147] Compressor 1612 pressurizes flow of refrigerant 1626, thereby increasing the temperature of refrigerant 1626. For example, if flow of refrigerant 1626 entering compressor 1612 is 128 psig/52° F./100% vapor, flow of refrigerant 1626 may be 340 psig/150° F./100% vapor as it leaves compressor 1612. Compressor 1612 receives flow of refrigerant 1626 from primary evaporator 1604 and supplies the pressurized flow of refrigerant 1626 to modulating valve 1602.

[0148] Modulating valve 1602 is operable to receive the pressurized flow of refrigerant 1626 from compressor 1612 and to direct the flow of refrigerant to primary condenser 1606, to alternate condenser 1620, or to both. In embodiments, the modulating valve 1602 may operate based, at least in part, on a pre-determined temperature set point for the dehumidified airflow 1630 and on an actual temperature of the dehumidified airflow 1630 output by dehumidification system 1600. Dehumidification system 1600 may utilize modulating valve 1602 to direct heat to be rejected from the flow of refrigerant 1626 away from the primary condenser 1606 and towards the alternate condenser 1620. Depending on a feedback loop comprising of the pre-determined temperature set point and the actual temperature of the dehumidified airflow 1630, modulating valve 1602 may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant 1626 to the alternate condenser 1620 and direct a remaining portion of the flow of refrigerant 1626 to the primary condenser 1606.

[0149] During operation of dehumidification system 1600, the modulating valve 1602 may direct the flow of refrigerant 1626 to primary condenser 1606 if the temperature of the dehumidified airflow 1630 output by the primary condenser 1606 does not exceed the pre-determined temperature set point monitored by the dehumidification system 1600. If the temperature of the dehumidified airflow 1630 is greater than the pre-determined temperature set point, the modulating

valve 1602 may be actuated to direct at least a portion of the flow of refrigerant 1626 to the alternate condenser 1620 and direct a remaining portion of the flow of refrigerant to the primary condenser 1606. As the dehumidification system 1600 operates, reduction in the volume of flow of refrigerant 1626 to primary condenser 1606 may reduce the available heat to be rejected into the dehumidified airflow 1630. With the reduced flow of refrigerant 1626 passing through primary condenser 1606 (for example, the remaining portion of the flow of refrigerant), the rate of heat transfer to the dehumidified airflow 1630 may subsequently be reduced, thereby producing a reduction in the temperature change of an incoming airflow and the output dehumidified airflow 1630. Once the temperature of the dehumidified airflow 1630 is lower than the pre-determined temperature set point, the modulating valve 1602 may be actuated to direct the at least a portion of the flow of refrigerant 1626 back to the primary condenser 1606. Any remaining refrigerant 1626 that had been directed to alternate condenser 1620 may combine with the flow of refrigerant 1626 further downstream.

[0150] With reference to FIGS. 16A and 16B, alternate condenser 1620 may be disposed in the external condenser unit 1624 and may be any type of coil (e.g., fin tube, micro channel, etc.) operable to receive flow of refrigerant 1626 from modulating valve 1602 and output flow of refrigerant 1626 at a lower temperature. Alternate condenser 1620 transfers heat from flow of refrigerant 1626, thereby causing flow of refrigerant 1626 to condense at least partially from gas to liquid. In some embodiments, alternate condenser 1620 completely condenses flow of refrigerant 1626 to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser 1620 partially condenses flow of refrigerant 1626 to a liquid (i.e., less than 100% liquid). As seen in FIG. 16A, the flow of refrigerant 1626 may be output to sub-cooling coil 1622 disposed adjacent to alternate condenser 1620 within the external condenser unit 1624. Alternate condenser 1620 and sub-cooling coil 1622 may receive a first outdoor airflow 1640 and output a second outdoor airflow 1642. Second outdoor airflow 1642 is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow 1640. In other embodiments, as shown in FIG. 16B, the first outdoor airflow 1640 may be received by the alternate condenser 1620 without previously flowing through sub-cooling coil 1622. In FIG. 16B, the external condenser unit 1624 may include the alternate condenser 1620 and a fan 1644 and may not include the sub-cooling coil 1622, wherein fan 1644 may be configured to facilitate flow of first outdoor airflow 1640 towards alternate condenser 1620.

[0151] With reference now to FIGS. 16C-16D, alternate condenser 1620 may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant 1626 to the flow of a fluid 1646, wherein the alternate condenser 1620 receives flow of refrigerant 1626 from modulating valve 1602 and outputs flow of refrigerant 1626 to sub-cooling coil 1622. In embodiments, the fluid 1646 may be any suitable fluid, such as water or a mixture of water and glycol. Alternate condenser 1620 receives both the flow of fluid 1646 and the flow of refrigerant 1626 during operation of dehumidification system 1600, wherein the alternate condenser 1620 is operable to transfer heat from the flow of refrigerant 1626, thereby causing flow of refrigerant 1626 to condense at least partially from gas to liquid. In some embodiments, alternate condenser 1620 completely

condenses flow of refrigerant **1626** to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser **1620** partially condenses flow of refrigerant **1626** to a liquid (i.e., less than 100% liquid).

[0152] As illustrated in FIGS. **16C-16D**, the dehumidification system **1600** may further comprise a first water pump **1648**. The first water pump **1648** may be disposed external to the alternate condenser **1620**. The first water pump may be any suitable device operable to provide for the flow of fluid **1646**. As depicted in FIG. **16C**, the first water pump **1648** may be disposed at any suitable location between the alternate condenser **1620** and a heat exchanger **1654** operable to cycle the flow of fluid **1646** between the heat exchanger **1654** and the alternate condenser **1620**. As depicted in FIG. **16D**, the first water pump **1648** may be disposed at any suitable location between the alternate condenser **1620** and an external source **1652** operable to cycle the flow of fluid **1646** between the external source **1652** and the alternate condenser **1620**.

[0153] With reference to FIG. **16C**, heat exchanger **1654** may receive the flow of fluid **1646** from alternate condenser **1620** and output flow of fluid **1646** after transferring heat away from the flow of fluid **1646**. Heat exchanger **1654** may be any suitable type of heat exchanger, such as a cooling tower or a dry cooler. Heat exchanger **1654** receives the flow of fluid **1646** and a first outdoor airflow **1656**, wherein heat is transferred between the flow of fluid **1646** and the first outdoor airflow **1656**. Heat exchanger **1654** may further output the flow of fluid **1646** and a second outdoor airflow **1658**, wherein the flow of fluid **1646** leaving the heat exchanger **1654** is at a lower temperature than the flow of fluid **1646** received by the heat exchanger **1654**, and the second outdoor airflow **1658** is at a greater temperature than the first outdoor airflow **1656**.

[0154] In embodiments wherein the heat exchanger **1654** is a cooling tower, the heat exchanger **1654** may be operable to dispense the flow of fluid **1646** within its internal structure, wherein the fluid **1646** directly contacts the first outdoor airflow **1656** as the fluid **1646** flows through the heat exchanger **1654** and transfers heat to the first outdoor airflow **1656**. At least a portion of the fluid **1646** may evaporate and exit to the atmosphere as the heat transfers from the fluid **1646** to the first outdoor airflow **1656**, and the heat exchanger **1654** may collect a remaining portion of the fluid **1646** after transferring heat to the first outdoor airflow **1656**, wherein the remaining portion of the fluid **1646** is at a lower temperature. In embodiments wherein the heat exchanger **1654** is a dry cooler, the heat exchanger **1654** may be operable to induce the first outdoor airflow **1656** to flow through the heat exchanger **1654** where heat transfers indirectly between the first outdoor airflow **1656** and the flow of fluid **1646**. In these embodiments, heat transfer would not result in loss of a portion of the fluid **1646** through evaporation to the atmosphere.

[0155] With reference to FIG. **16D**, external source **1652** may receive the flow of fluid **1646** and output flow of fluid **1646** to the alternate condenser **1620** via first water pump **1648**. External source **1652** may be configured to contain and/or store a volume of fluid **1646** to be used by alternate condenser **1620** to lower the temperature of the flow of refrigerant **1626** in the dehumidification system **1600**. Without limitations, the external source **1652** may be selected from a group consisting of a ground reservoir, a natatorium, an outdoor body of water, and any combinations thereof. In

embodiments wherein the external source **1652** is a ground reservoir, the external source **1652** may implement an open or closed ground water system, wherein the conduit providing for the flow of fluid **1646** within the ground reservoir may be disposed substantially parallel to a horizontal plane of the ground surface, substantially perpendicular to the horizontal plane of the ground surface, or combinations thereof.

[0156] In embodiments wherein the external source **1652** is a natatorium, the external source **1652** may be within a multi-loop system operable to contain and cool the flow of fluid **1646** before the alternate condenser **1620** uses the flow of fluid **1646** to lower the temperature of the flow of refrigerant **1626**. The external source **1652** may be configured to receive the flow of fluid **1646** from alternate condenser **1620** at a first temperature and output flow of fluid **1646** to alternate condenser **1620** at a second temperature after transferring heat away from the flow of fluid **1646**, wherein the second temperature is lower than the first temperature. External source **1652** receives the flow of fluid **1646** and may receive a flow of a secondary fluid (not shown), wherein heat is transferred between the flow of fluid **1646** and the flow of secondary fluid. External source **1652** may then output the flow of fluid **1646** and the flow of secondary fluid, wherein the flow of fluid **1646** leaving the external source **1652** is at a lower temperature than the flow of fluid **1646** received by the external source **1652**, and wherein the flow of secondary fluid leaving the external source **1652** is at a greater temperature than the flow of secondary fluid received by the external source **1652**.

[0157] The flow of secondary fluid may then be directed to a tertiary condenser (not shown). The tertiary condenser receives the flow of secondary fluid from external source **1652** and outputs flow of secondary fluid back to the external source **1652** at a lower temperature. The tertiary condenser may be any type of air-cooled or liquid-cooled heat exchanger operable to transfer heat away from the flow of secondary fluid. In embodiments, a second pump (not shown) may be at any suitable position in relation to the external source **1652** and the tertiary condenser operable to cycle the flow of secondary fluid between the external source **1652** and the tertiary condenser, wherein the second pump may be any suitable device operable to provide for the flow of secondary fluid.

[0158] Referring back to each of FIGS. **16A-16D**, fan **1618** may include any suitable components operable to draw inlet air **1628** into dehumidification system **1600** and through secondary evaporator **1608**, primary evaporator **1604**, secondary condenser **1610**, sub-cooling coil **1622**, and primary condenser **1606**. Fan **1618** may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan **1618** may be a backward inclined impeller positioned adjacent to primary condenser **1606** as illustrated in FIGS. **16A-16D**. While fan **1618** is depicted in FIGS. **16A-16D** as being located adjacent to primary condenser **1606**, it should be understood that fan **1618** may be located anywhere along the airflow path of dehumidification system **1600**. For example, fan **1618** may be positioned in the airflow path of any one of airflows **1628**, **1634**, **1632**, **1636**, **1638**, or **1630**. Moreover, dehumidification system **1600** may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, with reference to FIGS. **16A-16B**, while a fan **1644** of external condenser unit **1624** is depicted as being

located above alternate condenser **1620**, it should be understood that fan **1644** may be located anywhere (e.g., above, below, beside) with respect to alternate condenser **1620** and optional sub-cooling coil **1622**, so long as fan **1644** is appropriately positioned and configured to facilitate flow of first outdoor airflow **1640** towards alternate condenser **1620**.

[0159] Primary metering device **1614** and secondary metering device **1616** are any appropriate type of metering/expansion device. In some embodiments, primary metering device **1614** is a thermostatic expansion valve (TXV) and secondary metering device **1616** is a fixed orifice device (or vice versa). In certain embodiments, metering devices **1614** and **1616** remove pressure from flow of refrigerant **1626** to allow expansion or change of state from a liquid to a vapor in evaporators **1604** and **1608**. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices **1614** and **1616** is at a higher temperature than the liquid refrigerant **1626** leaving metering devices **1614** and **1616**. For example, if flow of refrigerant **1626** entering primary metering device **1614** is 340 psig/80° F./0% vapor, flow of refrigerant **1626** may be 196 psig/68° F./5% vapor as it leaves primary metering device **1614**. As another example, if flow of refrigerant **1626** entering secondary metering device **1616** is 196 psig/68° F./4% vapor, flow of refrigerant **1626** may be 128 psig/44° F./14% vapor as it leaves secondary metering device **1616**.

[0160] Refrigerant **1626** may be any suitable refrigerant such as R410a. In general, dehumidification system **1600** utilizes a closed refrigeration loop of refrigerant **1626** that passes from compressor **1612** through modulating valve **1602**, primary condenser **1612** and/or alternate condenser **1620**, (optionally) sub-cooling coil **1622**, primary metering device **1614**, secondary evaporator **1608**, secondary condenser **1610**, secondary metering device **1616**, and primary evaporator **1604**. Compressor **1612** pressurizes flow of refrigerant **1626**, thereby increasing the temperature of refrigerant **1626**. Primary and secondary condensers **1606** and **1610**, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant **1626** by facilitating heat transfer from the flow of refrigerant **1626** to the respective airflows passing through them (i.e., third or fourth airflow **1636**, **1638** and second airflow **1632**). Further, alternate condenser **1620**, which may include any suitable heat exchanger, cools the pressurized flow of refrigerant **1626** by facilitating heat transfer from the flow of refrigerant **1626** to either the airflow passing through it (i.e., first outdoor airflow **1640** as illustrated in FIGS. 16A-16B) or to the flow of fluid provided by the external source **1652** passing through it (i.e., flow of fluid **1646** as illustrated in FIGS. 16C-16D). The cooled flow of refrigerant **1626** leaving primary and/or alternate condensers **1606** and **1620** may enter primary metering device **1614**, which is operable to reduce the pressure of flow of refrigerant **1626**, thereby reducing the temperature of flow of refrigerant **1626**. The cooled flow of refrigerant **1626** leaving secondary condenser **1610** may enter secondary metering device **1616**, which is operable to reduce the pressure of flow of refrigerant **1626**, thereby reducing the temperature of flow of refrigerant **1626**. Primary and secondary evaporators **1604** and **1608**, which may include any suitable heat exchanger, receive flow of refrigerant **1626** from secondary metering device **1616** and primary metering device **1614**, respectively. Primary and secondary evaporators **1604** and **1608** facilitate the transfer of heat from the respective airflows passing through them

(i.e., inlet air **1628** and first airflow **1634**) to flow of refrigerant **1626**. Flow of refrigerant **1626**, after leaving primary evaporator **1604**, passes back to compressor **1612**, and the cycle is repeated.

[0161] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators **1604** and **1608** operate in a flooded state. In other words, flow of refrigerant **1626** may enter evaporators **1604** and **1608** in a liquid state, and a portion of flow of refrigerant **1626** may still be in a liquid state as it exits evaporators **1604** and **1608**. Accordingly, the phase change of flow of refrigerant **1626** (liquid to vapor as heat is transferred to flow of refrigerant **1626**) occurs across evaporators **1604** and **1608**, resulting in nearly constant pressure and temperature across the entire evaporators **1604** and **1608** (and, as a result, increased cooling capacity).

[0162] In operation of example embodiments of dehumidification system **1600**, inlet air **1628** may be drawn into dehumidification system **1600** by fan **1618**. Inlet air **1628** passes through secondary evaporator **1608** in which heat is transferred from inlet air **1628** to the cool flow of refrigerant **1626** passing through secondary evaporator **1608**. As a result, inlet air **1628** may be cooled. As an example, if inlet air **1628** is 80° F./60% humidity, secondary evaporator **1608** may output first airflow **1634** at 70° F./84% humidity. This may cause flow of refrigerant **1626** to partially vaporize within secondary evaporator **1608**. For example, if flow of refrigerant **1626** entering secondary evaporator **1608** is 196 psig/68° F./5% vapor, flow of refrigerant **1626** may be 196 psig/68° F./38% vapor as it leaves secondary evaporator **1608**.

[0163] The cooled inlet air **1628** leaves secondary evaporator **1608** as first airflow **1634** and enters primary evaporator **1604**. Like secondary evaporator **1608**, primary evaporator **1604** transfers heat from first airflow **1634** to the cool flow of refrigerant **1626** passing through primary evaporator **1604**. As a result, first airflow **1634** may be cooled to or below its dew point temperature, causing moisture in first airflow **1634** to condense (thereby reducing the absolute humidity of first airflow **1634**). As an example, if first airflow **1634** is 70° F./84% humidity, primary evaporator **1604** may output second airflow **1632** at 54° F./98% humidity. This may cause flow of refrigerant **1626** to partially or completely vaporize within primary evaporator **1604**. For example, if flow of refrigerant **1626** entering primary evaporator **1604** is 128 psig/44° F./14% vapor, flow of refrigerant **1626** may be 128 psig/52° F./100% vapor as it leaves primary evaporator **1604**.

[0164] The cooled first airflow **1634** leaves primary evaporator **1604** as second airflow **1632** and enters secondary condenser **1610**. Secondary condenser **1610** facilitates heat transfer from the hot flow of refrigerant **1626** passing through the secondary condenser **1610** to second airflow **1632**. This reheats second airflow **1632**, thereby decreasing the relative humidity of second airflow **1632**. As an example, if second airflow **1632** is 54° F./98% humidity, secondary condenser **1610** may output third airflow **1636** at 65° F./68% humidity. This may cause flow of refrigerant **1626** to partially or completely condense within secondary condenser **1610**. For example, if flow of refrigerant **1626** entering secondary condenser **1610** is 196 psig/68° F./38% vapor, flow of refrigerant **1626** may be 196 psig/68° F./4% vapor as it leaves secondary condenser **1610**.

[0165] In some embodiments, the dehumidified second airflow 1632 leaves secondary condenser 1610 as third airflow 1636 and enters primary condenser 1606, as illustrated in FIG. 16A. Primary condenser 1606 facilitates heat transfer from the hot flow of refrigerant 1626 passing through the primary condenser 1606 to third airflow 1636. This further heats third airflow 1636, thereby further decreasing the relative humidity of third airflow 1636. As an example, if third airflow 1636 is 65° F./68% humidity, primary condenser 1606 may output dehumidified air 1630 at 102° F./19% humidity. This may cause flow of refrigerant 1626 to partially or completely condense within primary condenser 1606. For example, if flow of refrigerant 1626 entering primary condenser 1606 is 340 psig/150° F./100% vapor, flow of refrigerant 1626 may be 340 psig/105° F./60% vapor as it leaves primary condenser 1606.

[0166] As described above, some embodiments of dehumidification system 1600 may include a sub-cooling coil 1622 in the airflow between secondary condenser 1610 and primary condenser 1606, as best seen in FIGS. 16B-16D. Sub-cooling coil 1622 facilitates heat transfer from the hot flow of refrigerant 1626 passing through sub-cooling coil 1622 to third airflow 1636. This further heats third airflow 1636, thereby further decreasing the relative humidity of third airflow 1636. As an example, if third airflow 1636 is 65° F./68% humidity, sub-cooling coil 1622 may output fourth airflow 1638 at 81° F./37% humidity. This may cause flow of refrigerant 1626 to partially or completely condense within sub-cooling coil 1622. For example, if flow of refrigerant 1626 entering sub-cooling coil 1622 is 340 psig/150° F./60% vapor, flow of refrigerant 1626 may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil 1622. In these embodiments, the fourth airflow 1638 may then undergo heat transfer in primary condenser 1606 to produce dehumidified airflow 1630.

[0167] Some embodiments of dehumidification system 1600 may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0168] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system 1600, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory, including, without limitation, magnetic media, optical

media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0169] Although particular implementations of dehumidification system 1600 are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system 1600, according to particular needs. Moreover, although various components of dehumidification system 1600 have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0170] FIGS. 17A, 17B, and 17C illustrate an example dehumidification system 1700 with a first modulating valve 1702 that may be used to control the ratio of sensible heat to latent heat for the surrounding airflow. First modulating valve 1702 may be configured to actuate between different modes of operation based on input received from a superheat control evaporator 1704. For example, operations of dehumidification system 1700 may be based, at least in part, on controlling the superheat from one or more evaporator coils of the dehumidification system 1700. Providing increased superheat control may prevent the cooling of an airflow already determined to be at a control or set temperature.

[0171] Dehumidification system 1700 may comprise the first modulating valve 1702, the superheat control evaporator 1704, a primary evaporator 1706, a primary condenser 1708, a secondary evaporator 1710, a secondary condenser 1712, a compressor 1714, a primary metering device 1716, a secondary metering device 1718, a fan 1720, a second modulating valve 1722, and an alternate condenser 1724. As illustrated, the alternate condenser 1724 may be disposed in an external condenser unit 1726 and may be air-cooled. In alternate embodiments, the alternate condenser 1724 may be liquid-cooled. In some embodiments, dehumidification system 1700 may additionally include an optional sub-cooling coil 1728. With reference to the figures, the optional sub-cooling coil 1728 may be disposed adjacent to the primary condenser 1708, wherein sub-cooling coil 1728 and primary condenser 1708 may be combined into a single coil. FIGS. 17A and 17B illustrate an embodiment of dehumidification system 1700 wherein superheat control evaporator 1704 is disposed separately from secondary evaporator 1710, and FIG. 17C illustrates an embodiment of dehumidification system 1700 wherein superheat control evaporator 1704 and secondary evaporator 1710 are jointly integrated into an intermixed coil unit 1730.

[0172] With reference to each of FIGS. 17A-17C, a flow of refrigerant 1732 is circulated through dehumidification system 1700, as illustrated. In general, dehumidification system 1700 may receive one or more inlet airflows 1734, remove water from the one or more inlet airflows 1734, and output a dischargeable airflow 1736. In embodiments, dischargeable airflow 1736 may be at least partially dehumidified and/or at a lower temperature than the one or more inlet airflows 1734. Water may be removed from the one or more inlet airflows 1734 using a refrigeration cycle of flow of refrigerant 1732. By including secondary evaporator 1710 and secondary condenser 1712, however, dehumidification system 1700 may cause at least part of the flow of refrigerant 1732 to evaporate and condense twice in a single refrigeration cycle. This may increase the refrigeration capacity over

typical systems without adding any additional power to the compressor, thereby increasing the overall dehumidification efficiency of the system.

[0173] In general, dehumidification system 1700 may attempt to match the saturating temperature of secondary evaporator 1710 to the saturating temperature of secondary condenser 1712. The saturating temperature of secondary evaporator 1710 and secondary condenser 1712 generally is controlled according to the equation: (temperature of inlet airflow 1734+temperature of an airflow received by secondary condenser 1712)/2. As the saturating temperature of secondary evaporator 1710 is lower than inlet airflow 1734, evaporation happens in secondary evaporator 1710. As the saturating temperature of secondary condenser 1712 is higher than an airflow received by secondary condenser 1712, condensation happens in the secondary condenser 1712. The amount of refrigerant 1732 evaporating in secondary evaporator 1710 may be substantially equal to that condensing in secondary condenser 1712.

[0174] Superheat control evaporator 1704 may receive a flow of refrigerant 1732 from primary evaporator 1706 and may output a flow of refrigerant 1732 to compressor 1714. Superheat control evaporator 1704 may be any type of coil (e.g., fin tube, micro channel, etc.). With reference to FIG. 17A, superheat control evaporator 1704 may receive inlet airflow 1734 and may output a first airflow 1738 to secondary evaporator 1710. First airflow 1738, in general, is at a cooler temperature than inlet airflow 1734. To cool incoming inlet airflow 1734, superheat control evaporator 1734 may transfer heat from inlet airflow 1734 to the flow of refrigerant 1732, thereby causing the flow of refrigerant 1732 to evaporate at least partially from liquid to gas.

[0175] Secondary evaporator 1710 may receive the flow of refrigerant 1732 from primary metering device 1716 and may output a flow of refrigerant 1732 to the first modulating valve 1702. Similar to superheat control evaporator 1704, secondary evaporator 1710 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary evaporator 1710 receives first airflow 1738 and outputs a second airflow 1740 to primary evaporator 1706. Second airflow 1740, in general, is at a cooler temperature than first airflow 1738. To cool incoming first airflow 1738, secondary evaporator 1710 transfers heat from first airflow 1738 to flow of refrigerant 1732, thereby causing flow of refrigerant 1732 to evaporate at least partially from liquid to gas.

[0176] In this embodiment, superheat control evaporator 1704 may be disposed upstream of both the secondary evaporator 1710 and primary evaporator 1706 with reference to the air flow through the dehumidification system 1700. For example, the superheat control evaporator 1704 may be disposed in series with the other coils of dehumidification system 1700 and may be the first component to receive one or more inlet airflows 1734. In other embodiments, superheat control evaporator 1704 may be disposed in various other configurations in view of the other components of dehumidification system 1700.

[0177] For example, with reference now to FIG. 17B, superheat control evaporator 1704 may be disposed parallel to the secondary evaporator 1710. Superheat control evaporator 1704 and secondary evaporator 1710 may receive one or more inlet airflows 1734 and may discharge treated air to primary evaporator 1706. In this embodiment, superheat control evaporator 1704 may receive a first inlet airflow 1734a, and secondary evaporator 1710 may receive a second

inlet airflow 1734b. Superheat control evaporator 1704 may transfer heat from first inlet airflow 1734a to the flow of refrigerant 1732 flowing through the superheat control evaporator 1704 and output the generated first airflow 1738 to primary evaporator 1706. Similarly, secondary evaporator 1710 may transfer heat from second inlet airflow 1734b to the flow of refrigerant 1732 flowing through the secondary evaporator 1710 and output second airflow 1740 to primary evaporator 1706. As illustrated, primary evaporator 1706 may receive both the first airflow 1738 and the second airflow 1740 for further operations within dehumidification system 1700. The present embodiment may provide the superheat control evaporator 1704 and secondary evaporator 1710 being disposed separate from each other and receiving separate inlet airflows 1734. Further, there may be separate ductwork and/or conduits directing the airflows received by the dehumidification system 1700 (i.e., inlet airflows 1734a, b) towards the primary evaporator 1706.

[0178] FIG. 17C illustrates another embodiment of dehumidification system 1700, wherein both the superheat control evaporator 1704 and secondary evaporator 1710 are jointly integrated into the intermixed coil unit 1730. In certain embodiments, the superheat control evaporator 1704 and secondary evaporator 1710 may be collectively referred to as an “intermixed coil unit 1730” when coupled together. The intermixed coil unit 1730 may comprise any suitable size, height, shape, and any combinations thereof. The intermixed coil unit 1730 may further comprise any suitable housing or containment equipment for the superheat control evaporator 1704 and the secondary evaporator 1710. For example, the superheat control evaporator 1704 may be physically coupled or secured to the secondary evaporator 1710 in order for both evaporator coils 1704, 1710 to receive the same inlet airflow 1734. While both the superheat control evaporator 1704 and secondary evaporator 1710 may be coupled together, there may be separate flowpaths for the refrigerant 1732 flowing through superheat control evaporator 1704 and secondary evaporator 1710 (as illustrated). In embodiments, the intermixed coil unit 1730 may receive the inlet airflow 1734 and may output the first airflow 1738 to the primary evaporator 1706. The first airflow 1738 may be generated by transferring heat from the inlet airflow 1734 to the flow of refrigerant 1732 within both the superheat control evaporator 1704 and secondary evaporator 1710 as the inlet airflow 1734 passes through both the superheat control evaporator 1704 and the secondary evaporator 1710.

[0179] Referring back to each of FIGS. 17A-17C, the primary evaporator 1706 may receive a flow of refrigerant 1732 and may output the flow of refrigerant 1732 to the superheat control evaporator 1704. As illustrated, the primary evaporator 1706 may receive the flow of refrigerant 1732 from secondary metering device 1718, wherein the secondary metering device 1718 may receive the flow of refrigerant 1732 from the secondary condenser 1712 and/or from the first modulating valve 1702. In other embodiments, the primary evaporator 1706 may receive the flow of refrigerant 1732 from the first modulating valve 1702, wherein the first modulating valve 1702 may direct refrigerant to bypass both secondary condenser 1712 and secondary metering device 1718, wherein the output of the first modulating valve 1702 is connected to a location downstream of the secondary metering device 1718 but upstream of the primary evaporator 1706.

[0180] Primary evaporator 1706 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary evaporator 1706 may be configured to receive first airflow 1738 and/or second airflow 1740 and generate a third airflow 1742 to be discharged. For example, with reference to FIG. 17A, primary evaporator 1706 may receive second airflow 1740 from secondary evaporator 1710. Referring to FIG. 17B, primary evaporator 1706 may receive second airflow 1740 from secondary evaporator 1710 and may receive first airflow 1738 from superheat control evaporator 1704. With reference to FIG. 17C, primary evaporator 1706 may receive first airflow 1738 from intermixed coil unit 1730. Third airflow 1742, in general, is at a cooler temperature than first airflow 1738 and/or second airflow 1740. In embodiments, primary evaporator 1706 may transfer heat from first airflow 1738 and/or second airflow 1740 to the flow of refrigerant 1732, thereby causing flow of refrigerant 1732 to evaporate at least partially from liquid to gas. This transfer of heat from first airflow 1738 and/or second airflow 1740 to flow of refrigerant 1732 may further remove water from first airflow 1738 and/or second airflow 1740.

[0181] Secondary condenser 1712 may receive the flow of refrigerant 1732 from the first modulating valve 1702 and may output the flow of refrigerant 1732 to secondary metering device 1718. Secondary condenser 1712 may be any type of coil (e.g., fin tube, micro channel, etc.). Secondary condenser 1712 may receive the third airflow 1742 from primary evaporator 1706 and may output a fourth airflow 1744. Fourth airflow 1744 may be, in general, warmer and drier (i.e., the dew point may be the same but relative humidity may be lower) than third airflow 1742. Secondary condenser 1712 may generate fourth airflow 1744 by transferring heat from flow of refrigerant 1732 to third airflow 1742, thereby causing flow of refrigerant 1732 to condense at least partially from gas to liquid.

[0182] First modulating valve 1702 may be configured to receive the flow of refrigerant 1732 from secondary evaporator 1710 and to direct the flow of refrigerant 1732 to secondary condenser 1712, to primary evaporator 1706, or to both. In embodiments, the first modulating valve 1702 may operate based, at least in part, on the superheat measured at one or more of the evaporator coils within dehumidification system 1700, such as at superheat control evaporator 1704. Dehumidification system 1700 may utilize first modulating valve 1702 to direct the flow of refrigerant 1732 to secondary condenser 1712, to bypass the secondary condenser 1712 and towards the primary evaporator 1706, or a combination thereof. Depending on a feedback loop, first modulating valve 1702 may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant 1732 to the secondary condenser 1712 and direct a remaining portion of the flow of refrigerant 1732 to the primary evaporator 1706.

[0183] In embodiments, dehumidification system 1700 may operate in a first mode of operation. During the first mode of operation, the first modulating valve 1702 may be actuated to direct the flow of refrigerant 1732 to secondary condenser 1712. As the refrigerant 1732 flows through the secondary condenser 1712, the secondary condenser 1712 may generate the fourth airflow 1744. The dehumidification system 1700 may operate in the first mode of operation to dehumidify or remove water from the air to be output as the dischargeable airflow 1736. In further embodiments, dehumidification system 1700 may operate in a second mode of

operation. During the second mode of operation, the first modulating valve 1702 may be actuated to direct the flow of refrigerant 1732 to primary evaporator 1706, thereby bypassing the secondary condenser 1712. As the refrigerant 1732 does not flow through the secondary condenser 1712, the secondary condenser 1712 may not be capable of transferring heat between the refrigerant 1732 and the received third airflow 1742. As a result, the resulting airflow passing through the secondary condenser 1712 (i.e., the fourth airflow 1744) may comprise approximately the same temperature and humidity as the third airflow 1742. The dehumidification system 1700 may operate in the second mode of operation to lower the temperature of the air to be output as the dischargeable airflow 1736 and not to dehumidify. In other embodiments, the dehumidification system 1700 may operate in a third mode of operation, wherein the first modulating valve 1702 is operable to direct a portion of the flow of refrigerant 1732 to the secondary condenser 1712 and a remaining portion of the flow of refrigerant 1732 to the primary evaporator 1706. As at least a portion of the refrigerant 1732 flows through the secondary condenser 1712, the secondary condenser 1712 may generate the fourth airflow 1744 by transferring heat from that portion of refrigerant 1732 to the received third airflow 1742. In embodiments, the fourth airflow 1744 of the third mode of operation may be more humid than the fourth airflow 1744 of the first mode of operation and less humid than the fourth airflow 1744 of the second mode of operation. Further, the fourth airflow 1744 of the third mode of operation may be at a greater temperature than the fourth airflow 1744 of the second mode of operation and at a lower temperature than the fourth airflow 1744 of the first mode of operation.

[0184] The primary condenser 1708 may be any type of coil (e.g., fin tube, micro channel, etc.). Primary condenser 1708 is operable to receive the flow of refrigerant 1732 from the second modulating valve 1722 and output the flow of refrigerant 1732 to either primary metering device 1716 or sub-cooling coil 1728. As illustrated, primary condenser 1708 may output the flow of refrigerant 1732 to the optional sub-cooling coil 1728 before the flow of refrigerant 1732 flows to primary metering device 1716. In these embodiments, sub-cooling coil 1728 may be optional for dehumidification system 1700, and primary condenser 1708 may alternatively direct the flow of refrigerant 1732 to the primary metering device 1716. Primary condenser 1708 may be configured to receive a fifth airflow 1746 generated by the sub-cooling coil 1728 and output dischargeable airflow 1736. With reference to each of FIGS. 17A-17C, dischargeable airflow 1736 may be, in general, warmer and drier (i.e., have a lower relative humidity) than either fourth airflow 1744 or fifth airflow 1746. Primary condenser 1708 may generate dischargeable airflow 1736 by transferring heat away from flow of refrigerant 1732, thereby causing flow of refrigerant 1732 to condense at least partially from gas to liquid. In some embodiments, primary condenser 1708 completely condenses flow of refrigerant 1732 to a liquid (i.e., 100% liquid). In other embodiments, primary condenser 1708 partially condenses flow of refrigerant 1732 to a liquid (i.e., less than 100% liquid).

[0185] Sub-cooling coil 1728, which is an optional component of dehumidification system 1700, may be configured to sub-cool the liquid refrigerant 1732 as it leaves the primary condenser 1708, the alternate condenser 1724, or combinations thereof. In embodiments wherein the sub-

cooling coil 1728 is disposed adjacent to the primary condenser 1708, the sub-cooling coil 1728 may receive refrigerant 1732 as it leaves the primary condenser 1708 and/or the alternate condenser 1724, as seen in FIGS. 17A-17C. This, in turn, may supply primary metering device 1716 with a liquid refrigerant that is up to 30 degrees (or more) cooler than before it enters sub-cooling coil 1728. For example, if flow of refrigerant 1732 entering sub-cooling coil 1728 is 340 psig/105° F./60% vapor, flow of refrigerant 1732 may be 340 psig/80° F./0% vapor as it leaves sub-cooling coil 1728. The sub-cooled refrigerant 1732 has a greater heat enthalpy factor as well as a greater density, which results in reduced cycle times and frequency of the evaporation cycle of flow of refrigerant 1732. This may result in greater efficiency and less energy use of dehumidification system 1700.

[0186] Compressor 1714 may be configured to pressurize the flow of refrigerant 1732, thereby increasing the temperature of refrigerant 1732. For example, if flow of refrigerant 1732 entering compressor 1714 is 128 psig/52° F./100% vapor, flow of refrigerant 1732 may be 340 psig/150° F./100% vapor as it leaves compressor 1714. Compressor 1714 may be configured to receive flow of refrigerant 1732 from superheat control evaporator 1704 and to supply the pressurized flow of refrigerant 1732 to the second modulating valve 1722.

[0187] Second modulating valve 1722 may be operable to receive the pressurized flow of refrigerant 1732 from compressor 1714 and to direct the flow of refrigerant 1732 to primary condenser 1708, to alternate condenser 1724, or to both. In embodiments, the second modulating valve 1722 may operate based, at least in part, on a pre-determined temperature set point for the dischargeable airflow 1736 and on an actual temperature of the dischargeable airflow 1736 output by dehumidification system 1700. Dehumidification system 1700 may utilize second modulating valve 1722 to direct heat to be rejected from the flow of refrigerant 1732 away from the primary condenser 1708 and towards the alternate condenser 1724. Depending on a feedback loop comprising of the pre-determined temperature set point and the actual temperature of the dischargeable airflow 1736, second modulating valve 1722 may be configured to partially open and/or close to direct at least a portion of the flow of refrigerant 1732 to the alternate condenser 1724 and direct a remaining portion of the flow of refrigerant 1732 to the primary condenser 1708.

[0188] During operation of dehumidification system 1700, the second modulating valve 1722 may direct the flow of refrigerant 1732 to primary condenser 1708 if the temperature of the dischargeable airflow 1736 output by the primary condenser 1708 does not exceed the pre-determined temperature set point monitored by the dehumidification system 1700. If the temperature of the dischargeable airflow 1736 is greater than the pre-determined temperature set point, the second modulating valve 1722 may be actuated to direct at least a portion of the flow of refrigerant 1732 to the alternate condenser 1724 and direct a remaining portion of the flow of refrigerant 1732 to the primary condenser 1708. As the dehumidification system 1700 operates, reduction in the volume of flow of refrigerant 1732 to primary condenser 1708 may reduce the available heat to be rejected into the dischargeable airflow 1736. With the reduced flow of refrigerant 1732 passing through primary condenser 1708 (for example, the remaining portion of the flow of refrigerant), the rate of heat transfer to the dischargeable airflow 1736

may subsequently be reduced, thereby producing a reduction in the temperature change of an incoming airflow and the output dischargeable airflow 1736. Once the temperature of the dischargeable airflow 1736 is lower than the pre-determined temperature set point, the second modulating valve 1722 may be actuated to direct the at least a portion of the flow of refrigerant 1732 back to the primary condenser 1708. Any remaining refrigerant 1732 that had been directed to alternate condenser 1724 may combine with the flow of refrigerant 1732 further downstream.

[0189] As illustrated, alternate condenser 1724 may be disposed in the external condenser unit 1726 and may be any type of coil (e.g., fin tube, micro channel, etc.) operable to receive flow of refrigerant 1732 from second modulating valve 1722 and output flow of refrigerant 1732 at a lower temperature. Alternate condenser 1724 may be configured to transfer heat away from flow of refrigerant 1732, thereby causing flow of refrigerant 1732 to condense at least partially from gas to liquid. In some embodiments, alternate condenser 1724 completely condenses flow of refrigerant 1732 to a liquid (i.e., 100% liquid). In other embodiments, alternate condenser 1724 partially condenses flow of refrigerant 1732 to a liquid (i.e., less than 100% liquid). Alternate condenser 1724 may receive a first outdoor airflow 1748 and output a second outdoor airflow 1750. Second outdoor airflow 1750 is, in general, warmer (i.e., have a lower relative humidity) than first outdoor airflow 1748. As illustrated, the external condenser unit 1726 may include the alternate condenser 1724 and a fan 1752, wherein fan 1752 may be configured to facilitate flow of first outdoor airflow 1748 towards alternate condenser 1724. While alternate condenser 1724 may be air-cooled, alternate condenser 1724 may alternatively be liquid-cooled. In one or more embodiments, alternate condenser 1724 may be any type of liquid-cooled heat exchanger operable to transfer heat from the flow of refrigerant 1732 to the flow of a suitable fluid, such as water or a mixture of water and glycol.

[0190] Referring back to each of FIGS. 17A-17C, fan 1720 may include any suitable components operable to draw one or more inlet airflows 1734 into dehumidification system 1700 and through superheat control evaporator 1704, secondary evaporator 1710, primary evaporator 1706, secondary condenser 1712, sub-cooling coil 1728, and primary condenser 1708. Fan 1720 may be any type of air mover (e.g., axial fan, forward inclined impeller, and backward inclined impeller, etc.). For example, fan 1720 may be a backward inclined impeller positioned adjacent to primary condenser 1720 as illustrated in FIGS. 17A-17C. While fan 1720 is depicted as being located adjacent to primary condenser 1708, it should be understood that fan 1720 may be located anywhere along the airflow path of dehumidification system 1700. For example, fan 1720 may be positioned in the airflow path of any one of airflows 1734, 1738, 1740, 1742, 1744, 1746, or 1736. Moreover, dehumidification system 1700 may include one or more additional fans positioned within any one or more of these airflow paths. Similarly, while the fan 1752 of external condenser unit 1726 is depicted as being located adjacent alternate condenser 1724, it should be understood that fan 1752 may be located anywhere (e.g., above, below, beside) with respect to alternate condenser 1724, so long as fan 1752 is appropriately positioned and configured to facilitate flow of first outdoor airflow 1748 towards alternate condenser 1724.

[0191] Primary metering device 1716 and secondary metering device 1718 are any appropriate type of metering/expansion device. In some embodiments, primary metering device 1716 is a thermostatic expansion valve (TXV) and secondary metering device 1718 is a fixed orifice device (or vice versa). In certain embodiments, metering devices 1716, 1718 remove pressure from flow of refrigerant 1732 to allow expansion or change of state from a liquid to a vapor in evaporators 1710, 1706. The high-pressure liquid (or mostly liquid) refrigerant entering metering devices 1716, 1718 is at a higher temperature than the liquid refrigerant 1732 leaving metering devices 1716, 1718. For example, if flow of refrigerant 1732 entering primary metering device 1716 is 340 psig/80° F./0% vapor, flow of refrigerant 1732 may be 196 psig/68° F./5% vapor as it leaves primary metering device 1716. As another example, if flow of refrigerant 1732 entering secondary metering device 1718 is 196 psig/68° F./4% vapor, flow of refrigerant 1732 may be 128 psig/44° F./14% vapor as it leaves secondary metering device 1718.

[0192] Refrigerant 1732 may be any suitable refrigerant such as R410a. In general, dehumidification system 1700 utilizes a closed refrigeration loop of refrigerant 1732 that passes from compressor 1714 through second modulating valve 1722, primary condenser 1708 and/or alternate condenser 1724, (optionally) sub-cooling coil 1728, primary metering device 1716, secondary evaporator 1710, first modulating valve 1702, secondary condenser 1712 and/or secondary metering device 1718 (where refrigerant 1732 may bypass secondary condenser 1712), primary evaporator 1704, and superheat control evaporator 1704. Compressor 1714 pressurizes flow of refrigerant 1732, thereby increasing the temperature of refrigerant 1732. Primary and secondary condensers 1708, 1712, which may include any suitable heat exchangers, cool the pressurized flow of refrigerant 1732 by facilitating heat transfer from the flow of refrigerant 1732 to the respective airflows passing through them (i.e., fourth or fifth airflow 1744, 1746 and third airflow 1742, respectively). Further, alternate condenser 1724, which may include any suitable heat exchanger, cools the pressurized flow of refrigerant 1732 by facilitating heat transfer from the flow of refrigerant 1732 to either the airflow passing through it (i.e., first outdoor airflow 1748) or to the flow of a fluid provided by an external source. The cooled flow of refrigerant 1732 leaving primary and/or alternate condensers 1708, 1724 may enter primary metering device 1716, which is operable to reduce the pressure of flow of refrigerant 1732, thereby reducing the temperature of flow of refrigerant 1732. In embodiments, the refrigerant 1732 may first flow through the optional sub-cooling coil 1728 before engaging with the primary metering device 1716. Depending on the mode of operation, the cooled flow of refrigerant 1732 leaving secondary condenser 1712 may enter secondary metering device 1718, which is operable to reduce the pressure of flow of refrigerant 1732, thereby reducing the temperature of flow of refrigerant 1732. The refrigerant 1732 may alternatively be received by the secondary metering device 1718 from the first modulating valve 1702, bypassing the secondary condenser 1712. Primary and secondary evaporators 1706, 1710, which may include any suitable heat exchanger, receive flow of refrigerant 1732 from secondary metering device 1718 and primary metering device 1716, respectively. Primary and secondary evaporators 1706, 1710 facilitate the transfer of heat from the respective airflows passing through them to the flow of

refrigerant 1732. Flow of refrigerant 1732, after leaving primary evaporator 1706, may be received by the superheat control evaporator 1704. The superheat control evaporator 1704 may facilitate transfer of heat from inlet airflow 1734 passing therethrough to the flow of refrigerant 1732. Then, the refrigerant 1732 may be directed back to compressor 1714, and the cycle is repeated.

[0193] In certain embodiments, the above-described refrigeration loop may be configured such that evaporators 1706, 1710 operate in a flooded state. In other words, flow of refrigerant 1732 may enter evaporators 1706, 1710 in a liquid state, and a portion of flow of refrigerant 1732 may still be in a liquid state as it exits evaporators 1706, 1710. Accordingly, the phase change of flow of refrigerant 1732 (liquid to vapor as heat is transferred to flow of refrigerant 1732) occurs across evaporators 1706, 1710, resulting in nearly constant pressure and temperature across the entire evaporators 1706, 1710 (and, as a result, increased cooling capacity). In these embodiments, superheat control evaporator 1704 may additionally operate in a flooded state.

[0194] In operation of example embodiments of dehumidification system 1700, one or more inlet airflows 1734 may be drawn into dehumidification system 1700 by fan 1720. The one or more inlet airflows 1734 may pass through superheat control evaporator 1704 and/or secondary evaporator 1710 in which heat is transferred from the one or more inlet airflows 1734 to the cooler flow of refrigerant 1732 passing through evaporators 1704, 1710. As a result, the one or more inlet airflows 1734 may be cooled. As an example, if one or more inlet airflows 1734 is 80° F./60% humidity, superheat control evaporator 1704 and/or secondary evaporator 1710 may output first airflow 1738 and/or second airflow 1740 at 70° F./84% humidity. This may cause flow of refrigerant 1732 to partially vaporize within superheat control evaporator 1704 and/or secondary evaporator 1710. For example, if flow of refrigerant 1732 entering superheat control evaporator 1704 and/or secondary evaporator 1710 is 196 psig/68° F./5% vapor, flow of refrigerant 1732 may be 196 psig/68° F./38% vapor as it leaves superheat control evaporator 1704 and/or secondary evaporator 1710.

[0195] The cooled one or more inlet airflows 1734 may be discharged from superheat control evaporator 1704 and/or secondary evaporator 1710 as first airflow 1738 and/or second airflow, respectively, and may enter primary evaporator 1706. Like superheat control evaporator 1704 and/or secondary evaporator 1710, primary evaporator 1706 may transfer heat from first airflow 1738 and/or second airflow 1740 to the cool flow of refrigerant 1732 passing through primary evaporator 1706. As a result, the air may be cooled to or below its dew point temperature, causing moisture in first airflow 1738 and/or second airflow 1740 to condense (thereby reducing the absolute humidity of first airflow 1738 and/or second airflow 1740). As an example, if first airflow 1738 and/or second airflow 1740 is 70° F./84% humidity, primary evaporator 1706 may output third airflow 1742 at 54° F./98% humidity. This may cause the flow of refrigerant 1732 to partially or completely vaporize within primary evaporator 1706. For example, if flow of refrigerant 1732 entering primary evaporator 1706 is 128 psig/44° F./14% vapor, flow of refrigerant 1732 may be 128 psig/52° F./100% vapor as it leaves primary evaporator 1706.

[0196] The third airflow 1742 may be discharged from primary evaporator 1706 and may enter secondary condenser 1712. Secondary condenser 1712 may be configured

to facilitate heat transfer from the hot flow of refrigerant 1732 passing through the secondary condenser 1712 to third airflow 1742, depending on the mode of operation. This reheats third airflow 1742, thereby decreasing the relative humidity of third airflow 1742. As an example, if third airflow 1742 is 54° F./98% humidity, secondary condenser 1712 may output fourth airflow 1744 at 65° F./68% humidity. This may cause flow of refrigerant 1732 to partially or completely condense within secondary condenser 1712. For example, if flow of refrigerant 1732 entering secondary condenser 1712 is 196 psig/68° F./38% vapor, flow of refrigerant 1732 may be 196 psig/68° F./4% vapor as it leaves secondary condenser 1712.

[0197] In some embodiments, the fourth airflow 1744 may be discharged and may enter the optional sub-cooling coil 1728. Sub-cooling coil 1728 facilitates heat transfer from the hot flow of refrigerant 1732 passing through sub-cooling coil 1728 to fourth airflow 1744 to generate the fifth airflow 1746 to be output to the primary condenser 1708. In other embodiments, the fourth airflow 1744 may be discharged and may enter the primary condenser 1708 without flowing through the sub-cooling coil 1728. Primary condenser 1708 facilitates heat transfer from the hot flow of refrigerant 1732 passing through the primary condenser 1708 to fourth airflow 1744 or fifth airflow 1746. This further heats fourth airflow 1744 or fifth airflow 1746, thereby further decreasing the relative humidity of fourth airflow 1744 or fifth airflow 1746. As an example, if fourth airflow 1744 or fifth airflow 1746 is 65° F./68% humidity, primary condenser 1708 may output dischargeable airflow 1736 at 102° F./19% humidity. This may cause flow of refrigerant 1732 to partially or completely condense within primary condenser 1708. For example, if flow of refrigerant 1732 entering primary condenser 1708 is 340 psig/150° F./100% vapor, flow of refrigerant 1732 may be 340 psig/105° F./60% vapor as it leaves primary condenser 1708.

[0198] Some embodiments of dehumidification system 1700 may include a controller that may include one or more computer systems at one or more locations. Each computer system may include any appropriate input devices (such as a keypad, touch screen, mouse, or other device that can accept information), output devices, mass storage media, or other suitable components for receiving, processing, storing, and communicating data. Both the input devices and output devices may include fixed or removable storage media such as a magnetic computer disk, CD-ROM, or other suitable media to both receive input from and provide output to a user. Each computer system may include a personal computer, workstation, network computer, kiosk, wireless data port, personal data assistant (PDA), one or more processors within these or other devices, or any other suitable processing device. In short, the controller may include any suitable combination of software, firmware, and hardware.

[0199] The controller may additionally include one or more processing modules. Each processing module may each include one or more microprocessors, controllers, or any other suitable computing devices or resources and may work, either alone or with other components of dehumidification system 1700, to provide a portion or all of the functionality described herein. The controller may additionally include (or be communicatively coupled to via wireless or wireline communication) computer memory. The memory may include any memory or database module and may take the form of volatile or non-volatile memory,

including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), removable media, or any other suitable local or remote memory component.

[0200] Although particular implementations of dehumidification system 1700 are illustrated and primarily described, the present disclosure contemplates any suitable implementation of dehumidification system 1700, according to particular needs. Moreover, although various components of dehumidification system 1700 have been depicted as being located at particular positions and relative to one another, the present disclosure contemplates those components being positioned at any suitable location, according to particular needs.

[0201] Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such as, for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0202] Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

[0203] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

What is claimed is:

1. A dehumidification system comprising:
 - a primary metering device;
 - a secondary metering device;
 - an intermixed coil unit operable to:
 - receive an inlet airflow and output a first airflow, the first airflow comprising cooler air than the first inlet airflow, the first airflow generated by transferring heat from the inlet airflow to a flow of refrigerant within a superheat control evaporator and a secondary evaporator as the inlet airflow passes through both the superheat control evaporator and the secondary evaporator,
 - wherein the intermixed coil unit comprises:
 - the superheat control evaporator operable to receive a flow of refrigerant from a primary evaporator; and
 - the secondary evaporator operable to receive the flow of refrigerant from the primary metering device;
 - a primary evaporator operable to:
 - receive the flow of refrigerant from a first modulating valve and/or a secondary condenser; and
 - receive the first airflow and output a second airflow, the second airflow comprising cooler air than the first airflow, the second airflow generated by transferring heat from the first airflow to the flow of refrigerant as the first airflow pass through the primary evaporator;
 - the secondary condenser operable to:
 - receive the flow of refrigerant from the first modulating valve; and
 - receive the second airflow and output a third airflow;
 - the first modulating valve operable to:
 - receive the flow of refrigerant from the secondary evaporator;
 - direct the flow of refrigerant to the secondary condenser during a first mode of operation;
 - direct the flow of refrigerant to the primary evaporator during a second mode of operation, wherein the flow of refrigerant bypasses the secondary condenser; and
 - direct a portion of the flow of refrigerant to the secondary condenser and a remaining portion of the flow of refrigerant to the primary evaporator during a third mode of operation;
 - a compressor configured to:
 - receive the flow of refrigerant from the superheat control evaporator and discharge the flow of refrigerant at a higher pressure than the flow of refrigerant received at the compressor; and
 - a primary condenser operable to:
 - receive the flow of refrigerant discharged from the compressor;
 - in response to receiving the flow of refrigerant from the compressor, output a dischargeable airflow.
2. The dehumidification system of claim 1, further comprising a sub-cooling coil operable to:
 - receive the flow of refrigerant from the primary condenser;
 - output the flow of refrigerant to the primary metering device; and
 - receive the third airflow and output a fourth airflow, the fourth airflow generated by transferring heat from the flow of refrigerant to the third airflow as the third airflow passes through the sub-cooling coil.
3. The dehumidification system of claim 2, wherein the sub-cooling coil and primary condenser are combined in a single coil unit.
4. The dehumidification system of claim 2, wherein the primary condenser is further operable to:
 - receive the fourth airflow and generate the dischargeable airflow based on the received fourth airflow, the dischargeable airflow generated by transferring heat from the flow of refrigerant to the fourth airflow as the fourth airflow contacts the primary condenser.
5. The dehumidification system of claim 1, wherein the primary condenser is further operable to:
 - receive the third airflow and generate the dischargeable airflow based on the received third airflow, the dischargeable airflow generated by transferring heat from the flow of refrigerant to the third airflow as the third airflow contacts the primary condenser.
6. The dehumidification system of claim 1, wherein during the first mode of operation:
 - the third airflow comprises warmer and less humid air than the second airflow, the third airflow generated by transferring heat from the flow of refrigerant to the second airflow as the second airflow passes through the secondary condenser.
7. The dehumidification system of claim 1, wherein during the second mode of operation, the third airflow comprises approximately the same temperature and humidity as the second airflow.
8. The dehumidification system of claim 1, further comprising an alternate condenser disposed in an external condenser unit, operable to:
 - receive the flow of refrigerant discharged from the compressor;
 - in response to receiving the flow of refrigerant from the compressor, output a second dischargeable airflow, the second dischargeable airflow generated by transferring heat away from the flow of refrigerant received by the compressor.
9. The dehumidification system of claim 1, wherein two or more members selected from the group consisting of the secondary evaporator, the primary evaporator, and the secondary condenser are combined in a single coil pack.
10. The dehumidification system of claim 1, wherein at least one of the primary evaporator and the secondary evaporator comprises two or more circuits for the flow of refrigerant.

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